

Trainer's Guide

Junior Software Developer



Qualifications Pack-Junior Software Developer

- ◆ **Sector** : IT-ITes
- ◆ **Sub-sector** : IT Services
- ◆ **Occupation** : Application Development
- ◆ **Reference ID** : SSC/Q0508

Junior Software Developer Trainer's Guide



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Welcome Note

Dear Trainer,

Welcome to this training programme on Junior Software Developer. This training programme has been developed in response to the need of skilled programmer in the IT industry and IT enabled services. An organised sector such as this, calls for professionals with in-depth knowledge of the subject and work ethics. Each participant you train can go on to join as a junior software programmer in any prestigious IT company.

As a trainer you will have to ensure quality training and must have the willingness and ability to pass on your knowledge and skills. It is crucial for the programme's success that the trainees interact with each other and share knowledge. Your role, therefore, is more of a facilitator who will guide and add value to the learning environment because of your knowledge, skill, experience and personality.

The total duration of the Training Programme is 50 working days. Its terminal goal is to enable the participants to develop Web applications applying the technical knowledge and skills gained through this programme.

This programme allows the participants to elect any one of the two Web technologies namely, Visual Studio .NET and Java2 Enterprise Edition (J2EE), after going through a basic training and hands-on practice in applications such as MS Office and programming with C and C++. Moreover, it covers the Relational Database Management System (RDBMS) and Networking concepts.

At the end of the training programme the participants are assigned a project work to be developed in a workshop mode, as a team. This project work will help the participants to study, analyse and design software applications.

Resources provided:

1. Participant Handbook

Part I – [Basics of Information Technology, MS Office, Internet, C, OOP using C++, Data Structures using C++, HTML]

Part II – Elective 1: Visual Studio Dot Net; Elective 2: Java2 Enterprise Edition (J2EE)

Part III– Relational Database Management System (RDBMS); Basics of Hardware and Networking

2. Practical Guide

Part I – [C, OOP using C++, Data Structures using C++, HTML]

Part II – Elective 1: Visual Studio Dot Net; Elective 2: Java2 Enterprise Edition (J2EE)

Part III – Relational Database Management System (RDBMS)

3. Multimedia Content
4. Trainer's Guide
5. PowerPoint Presentation
6. Session Plan
7. Syllabus
8. Assessment Guide

Install the following software for the training programme:

1. Windows Operating System
2. MS Office 2007
3. Turbo C 3.0
4. Notepad
5. Dreamweaver
6. Mozilla Fire Fox or Internet Explorer (Browser)
7. J2 EE JDK, Netbeans/Eclipse
8. Visual Studio .NET
9. Oracle and SQL

We hope that you too will gain from this programme and help us take it to a higher level through your delivery!

Skills acquired through the Training Programme

The IT industry has a positive impact on the socio-economic climate, directly or indirectly. The IT industry is diverse where companies look for multi-skilled, tech-savvy workforce. Today IT skills often prove to be essential for employment even in other (non-IT) sectors.

India is growing as a preferred destination for companies looking for workforce for their offshore back-office jobs mainly due to cost advantage and availability of human resource. Availability of skilled human resource supported by appropriate skill building initiatives will be key to this growth.

The profile of people employed at various levels in this occupation depends on their functional skills, industry experience and job requirements. Graduate or Diploma holders from non-engineering field, able to communicate in English are eligible for this Training Programme.

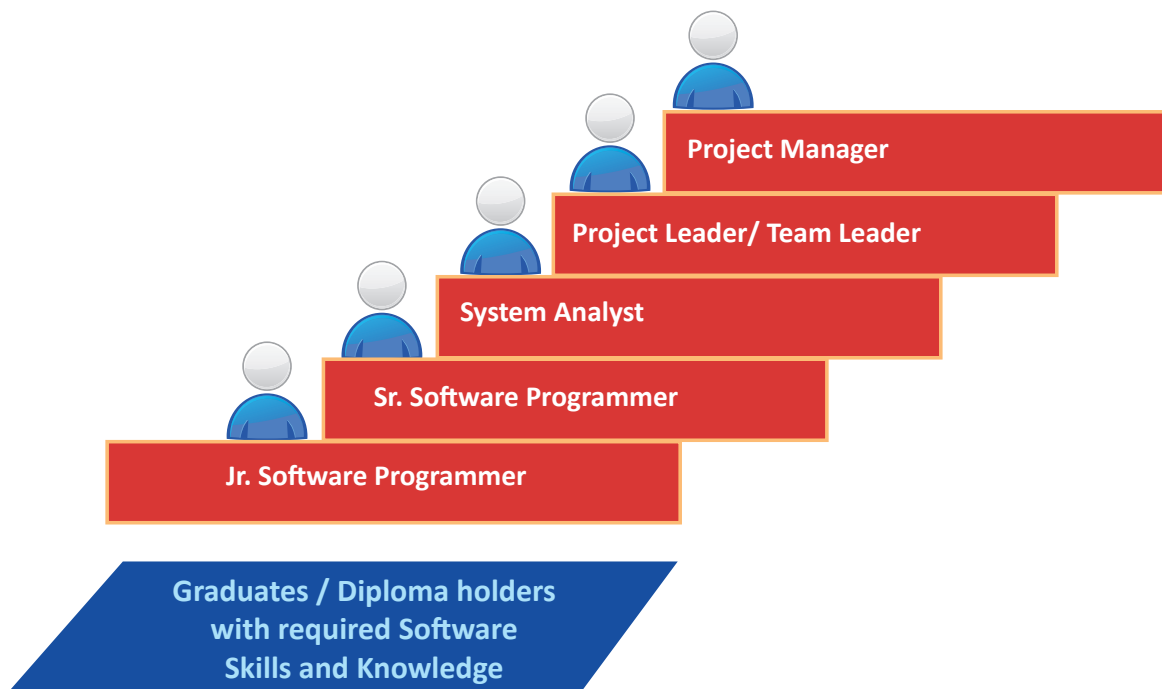
This Training Programme aims to prepare the participants in the following skills:

- ◆ Basic programming skills – with proficiency in programming languages like C++ and Java
- ◆ Adequate technical skills/Expertise in Technology – Domain/Platform and software knowledge
- ◆ Competency to transform functional specifications to design and system specifications
- ◆ Knowledge of process and quality compliance
- ◆ Lead and work with a team
- ◆ Logical and analytical thinking ability

- ◆ Communication skills
- ◆ Ability to understand customer needs
- ◆ Ability to plan, set targets and manage resources and process

With the functional knowledge, skills and hands-on experience, this field offers a promising future with growth prospects.

The Career Path



Role of the Trainer

As a trainer, keep in mind the following guidelines:

Know your job thoroughly

Before you begin your training programme, you must be well-versed with the theory and practical aspects of the job. Prepare well before each session. Study and teach each exercise/activity with respect to these six key points:

- ◆ Method – the best practices
- ◆ Quality – the awareness of quality
- ◆ Waste – minimisation of waste
- ◆ Safety – the practice of safety measures
- ◆ Ergonomics – the care given to body posture and other work place ergonomics
- ◆ Housekeeping – the importance of cleanliness at workplace
- ◆ Productivity

Know your trainees

You are going to be conducting the training over a period of 42 days. For effective training, it is very important for you to build a good rapport with the trainees. In order to have a good working relationship with each of them, it is always best to know them first.

- ◆ Who are they?
- ◆ Are they married? Do they have children?
- ◆ How is their performance on the job?
- ◆ What can you say about their attitude?
- ◆ How do they perceive you as a Trainer?

Practise these common courtesies

- ◆ Greet the trainees.
- ◆ Be warm and friendly. Watch your tone of voice; do not scare them.
- ◆ Introduce yourself.
- ◆ Ask their names.
- ◆ Explain the purpose and goals of the training programme.
- ◆ Ask them what their expectations are.
- ◆ Always ask questions.
- ◆ Listen to them patiently and answer their questions.
- ◆ In case you are unable to respond to a query say that you will get back to them. Do remember to follow up; do not make promises that you cannot keep.
- ◆ Respect the job of the trainees.
- ◆ Do not lean over them, their work, or get in their work space.

- ◆ Do not take their work or move it without asking for their permission.
- ◆ Be a good observer. Watch the trainees while they are at work. Make notes of their performances.
- ◆ Offer remediation for weaker trainees.
- ◆ Correct the faulty practices of trainees on the job before they turn into habits.
- ◆ Do not criticise. Be positive while giving feedback.
- ◆ Show appreciation where it is due.
- ◆ Always say **“Please”, “Thank you”** and **“Sorry”**. This helps create a good impression.
- ◆ Be a mentor to the trainees.

It has been found that those who are already in the industry do not follow standardised practices. Different methods/techniques are practiced by different operators across the country for the same task/job. To add to this, there is little or no quality consciousness. The challenges are many, be it to learn, unlearn, relearn the skills or to be able to build the right attitude, work-place etiquette and behaviour. Therefore, in order to develop the learner holistically, a blend of both technical as well as soft skills is a must and will go a long way in helping the learner find a place for themselves in the industry.

Use of Multimedia Content on K-Yan

About K-Yan

K-Yan is an innovative teaching aid that eliminates the need for other teaching aids such as computers, projectors and others. The K-Yan, developed and patented by IL&FS Education and Technology Services Ltd. (IETS), is a fully integrated unit that contains a high-performance computer, projector, CD/DVD writer and audio system all rolled into one.

Multimedia Content

Multimedia is a combination of multiple media such as text, audio, video, graphics and sound.

Advantages of Multimedia Content:

- ◆ Video-Based Step-By-Step Learning Process: The entire training content has been presented through video demonstrations. Each task/process has been broken down into a series of very simple steps supported by text and voice. Learning through practical video demonstrations has been found to be the most effective tried and tested method of learning a skill such as this.
- ◆ Standardisation of Content: The multimedia content has been developed keeping in mind the best practices of the trade. Therefore, trainees from all regions are exposed to standardised best practices.
- ◆ Increased Training: Multimedia content, when projected through K-Yan can reach out to more number of trainees at a given point in time.
- ◆ Flexibility: The content has been developed with adequate controls given to you to play, replay, or pause the content depending on the learning capabilities of your trainees.
- ◆ Language of Your Choice: The content is available in 9 regional languages. You may choose the language depending on your audience.

Suggested Batch Size

The suggested batch size for this course is around 25 participants.

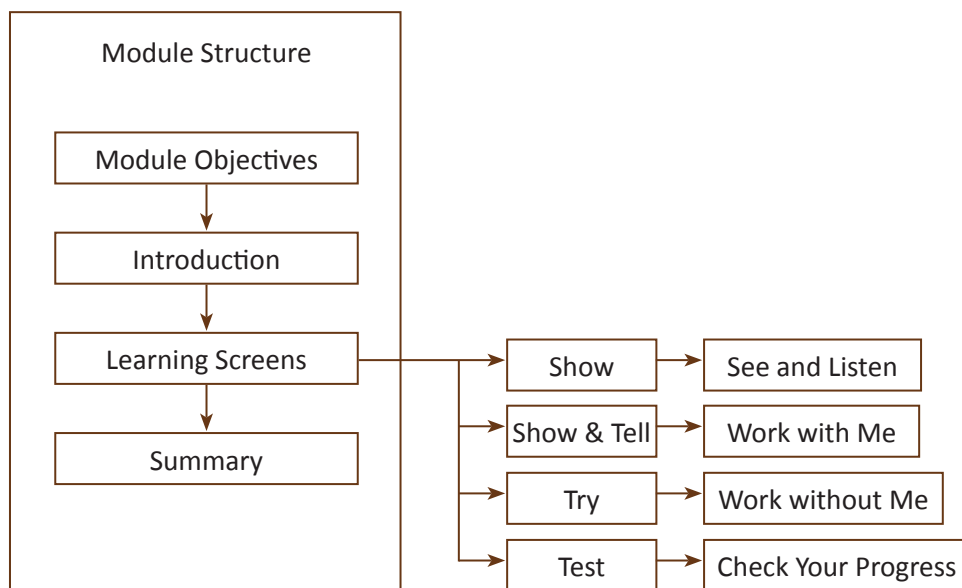
Methodology

It is recommended that you follow a particular method while conducting the training. You should blend your classroom and practical sessions with the training content that has been provided to you. Technical content provided to you has been developed primarily using multimedia technology. Multimedia uses multiple media mixes such as video, text, voice and varied patterns of learning to gain a meaningful learning experience. While you play the multimedia content in the classroom, you, as the Trainer, must provide supporting inputs. Therefore, your contribution in the classroom while the multimedia is being played is very crucial.

All the multimedia modules have been organised in the following manner beginning with objectives and ending with a summary:

- ◆ Objectives
- ◆ Introduction
- ◆ Show
- ◆ Tell
- ◆ Try
- ◆ Test
- ◆ Summary

The detailed module structure is as follows:



Show (See)

This screen, as the name suggests, shows in one glance the entire procedure. The idea is to give you and the learner the big picture in one glance. While you play this screen, observe the learners and gauge them through non-verbal cues. Once the screen ends, ask them whether they have understood the content and respond to their queries, if required.

Tell (Work with me)

After the learners are familiar with the procedure, they must be shown the entire procedure once again, but this time step by step. The screen that follows the “Show” screen is the “Tell” screen. Observe that this screen gives a step-by-step breakup of the entire procedure supported with text, voice and video. Replay the step, if need be. Proceed to the next step only when you are convinced that all participants have understood.

Try (Work under guidance)

Next is the “Try” screen. This screen will have instructions as to what the trainees must do during their practical session. During the practical session, ensure that trainees practice on what they have seen during the classroom sessions. They must be able to perform the steps that were outlined in the “Tell” screen. Your role will be that of a mentor. Observe each participant, record the number of participants who could perform accurately at the first instance, those who performed independently, those that needed guidance, those who could perform within the set productivity norms and so on. For those lagging behind, provide more practice. Play the “Tell” screen of the multimedia content once again, if needed.

Test (Check your progress)

Each module will have a brief test at the end to confirm learning. This could be in the form of theory as well as practical. Theory questions can be tested through a quiz which is part of the multimedia content. You may test the practical aspect during the practical session. Ensure that the participants do this independently without assistance.

Session Plan

Program Name	Junior Software Developer		
Name of Client	NSDC		
Version No.	1.0	Version Update Date	05-10-12
Pre-requisites to Training	Graduate or Diploma Holders in any discipline		
Training Outcomes	After completing this program, participants will be able to: ◆ Code and execute programs in C ◆ Code and execute programs in C++ using the Object Oriented Programming (OOP) ◆ Develop Web Applications using Java2 Enterprise Edition (J2EE) or Visual Studio Dot Net ◆ Create and manipulate database		

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Day 1						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ -Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser Booklet	10 minutes
2	Icebreaker	Icebreaker	To introduce each other and build rapport with fellow participants and Trainer	- Activity - Group participation	Trainer's Guide	50 minutes
3	Basics of Information Technology	Introduction to Computers and Windows Operating System	- State the applications of computers - Identify the Input/ Output devices - State the features of File Management System	- Multimedia based learning - Trainer-led discussion - Hands on practice	- Multimedia (K-yan) - Trainer's Guide - Participant Handbook	1 hour
4	MS Office	Word Processor (MS Word 2007)	- Create and format a document using MS Word 2007 - Insert tables in a MS Word document - Print a MS Word document	- Multimedia based learning - Trainer-led discussion - Hands on practice	- Multimedia (K-yan) - Trainer's Guide - Participant Handbook	1 hour 30 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
5	MS Office	Spreadsheet (MS Excel 2007)	◆ Create and format a spreadsheet using MS Excel 2007 ◆ Create and format charts in MS Excel 2007	◆ Multimedia based learning ◆ Trainer-led discussion ◆ Hands on practice	◆ Multimedia (K-yan) ◆ Trainer's Guide ◆ Participant Handbook	2 hours 30 minutes
6	MS Office	Presentation (MS Power Point 2007)	◆ Create a Presentation using MS PowerPoint 2007	◆ Multimedia based learning ◆ Trainer-led discussion ◆ Hands on practice	◆ Multimedia (K-yan) ◆ Trainer's Guide ◆ Participant Handbook	1 hour 30 minutes
7	Internet	Internet	◆ Browse the internet using a Web Browser	◆ Multimedia based learning ◆ Trainer-led discussion	◆ Multimedia (K-yan) ◆ Trainer's Guide ◆ Participant Handbook	30 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Day 2						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser Booklet	10 minutes
2	Recap	Recap	To revise previous day's learning	- Teach back session - Trainer-led discussion - Reinforcement	- Trainer's Guide - Multimedia (K-Yan)	20 minutes
3	MS Office	Practice Session	To practice previous day's learning	Hands on practice	- Multimedia (K-Yan) - Trainer's Guide - Participant Handbook	3 hours
4	Basic Principles of Programming	Algorithm and Flowchart	- To create an algorithm for solving the given problem - To create a flowchart for solving the given problem	- Multimedia based learning - Trainer-led discussion	- Multimedia (K-Yan) - Trainer's Guide - Participant Handbook	2 hours
5	'C' Programming Language	Basics of 'C' Programming	To state the keywords, operators and data types in 'C'.	Trainer-led discussion	- Trainer's Guide - Participant Handbook	1 hour
6	'C' Programming Language	First 'C' Program	To write a simple program in 'C'.	- Multimedia based learning - Trainer-led discussion	- Multimedia (K-Yan) - Trainer's Guide - Participant Handbook	1 hour 30 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Day 3						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser Booklet	10 minutes
2	Recap	Recap	To revise previous day's learning	- Teach back session - Reinforcement	Trainer's Guide	10 minutes
3	'C' Programming Language	Turbo 'C' editor	- Code a 'C' program using the editor - Compile and fix errors - Run or execute the program	- Multimedia based learning - Trainer-led discussion - Hands on practice	- Multimedia (K-Yan) - Trainer's Guide - Participant Handbook - Practical Guide	1 hour 30 minutes
4	'C' Programming Language	Decision making and branching	To write a program in 'C' using the "if – else" construct	- Multimedia based learning - Trainer-led discussion - Hands on practice	- Multimedia (K-Yan) - Trainer's Guide - Participant Handbook - Practical Guide	1 hour
5	'C' Programming Language	Nested if-else	To write a program in 'C' using multiple if " construct	- Multimedia based learning - Trainer-led discussion - Hands on practice	- Multimedia (K-Yan) - Trainer's Guide - Participant Handbook - Practical Guide	1 hour
6	'C' Programming Language	Logical Operator - AND	To use the logical operator "AND" for testing a combination of conditions	- Multimedia based learning - Trainer-led discussion - Hands on practice	- Multimedia (K-Yan) - Trainer's Guide - Participant Handbook - Practical Guide	30 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
7	'C' Programming Language	Logical Operators - OR and NOT	◆ To use the "OR" logical operator for testing a compound condition ◆ To use the "NOT" logical operator for testing a condition	◆ Multimedia based learning ◆ Trainer-led discussion ◆ Hands on practice	◆ Multimedia (K-Yan) ◆ Trainer's Guide ◆ Participant Handbook ◆ Practical Guide	30 minutes
8	'C' Programming Language	Switch case	◆ To write a program in 'C' using "Switch case" construct	◆ Trainer-led discussion ◆ Hands on practice	◆ Trainer's Guide ◆ Participant Handbook ◆ Practical Guide	35 minutes
9	'C' Programming Language	The while and the do while loop	◆ To repeat a set of program instructions using the "while" and the "do while" loops.	◆ Multimedia based learning ◆ Trainer-led discussion ◆ Hands on practice	◆ Multimedia (K-Yan) ◆ Trainer's Guide ◆ Participant Handbook ◆ Practical Guide	1 hour
10	'C' Programming Language	The for Loop	◆ To use the 'for' loop to repeat a set of program instructions	◆ Multimedia based learning ◆ Trainer-led discussion ◆ Hands on practice	◆ Multimedia (K-Yan) ◆ Trainer's Guide ◆ Participant Handbook ◆ Practical Guide	1 hour 35 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Day 4						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser Booklet	10 minutes
2	Recap	Recap	To revise previous day's learning	- Teach back session - Trainer-led discussion - Reinforcement	- Trainer's Guide - Multimedia (K-Yan) - Participant Handbook	20 minutes
3	'C' Programming Language	Introduction to Functions	- To state the need for using functions - To define functions - Use functions in 'C' program	- Multimedia based learning - Trainer-led discussion	- Multimedia (K-Yan) - Trainer's Guide - Participant Handbook	30 minutes
4	'C' Programming Language	Writing C Functions	To write functions using 'C' programming language	- Multimedia based learning - Trainer-led discussion - Hands on practice	- Multimedia (K-Yan) - Trainer's Guide - Participant Handbook - Practical Guide	1 hour
5	'C' Programming Language	Function Categories	- To state the categories of functions in 'C' - To use the different function category in 'C' programs	- Multimedia based learning - Trainer-led discussion - Hands on practice	- Multimedia (K-Yan) - Trainer's Guide - Participant Handbook	1 hour

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
6	'C' Programming Language	Function Calls	◆ To call functions by value ◆ To call functions by reference	◆ Multimedia based learning ◆ Trainer-led discussion ◆ Hands on practice	◆ Multimedia (K-Yan) ◆ Trainer's Guide ◆ Participant Handbook ◆ Practical Guide	1 hour
7	'C' Programming Language	Introduction to Arrays	◆ To construct an array ◆ To access the array elements ◆ To use array in functions	◆ Multimedia based learning ◆ Trainer-led discussion ◆ Hands on practice	◆ Multimedia (K-Yan) ◆ Trainer's Guide ◆ Participant Handbook ◆ Practical Guide	1 hour
8	'C' Programming Language	Two Dimensional Arrays	◆ To use a two dimensional array in your "C" Program	◆ Multimedia based learning ◆ Trainer-led discussion ◆ Hands on practice	◆ Multimedia (K-Yan) ◆ Trainer's Guide ◆ Participant Handbook ◆ Practical Guide	1 hour
9	'C' Programming Language	More on Arrays	◆ To construct a multi dimensional array ◆ To access the multi dimensional array elements	◆ Trainer-led discussion ◆ Hands on practice	◆ Trainer's Guide ◆ Participant Handbook ◆ Practical Guide	30 minutes
10	'C' Programming Language	Strings	◆ To use string as an array of characters ◆ To use string functions to manipulate strings	◆ Multimedia based learning ◆ Trainer-led discussion ◆ Hands on practice	◆ Multimedia (K-Yan) ◆ Trainer's Guide ◆ Participant Handbook ◆ Practical Guide	30 minutes
11	'C' Programming Language	Structures	◆ To use "structures" to handle a collection of related dissimilar data types	◆ Multimedia based learning ◆ Trainer-led discussion ◆ Hands on practice	◆ Multimedia (K-Yan) ◆ Trainer's Guide ◆ Participant Handbook ◆ Practical Guide	1 hour

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Day 5						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	◆ To energise the participants to take on the day	◆ Group participation	◆ Trainer's Guide ◆ Morning Energiser Booklet	10 minutes
2	Recap`	Recap	◆ To revise previous day's learning	◆ Teach back session ◆ Question based discussion ◆ Reinforcement	◆ Trainer's Guide	20 minutes
3	Object Oriented Programming (OOP) using C++	Concept of OOP	◆ To explain Object Oriented Programming	◆ Trainer-led discussion	◆ Trainer's Guide ◆ Participant Handbook ◆ Participant Handbook ◆ Presentation	30 minutes
4	Object Oriented Programming (OOP) using C++	Programming with C++	◆ To use the OOP features in C++ Programs	◆ Trainer-led discussion ◆ Hands on Practice	◆ Trainer's Guide ◆ Participant Handbook ◆ Practical Guide ◆ Presentation	1 hour
5	Object Oriented Programming (OOP) using C++	Objects and Classes	◆ To use the OOP features in C++ Programs	◆ Trainer-led discussion ◆ Hands on Practice	◆ Trainer's Guide ◆ Participant Handbook ◆ Practical Guide	1 hour
6	Object Oriented Programming (OOP) using C++	Member Functions	◆ To use the OOP features in C++ Programs	◆ Trainer-led discussion ◆ Hands on Practice	◆ Trainer's Guide ◆ Participant Handbook ◆ Practical Guide ◆ Presentation	30 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
7	Object Oriented Programming (OOP) using C++	Constructors and Destructors	To use the OOP features in C++ Programs	- Multimedia based learning - Trainer-led discussion - Hands on practice	- Multimedia (K-Yan) - Trainer's Guide - Participant Handbook - Practical Guide	1 hour
8	Object Oriented Programming (OOP) using C++	Operator Overloading and Function Overloading	To use the OOP features in C++ Programs	- Multimedia based learning - Trainer-led discussion - Hands on practice	- Multimedia (K-Yan) - Trainer's Guide - Participant Handbook - Practical Guide	2 hours
9	Object Oriented Programming (OOP) using C++	Inheritance	To use the OOP features in C++ Programs	- Multimedia based learning - Trainer-led discussion - Hands on practice	- Multimedia (K-Yan) - Trainer's Guide - Participant Handbook - Practical Guide	1 hour 30 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Day 6						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser Booklet	10 minutes
2	Recap	Recap	To revise previous day's learning	- Teach back session - Quiz - Reinforcement	- Trainer's Guide - Presentation	20 minutes
3	Object Oriented Programming (OOP) using C++	Understanding Protected Members	To use protected members in C++ Programs	- Trainer-led discussion - Hands on practice	- Trainer's Guide - Participant Handbook - Presentation	1 hour
4	Object Oriented Programming (OOP) using C++	Pointers in C++	To use pointers in C++ Programs	- Multimedia based learning - Trainer-led discussion	- Multimedia (K-Yan) - Trainer's Guide - Participant Handbook - Presentation	1 hour
5	Object Oriented Programming (OOP) using C++	Pointers in C++ - Practice	To use pointers in C++ Programs	- Trainer-led discussion - Hands on practice	- Trainer's Guide - Participant Handbook - Practical Guide - Presentation	1 hour 30 minutes
6	Data Structures Using C++	Introduction to Data Structure	- To explain the organisation of Data in memory - To state the different types of Data Structures	Trainer-led discussion	- Trainer's Guide - Participant Handbook - Presentation	30 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
7	Data Structures Using C++	Data Types	- To list the fundamental and derived data types - To use the data types in C++ Programs	Trainer-led discussion	- Trainer's Guide - Participant Handbook - Presentation	30 minutes
8	Data Structures Using C++	Operations on Data Structure	- To state the various operations performed on Data Structures. - To implement Data Structures in C++ Programs	Trainer-led discussion	- Trainer's Guide - Participant Handbook - Presentation	1 hour
9	Data Structures Using C++	Bubble Sort	To implement bubble sort algorithm in C++ Programs	- Multimedia based learning - Trainer-led discussion - Hands on practice	- Multimedia (K-yan) - Trainer's Guide - Participant Handbook - Practical Guide	1 hour
10	Data Structures Using C++	Searching	To implement search algorithm in C++ Programs	- Trainer-led discussion - Hands on practice	- Trainer's Guide - Participant Handbook - Practical Guide - Presentation	1 hour

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Day 7						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser Booklet	10 minutes
2	Recap	Recap	To revise previous day's learning	- Teach back session - Question based discussion - Reinforcement	Trainer's Guide	20 minutes
3	Data Structures Using C++	Stacks	- To state the use of stacks - To implement stacks in C++ Programs	- Trainer-led discussion - Hands on practice	- Trainer's Guide - Participant Handbook - Practical Guide - Presentation	1 hour
4	Data Structures Using C++	Queues	- To state the use of queues in Programming - To implement queues in C++ Programs	- Trainer-led discussion - Hands on practice	- Trainer's Guide - Participant Handbook - Practical Guide - Presentation	1 hour
5	Data Structures Using C++	Types and Applications of Queues	- To state the types and applications of queues - To implement queues in C++ Programs	- Trainer-led discussion - Hands on practice	- Trainer's Guide - Participant Handbook - Practical Guide - Presentation	1 hour

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
6	Data Structures Using C++	Linked Lists	◆ To insert, find, delete an element from a Linked Lists ◆ To state the difference between an array and a Linked Lists	◆ Trainer-led discussion ◆ Hands on practice ◆ Practical	◆ Trainer's Guide ◆ Participant Handbook ◆ Practical Guide ◆ Presentation	1 hour
7	Data Structures Using C++	Types of Linked Lists	◆ To state the types of Linked Lists ◆ To state the applications of Linked Lists	◆ Trainer-led discussion	◆ Trainer's Guide ◆ Participant Handbook ◆ Presentation	1 hour
8	Data Structures Using C++	Trees	◆ To describe trees and its types ◆ Use trees in Data Structures	◆ Trainer-led discussion	◆ Trainer's Guide ◆ Participant Handbook ◆ Presentation	1 hour
9	Data Structures Using C++	Operations on Binary Tree	◆ To state the Operations performed on Binary Tree	◆ Trainer-led discussion	◆ Trainer's Guide ◆ Participant Handbook ◆ Presentation	1 hour
10	Data Structures Using C++	Graphs	◆ To state the different types of Graphs ◆ To state the Operations performed on different types of Graphs	◆ Trainer-led discussion	◆ Trainer's Guide ◆ Participant Handbook ◆ Presentation	30 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Day 8						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser Booklet	10 minutes
2	Recap	Recap	To revise previous day's learning	- Teach back session - Question based discussion - Reinforcement	- Trainer's Guide - Participant Handbook	20 minutes
3	Writing HTML Codes	Introduction to HTML	- Create web pages using HTML codes - Use HTML tags for inserting List, Table, Frames, Form and CSS	- Multimedia based learning - Trainer-led discussion	- Multimedia (K-Yan) - Trainer's Guide - Participant Handbook - Presentation	2 hours 30 minutes
4	Writing HTML Codes	Creating HTML Page	Create web pages using HTML codes	- Multimedia based learning - Trainer-led discussion - Hands on practice	- Multimedia (K-Yan) - Trainer's Guide - Participant Handbook - Practical Guide	45 minutes
5	Writing HTML Codes	Creating Lists	Create HTML web pages using lists	- Trainer-led discussion - Hands on practice	- Trainer's Guide - Participant Handbook - Practical Guide	45 minutes
6	Writing HTML Codes	Images, Colours and Background	Add images and Colours to web pages.	- Trainer-led discussion - Hands on practice	- Trainer's Guide - Participant Handbook - Practical Guide	30 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
7	Writing HTML Codes	Tables and Frames	Create tables and frames using HTML tags	- Multimedia based learning - Trainer-led discussion - Hands on practice	- Multimedia (K-yan) - Trainer's Guide - Participant Handbook - Practical Guide	30 minutes
8	Writing HTML Codes	Forms	Create web forms using HTML	- Trainer-led discussion - Hands on practice	- Trainer's Guide - Participant Handbook - Practical Guide	30 minutes
9	Writing HTML Codes	Cascading Style Sheets (CSS)	Add style sheets to HTML documents	- Trainer-led discussion - Hands on practice	- Trainer's Guide - Participant Handbook - Practical Guide	1 hour 50 minutes

Day 9 to Day 32

Elective 1: Visual Studio .Net

Elective 2: J2EE

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 1 - Day 9						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Formative Assessment	Test 1	To test the participants on skill acquired in C Programming	- Theory - Practical	- Assessment Guide - Trainer's Guide	2 hours
3	Building Web Application Using Visual Studio Dot Net	Introduction to Dot Net Framework	- Describe Microsoft .Net Framework and net technologies - State the difference between .Net and .Net Framework	Trainer-led discussion	- Trainer's Guide - Participant Handbook - Presentation	1 hour
4	Building Web Application Using Visual Studio Dot Net	VB Net and its Functionality	- State the features of VB .Net - State the use of VB .Net	Trainer-led discussion	- Trainer's Guide - Participant Handbook - Presentation	1 hour
5	Building Web Application Using Visual Studio Dot Net	Programming using VB Net	- Describe the Integrated Development Environment (IDE) - State the different data types in VB .Net - State the different data types in VB .Net	Trainer-led discussion	- Trainer's Guide - Participant andbook - Practical Guide - Presentation	2 hours

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
6	Building Web Application Using Visual Studio Dot Net	Programming using VB Net	◆ Write programs in VB .Net using control structures and functions ◆ Write programs in VB .Net	◆ Trainer-led discussion ◆ Hands on practice	◆ Trainer's Guide ◆ Participant Handbook ◆ Practical Guide	1 hour 40 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 2 - Day 9						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Formative Assessment	Test – 1	To test the participant on skills acquired in "C" Programming	- Theory - Practical	- Assessment Guide - Trainer's Guide	2 hours
3	Building Web Application Using J2EE	Introduction to Core Java Programming	To state the features of OOP in Java	Trainer-led discussion	- Trainer's Guide - Participant Handbook - Presenation	1 hour
4	Building Web Application Using J2EE	Fundamentals of Java Programming	- To state the different data types in java - To state the use of looping - To difine exception handling	Trainer-led discussion	- Trainer's Guide - PPT - Participant Handbook	1 hour
5	Building Web Application Using J2EE	Objects and Classes	Write programs using classes and objects in Java	Trainer-led discussion	- Trainer's Guide - Presenation - Participant Handbook	2 hours
6	Building Web Application Using J2EE	Introduction to Core Java Programming	To write the programs in java and execute them	- Trainer-led discussion - Hands on practice	- Trainer's Guide - Practical Guide	1 hour 50 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 1 - Day 10						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Soft Skill	Grooming	To state the importance of grooming	- Multimedia based learning - Trainer-led discussion - Group participation	- Multimedia (K-yan) - Trainer's Guide - Participant Handbook	2 hours 30 minutes
3	Building Web Application Using Visual Studio Dot Net	Creating Forms in VB .Net	- To create forms in VB .Net - To define variables, user Defined Data types and Exception Handling	Trainer-led discussion	- Trainer's Guide - Participant Handbook - Presentation	2 hours
4	Building Web Application Using Visual Studio Dot Net	Creating Forms in VB .Net	To create forms in VB .Net	- Trainer-led discussion - Hands on practice	- Trainer's Guide - Practical Guide	1 hour 30 minutes
5	Recap	Recap	To revise the day's learning	Trainer-led discussion	Trainer's Guide	20 minutes
6	Building Web Application Using Visual Studio Dot Net	Creating Forms in VB .Net	To create forms in VB .Net	- Trainer-led discussion - Hands on practice	- Trainer's Guide - Practical Guide	2 hours

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 2 - Day 10						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Soft Skill	Grooming	To state the importance of grooming	- Multimedia based learning - Trainer-led discussion - Group participation	- Presentation - Trainer's Guide - Participant Handbook - Multimedia (K-Yan)	2 hours 30 minutes
3	Building Web Application Using J2EE	Arrays and Strings	- To define an array - To define strings	Trainer-led discussion	- Trainer's Guide - PPT - Participant Handbook	2 hours
4	Building Web Application Using J2EE	Arrays and Strings	To use array and string in Java programs	- Trainer-led discussion - Hands on practice	- Trainer's Guide - Practical Guide	1 hour 30 minutes
5	Recap	Recap	To revise the day's learning	Reinforcement	Trainer's Guide	20 minutes
6	Building Web Application Using J2EE	Arrays and Strings	To use arrays and Strings in Java programs	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide	1 hour 30 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 1 - Day 11						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Soft Skills	Discipline	To state the importance of being disciplined	- Multimedia based learning - Trainer-led Discussion - Group participation	- Multimedia (K-yan) - Trainer's Guide - Participant Handbook	2 hours 30 minutes
3	Building Web Application Using Visual Studio Dot Net	Form Controls	To use the different form controls	Trainer-led Discussion	- Trainer's Guide - Participant Handbook - Presentation	2 hours
4	Building Web Application Using Visual Studio Dot Net	Form Controls	To use the different form controls	Hands on Practice	- Trainer's Guide - Practical Guide	2 hours
5	Recap	Recap	To revise the day's learning	- Teachback session - Reinforcement	Trainer's Guide	20 minutes
6	Building Web Application Using Visual Studio Dot Net	Form Controls	To use the different form controls	Hands on Practice	- Trainer's Guide - Practical Guide	1 hour

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 2 - Day 11						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Soft Skills	Discipline	To state the importance of being disciplined	- Multimedia based learning - Group participation	- Multimedia (K-yan) - Trainer's Guide - Participant Handbook	2 hours 30 minutes
3	Building Web Application Using J2EE	Inheritance in Java Programming	- To define inheritance - To state how to implement inheritance in Java programs	Trainer-led Discussion	- Trainer's Guide - Participant Handbook - Presentation	1 hour 30 minutes
4	Building Web Application Using J2EE	Inheritance in Java Programming	To implement inheritance in Java programs	- Trainer-led Discussion - Hands on Practice	- Trainer's Guide - Practical Guide	2 hours 30 minutes
5	Recap	Recap	To revise the day's learning	Reinforcement	- Trainer's Guide - Presentation	20 minutes
6	Building Web Application Using J2EE	Inheritance in Java Programming	To implement inheritance in Java programs	- Trainer-led Discussion - Hands on Practice	- Trainer's Guide - Practical Guide	1 hour

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 1 - Day 12						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	◆ To energise the participants to take on the day	◆ Group participation	◆ Trainer's Guide ◆ Morning Energiser	10 minutes
2	Soft Skills	Time Management	◆ To manage your time better	◆ Multimedia based learning ◆ Group participation	◆ Multimedia (K-yan) ◆ Trainer's Guide ◆ Participant Handbook	2 hours
3	Building Web Application Using Visual Studio Dot Net	OOP using VB .Net	◆ To implement OOP features in VB .Net	◆ Trainer-led discussion	◆ Trainer's Guide ◆ Participant Handbook ◆ Presentation	1 hour 30 minutes
4	Recap	Recap	◆ To revise the day's learning	◆ Reinforcement	◆ Trainer's Guide	20 minutes
5	Building Web Application Using Visual Studio Dot Net	OOP using VB .Net	◆ To implement OOP features in VB .Net	◆ Hands on Practice	◆ Trainer's Guide ◆ Practical Guide	2 hours
6	Building Web Application Using Visual Studio Dot Net	OOP using VB .Net	◆ To implement OOP features in VB .Net	◆ Hands on Practice	◆ Trainer's Guide ◆ Practical Guide	2 hours

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 2 - Day 12						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Soft Skills	Time Management	To manage your time better	- Multimedia based learning - Group participation	- Multimedia (K-yan) - Trainer's Guide - Participant Handbook	2 hours
3	Building Web Application Using J2EE	Files and Streams	To explain file processing using input/output streams	Trainer-led discussion	- Trainer's Guide - Presentation - Participant Handbook	1 hour 10 minutes
4	Recap	Recap	To revise the day's learning	Reinforcement	Trainer's Guide	10 minutes
5	Building Web Application Using J2EE	Files and Streams	To use files processing in Java programs	Hands on Practice	- Trainer's Guide - Practical Guide	2 hours 30 minutes
6	Building Web Application Using J2EE	Files and Streams	To use files processing in java programs	Hands on Practice	- Trainer's Guide - Practical Guide	2 hours

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 1 - Day 13						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Soft Skills	Attitude at Work	To improve your attitude at work	- Multimedia based learning - Group Participation	- Multimedia (K-yan) - Trainer's Guide - Participant Handbook	2 hours
3	Building Web Application Using Visual Studio Dot Net	Component based Programming	- To state the features of component based programming - To use .Net Component in programs	Trainer-led Discussion	- Trainer's Guide - Participant Handbook - Presentation	1 hour 30 minutes
4	Building Web Application Using Visual Studio Dot Net	Component based Programming	To use .Net Component in programs	- Trainer-led Discussion - Hands on Practice	- Trainer's Guide - Practical Guide	2 hours
5	Recap	Recap	To revise the day's learning	Reinforcement	Trainer's Guide	20 minutes
6	Building Web Application Using Visual Studio Dot Net	Component based Programming	To use .Net Component in programs	Hands on Practice	- Trainer's Guide - Practical Guide	2 hours

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 2 - Day 13						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Soft Skill	Attitude at Work	To improve your attitude at work	- Multimedia based learning - Group participation - Activity	- Multimedia (K-yan) - Trainer's Guide - Participant Handbook	2 hours
3	Building Web Application Using J2EE	Types of Streams	To state the different types of streams and their uses	Trainer-led Discussion	- Trainer's Guide - Presentation - Participant Handbook	2 hours
4	Building Web Application Using J2EE	Types of Streams	To write Java programs using files and streams	- Trainer-led Discussion - Hands on Practice	- Trainer's Guide - Practical Guide	2 hours 30 minutes
5	Recap	Recap	To revise the day's learning	Reinforcement	- Trainer's Guide - Presentation	20 minutes
6	Building Web Application Using J2EE	Types of Streams	To write Java programs using files and streams	- Trainer-led Discussion - Hands on Practice	- Trainer's Guide - Practical Guide	1 hour

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 1 - Day 14						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Building Web Application Using Visual Studio Dot Net	Programming with ASP .Net	- To state the features of ASP .Net - To use the web server control and HTML control features	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Participant Handbook - Presentation	4 hours
3	Recap	Recap	To revise the day's learning	Reinforcement	Trainer's Guide	25 minutes
4	Building Web Application Using Visual Studio Dot Net	Programming with ASP .Net	To use the web server control and HTML control features in ASP .Net programs	- Trainer-led discussion - Hands on practice	- Trainer's Guide - Practical Guide	1 hour
5	Building Web Application Using Visual Studio Dot Net	Programming with ASP .Net	To use the web server control and HTML control features in ASP .Net programs	- Trainer-led discussion - Hands on practice	- Trainer's Guide - Practical Guide	2 hours 25 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 2 - Day 14						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Building Web Application Using J2EE	AWT(Abstract Windowing Toolkit)	- To describe AWT components and its hierarchy - To create window and frames in Java using AWT	- Trainer-led discussion - Hands on practice	- Trainer's Guide - Presentation - Participant Handbook	4 hour
3	Recap	Recap	To revise the day's learning	Reinforcement	- Trainer's Guide - Presentation	25 minutes
4	Building Web Application Using J2EE	AWT(Abstract Windowing Toolkit)	To create window and frames using AWT	- Trainer's Guide - Hands on practice	- Trainer's Guide - Practical Guide	3 hours 25 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 1 - Day 15						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Recap	Recap	To revise the previous day's learning	Reinforcement	Trainer's Guide	20 minutes
3	Building Web Application Using Visual Studio Dot Net	ASP .Net Web Applications	- To state the elements and features of ASP .Net web applications - To create ASP .Net web applications	Trainer-led discussion	- Trainer's Guide - Participant Handbook - Presentation	2 hours
4	Building Web Application Using Visual Studio Dot Net	ASP .Net Web Applications	To create ASP .Net web applications	- Trainer-led discussion - Hands on practice	- Trainer's Guide - Practical Guide	2 hours
5	Games and Activities	Games and Activities	To refresh the Participants	- Group Participation - Activity	- Trainer's Guide - Activities and Games Booklet	30 minutes
6	Building Web Application Using Visual Studio Dot Net	ASP .Net Web Applications	To create ASP .Net web applications	- Trainer-led discussion - Hands on practice	- Trainer's Guide - Practical Guide	3 hours

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 2 - Day 15						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Recap	Recap	To revise the previous day's learning	Reinforcement	Trainer's Guide	20 minutes
3	Building Web Application Using J2EE	Threads in Java	- Create threads in Java - Use thread scheduling in Java	Trainer-led discussion	- Trainer's Guide - Presentation - Participant Handbook	2 hours
4	Recap	Threads in Java	- Create threads in Java - Use thread scheduling in Java	Hand on Practice	- Trainer's Guide - Participant Handbook - Practical Guide	2 hours
5	Games and Activities	Games and Activities	To refresh the Participants	Group Participation Activity	- Trainer's Guide - Activities and Games Booklet	30 minutes
6	Building Web Application Using J2EE	Threads in Java	- Create threads in Java - Use thread scheduling in Java	Hands on Practice	- Trainer's Guide - Participant Handbook - Practical Guide	3 hours

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 1 - Day 16						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Building Web Application Using Visual Studio Dot Net	ASP .Net IDE and Web Forms	- To state the features of ASP .Net IDE - To list the visual C tools and its functions - To create web forms with its controls	Trainer-led discussion	- Trainer's Guide - Participant Handbook - Presentation	2 hours 30 minutes
3	Recap	Recap	To revise the previous day's learning	Reinforcement	Trainer's Guide	20 minutes
4	Building Web Application Using Visual Studio Dot Net	ASP .Net IDE and Web Forms	To create web form with its controls	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Participant Handbook - Practical Guide	3 hours
5	Building Web Application Using Visual Studio Dot Net	ASP .Net IDE and Web Forms	To create web form with its controls	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Participant Handbook - Practical Guide	2 hours

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 2 - Day 16						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Building Web Application Using J2EE	Networking	To use Java for network programming	Trainer-led discussion	- Trainer's Guide - Participant Handbook - Presentation	2 hours 30 minutes
3	Recap	Recap	To revise the previous day's learning	Reinforcement	- Trainer's Guide - Presentation	20 minutes
4	Building Web Application Using J2EE	Networking	To use Java for network programming	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Participant Handbook - Practical Guide	3 hours
5	Building Web Application Using J2EE	Networking	To use Java for network programming	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Participant Handbook - Practical Guide	2 hours

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 1 - Day 17						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Building Web Application Using Visual Studio Dot Net	Making Web Forms	- To create web forms with web controls - To implement events in web forms	Trainer-led discussion	- Trainer's Guide - Participant Handbook - Presentation	2 hours
3	Building Web Application Using Visual Studio Dot Net	Making Web Forms	- To create web forms with web controls - To implement events in web forms	Hands on Practice	- Trainer's Guide - Practical Guide	3 hours
4	Recap	Recap	To revise the previous day's learning	Reinforcement	- Trainer's Guide - Presentation	20 minutes
5	Building Web Application Using Visual Studio Dot Net	Making Web Forms	- To create web forms with web controls - To implement events in web forms	Hands on Practice	- Trainer's Guide - Practical Guide	2 hours 30 minutes

Elective 2 - Day 17					
Sr. No.	Module	Session	Objectives	Methodology	Tools
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser
2	Building Web Application Using J2EE	SWING	Use SWING components and its features in Java programming	Trainer-led discussion	- Trainer's Guide - Participant Handbook - Presentation
3	Building Web Application Using J2EE	SWING	Use SWING components and its features in Java programming	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide
4	Recap	Recap	To revise the previous day's learning	Reinforcement	- Trainer's Guide - Presentation
5	Building Web Application Using J2EE	SWING	Use SWING components and its features in Java programming	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 1 - Day 18						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Building Web Application Using Visual Studio Dot Net	Session Objects and Events	To describe session objects and events	Trainer-led discussion	- Trainer's Guide - Participant Handbook - Presentation	2 hours
3	Building Web Application Using Visual Studio Dot Net	Session Objects and Events	To program using session objects and events	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide	3 hours
4	Building Web Application Using Visual Studio Dot Net	Session Objects and Events	To program using session objects and events	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide	2 hours
5	Recap	Recap	To revise the previous day's learning	Reinforcement	Trainer's Guide	20 minutes
6	Games and Activities	Games and Activities	To refresh the participants	- Activity - Group participation	- Trainer's Guide - Activities and Games Booklet	30 minutes

Elective 2 - Day 18					
Sr. No.	Module	Session	Objectives	Methodology	Tools
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	◆ To energise the participants to take on the day	◆ Group participation	◆ Trainer's Guide ◆ Morning Energiser
2	Building Web Application Using J2EE	Handling GUI Generated Events	◆ To program GUI generated events	◆ Trainer-led discussion	◆ Trainer's Guide ◆ Participant Handbook ◆ Presentation
3	Building Web Application Using J2EE	Handling GUI Generated Events	◆ To program GUI generated events	◆ Trainer-led discussion ◆ Hands on Practice	◆ Trainer's Guide ◆ Practical Guide
4	Building Web Application Using J2EE	Handling GUI Generated Events	◆ To program GUI generated events	◆ Trainer-led discussion ◆ Hands on Practice	◆ Trainer's Guide

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 1 - Day 19						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Soft Skills	Team Work	To state the importance of team work	- Multimedia based learning - Trainer-led discussion - Group participation - Activities	- Multimedia (K-yan) - Trainer's Guide - Participant Handbook	2 hours
3	Building Web Application Using Visual Studio Dot Net	Session Objects and Cookies	To describe session objects and cookies	Trainer-led discussion	- Trainer's Guide - Participant Handbook - Presentation	2 hours 30 minutes
4	Recap	Recap	To revise the previous day's learning	Reinforcement	Trainer's Guide	20 minutes
5	Building Web Application Using Visual Studio Dot Net	Session Objects and Cookies	To program using session objects and cookies	- Trainer-led discussion - Hands on Practice	Trainer's Guide	5 hours

Elective 2 - Day 19					
Sr. No.	Module	Session	Objectives	Methodology	Tools
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser
2	Soft Skill`	Team Work	To state use importance of team work	- Multimedia based learning - Trainer-led discussion - Group participation - Activities	- Multimedia (K-yan) - Trainer's Guide - Participant Handbook
3	Building Web Application Using J2EE	Comparison between AWT and SWING	To state the difference between AWT and SWING	Trainer-led discussion	- Trainer's Guide - Participant Handbook - Presentation
4	Recap	Recap	To revise the previous day's learning	Reinforcement	Trainer's Guide
5	Building Web Application Using J2EE	SWING Components	Use SWING components in Java programming	- Trainer-led discussion - Hands on Practice	Trainer's Guide
6	Building Web Application Using J2EE	SWING Components	Use SWING components in Java programming	- Trainer-led discussion - Hands on Practice	Trainer's Guide

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 1 - Day 20						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Formative Assessment	Test – 2	To test the participant on skills acquired in C++ and Data Structures	- Trainer-led discussion - Hands on Practice	Assessment Guide	2 hours
3	Building Web Application Using Visual Studio Dot Net	Programming with C#	- To state the features of C# as programming language - Write programs in C#	Trainer-led discussion	- Trainer's Guide - Participant Handbook - Presentation	2 hours 20 minutes
4	Building Web Application Using Visual Studio Dot Net	Programming with C#	Write programs in C#	Hands on Practice	- Trainer's Guide - Practical Guide	1 hour 30 minutes
5	Building Web Application Using Visual Studio Dot Net	Programming with C#	Write programs in C#	Hands on Practice	- Trainer's Guide - Practical Guide	2 hours

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 2 - Day 20						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Formative Assessment	Test – 2	To test the participants on skills acquired in C++ and data structures	- Theory - Practical	- Trainer's Guide - Assessment Guide	2 hours
3	Building Web Application Using J2EE	Applets	- To describe applets and its advantages - To create and run Applets	Trainer-led discussion	- Trainer's Guide - Participant Handbook - Presentation	2 hours 30 minutes
4	Building Web Application Using J2EE	Applets	To create and run Applets	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide	2 hours 30 minutes
5	Recap	Recap	To revise the previous day's learning	Reinforcement	Trainer's Guide	20 minutes
6	Games and Activities	Games and Activities	To refresh the participants	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Games and Activities	30 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 1 - Day 21						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Recap	Recap	To revise the previous day's learning	Reinforcement	Trainer's Guide	20 minutes
3	Building Web Application Using Visual Studio Dot Net	Data types and Classes in C#	- To state the different data types in C# - To write program in C# using data types and classes	Trainer-led discussion	- Trainer's Guide - Participant Handbook - Presentation	2 hours 30 minutes
4	Building Web Application Using Visual Studio Dot Net	Data types and Classes in C#	To write program in C# using data types and classes	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide	2 hours 30 minutes
5	Building Web Application Using Visual Studio Dot Net	Data types and Classes in C#	To write program in C# using data types and classes	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide	2 hours 30 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 2 - Day 21						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Recap	Recap	To revise the previous day's learning	Reinforcement	Trainer's Guide	20 minutes
3	Building Web Application Using J2EE	Java Script	- To state the areas of application of Java Script - To write programs using Java Script	Trainer-led discussion	- Trainer's Guide - Participant Handbook - Presentation	2 hours
4	Building Web Application Using J2EE	Java Script	To write programs using Java Script	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide	3 hours
5	Building Web Application Using J2EE	Java Script	To write programs using Java Script	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide	2 hours 30 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 1 - Day 22						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Building Web Application Using Visual Studio Dot Net	Control Statements and Loops in C#	To write programs in C# using control structures and loops	Trainer-led discussion	- Trainer's Guide - Participant Handbook - Presentation	2 hours 30 minutes
3	Building Web Application Using Visual Studio Dot Net	Control Statements and Loops in C#	To write programs in C# using control structures and loops	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide	2 hours 30 minutes
4	Recap	Recap	To revise the day's learning	Reinforcement	- Trainer's Guide - Practical Guide	20 minutes
5	Building Web Application Using Visual Studio Dot Net	Control Statements and Loops in C#	To write programs in C# using control structures and loops	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide	2 hours 30 minutes

Elective 2 - Day 22					
Sr. No.	Module	Session	Objectives	Methodology	Tools
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser
2	Building Web Application Using J2EE	JavaScript – Operators and Conditional Statements	- To state the different operators of Java Script - To use the conditional statements of Java Script	Trainer-led discussion	- Trainer's Guide - Presentation - Participant Handbook
3	Building Web Application Using J2EE	JavaScript – Operators and Conditional Statements	To use the conditional statements of Java Script	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide
4	Building Web Application Using J2EE	JavaScript- Operators and Conditional Statements	To write codes in Java Script using the operators and conditional statements	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 1 - Day 23						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	◆ To energise the participants to take on the day	◆ Group participation	◆ Trainer's Guide ◆ Morning Energiser	10 minutes
2	Building Web Application Using Visual Studio Dot Net	Using Control Statements and Loops in C#	◆ To write programs in C# using contro and loops	◆ Trainer-led discussion	◆ Trainer's Guide ◆ Participant Handbook ◆ Presentation	2 hours 30 minutes
3	Building Web Application Using Visual Studio Dot Net	Using Control Statements and Loops in C#	◆ To write programs in C# using contro and loops	◆ Trainer-led discussion ◆ Hands on Practice	◆ Trainer's Guide ◆ Practical Guide	2 hours 30 minutes
4	Recap	Recap	◆ To revise the day's learning	◆ Reinforcement	◆ Trainer's Guide	20 minutes
5	Building Web Application Using Visual Studio Dot Net	Using Control Statements and Loops in C#	◆ To write programs in C# using contro and loops	◆ Trainer-led discussion ◆ Hands on Practice	◆ Trainer's Guide ◆ Practical Guide	2 hours 30 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 2 - Day 23						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Building Web Application Using J2EE	JavaScript – Objects	- State the types of objects, methods and boxes available in Java Script - Use document properties and array in Java Script	Trainer-led discussion	- Trainer's Guide - Participant Handbook - Presentation	2 hours 30 minutes
3	Building Web Application Using J2EE	JavaScript – Objects	Write Java Script codes using objects, methods, boxes, document properties and array	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide	2 hours 30 minutes
4	Recap	Recap	To revise the day's learning	Reinforcement	Trainer's Guide	15 minutes
5	Building Web Application Using J2EE	JavaScript – Objects	Write Java Script codes using objects, methods, boxes, document properties and array	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide	2 hours 30 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 1 - Day 24						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Soft Skill	Communication	To develop effective Communication Skills	- Multimedia based learning - Group participation - Activities - Trainer-led discussion	- Multimedia (K-yan) - Traine's Guide - Participant Handbook	2 hours 30 minutes
3	Building Web Application Using Visual Studio Dot Net	Programming with C#	Write programs using C#	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Participant Handbook	2 hours
4	Recap	Programming with C#	To revise the day's learning	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide	20 minutes
5	Building Web Application Using Visual Studio Dot Net	Programming with C#	Write programs using C#	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide	3 hour

Elective 2 - Day 24					
Sr. No.	Module	Session	Objectives	Methodology	Tools
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser
2	Soft Skill	Communication	To develop effective communication skills	- Multimedia based learning - Group participation - Activities - Trainer-led discussion	- Multimedia (K-yan) - Trainer's Guide - Participant Handbook
3	Building Web Application Using J2EE	JavaScript – Objects	Write java script code using object, methods, boxes, document properties and array	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide
4	Building Web Application Using J2EE	JavaScript – Objects	Write java script code using object, methods, boxes, document properties and array	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide
5	Building Web Application Using J2EE	JavaScript – Objects	Write java script code using object, methods, boxes, document properties and array	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide
6	Recap	Recap	To revise the day's learning	Reinforcement	Trainer's Guide

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 1 - Day 25						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Building Web Application Using Visual Studio Dot Net	Programming with C#	Write Programs using C#	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide	2 hours 30 minutes
3	Building Web Application Using Visual Studio Dot Net	Programming with C#	Write Programs using C#	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide	2 hours 30 minutes
4	Recap	Recap	To revise the day's learning	Reinforcement	Trainer's Guide	20 minutes
5	Building Web Application Using Visual Studio Dot Net	Programming with C#	Write Programs using C#	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide	2 hours 30 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 2 - Day 25						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Building Web Application Using J2EE	Java Servlets	- To state the advantages of Java servlets - To load a servlet and create an instance	Trainer-led discussion	- Trainer's Guide - Participant Handbook - Presentation	3 hours
3	Building Web Application Using J2EE	Java Servlets	To write servlets codes	Hands on Practice	- Trainer's Guide - Practical Guide	2 hours 30 minutes
4	Recap	RECAP	To revise the day's learning	Reinforcement	Trainer's Guide	20 minutes
5	Building Web Application Using J2EE	Java Servlets	To write servlets codes	Hands on Practice	- Practical Guide - Trainer's Guide	2 hours

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 1 - Day 26						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Building Web Application Using Visual Studio Dot Net	Database Connectivity	- Explain database connectivity in ADO.Net - Use database connectivity in .Net applications	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Participant Handbook - Presentation	2 hours 30 minutes
3	Building Web Application Using Visual Studio Dot Net	Database Connectivity	Use database connectivity in .Net applications	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide	2 hours 30 minutes
4	Recap	Recap	To revise the day's learning	Reinforcement	Trainer's Guide	20 minutes
5	Building Web Application Using Visual Studio Dot Net	Database Connectivity	Use database connectivity in .Net applications	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide	2 hours 30 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 2 - Day 26						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Building Web Application Using J2EE	Java Servlets	To write servlets codes	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Participant Handbook - Practical Guide	2 hours 30 minutes
3	Building Web Application Using J2EE	Java Servlets	To write servlets codes	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide	2 hours 30 minutes
4	Recap	Recap	To revise the day's learning	Reinforcement	Trainer's Guide	20 minutes
5	Building Web Application Using J2EE	Java Servlets	To write servlets codes	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide	2 hours 30 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 1 - Day 27						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Soft Skill	Problem Solving	To define a problem and identify the causes	- Multimedia based learning - Group participation	- Multimedia (K-yan) - Trainer's Guide - Participant Handbook	2 hours 30 minutes
3	Building Web Application Using Visual Studio Dot Net	XML	To create XML document	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Participant Handbook - Presentation	2 hours
4	Building Web Application Using Visual Studio Dot Net	XML	To create XML document	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide	1 hour 30 minutes
5	Recap	Recap	To revise the day's learning	Reinforcement	Trainer's Guide	20 minutes
6	Building Web Application Using Visual Studio Dot Net	XML	To create XML document	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide	1 hour 30 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 2 - Day 27						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Soft Skill	Problem Solving	To define a problem and identify the causes	- Multimedia based learning - Group participation	- Multimedia (K-yan) - Participant Handbook - Trainer's Guide	2 hours 30 minutes
3	Building Web Application Using J2EE	Java Server Pages (JSP)	- To explain the structure, declaration and processing of JSP - To develop JSP	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Participant Handbook - Presentation - Practical Guide	2 hours
3	Building Web Application Using J2EE	Java Server Pages (JSP)	To develop JSP	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide	2 hours
4	Recap	Recap	To revise the day's learning	Reinforcement	- Trainer's Guide - Presentation	20 minutes
5	Building Web Application Using J2EE	Java Server Pages (JSP)	To develop JSP	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide	1 hour

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 1 - Day 28						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Soft Skill	Stress Management	To manage and deal with stress	- Multimedia based learning - Group participation	- Multimedia (K-yan) - Trainer's Guide - Participant Handbook	2 hours 30 minutes
3	Building Web Application Using Visual Studio Dot Net	XML – Electronic Data Exchange	- To state the advantages and disadvantages of electronic data interchange - To state the role of XML in data exchange	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Participant Handbook	2 hours
4	Recap	RECAP	To revise the day's learning	Reinforcement	Trainer's Guide	20 minutes
5	Building Web Application Using Visual Studio Dot Net	XML – Electronic Data Exchange	To use edata exchange in XML	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide	1 hour 30 minutes
6	Building Web Application Using Visual Studio Dot Net	XML – Electronic Data Exchange	To use edata exchange in XML	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide	1 hour 30 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 2 - Day 28						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Soft Skill	Stress Management	To manage and deal with stress	- Multimedia based learning - Group participation	- Multimedia (K-yan) - Trainer's Guide - Participant Handbook	2 hours 30 minutes
3	Building Web Application Using J2EE	JSP Objects, Actions and Directives	To build JSP using objects, actions and directives	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Participant Handbook - Presentation	2 hours 30 minutes
4	Recap	Recap	To revise the day's learning	Reinforcement	- Trainer's Guide - Practical Guide	20 minutes
5	Building Web Application Using J2EE	JSP Objects, Actions and Directives	To build JSP using objects, actions and directives	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide	1 hour
6	Building Web Application Using J2EE	JSP Objects, Actions and Directives	To build JSP using objects, actions and directives	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide	1 hour 30 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 1 - Day 29						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Recap	Recap	To revise the day's learning	Reinforcement	Trainer's Guide	20 minutes
3	Building Web Application Using Visual Studio Dot Net	XML – DTD and Schema	- To define DTD and s - To list the use of DTD and schema	Trainer-led discussion	- Trainer's Guide - Participant Handbook - Presentation	2 hours 30 minutes
4	Building Web Application Using Visual Studio Dot Net	XML – DTD and Schema	To use the features of DTD and schemas	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide	2 hours 30 minutes
5	Building Web Application Using Visual Studio Dot Net	XML – DTD and Schema	To use the features of DTD and schema	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide	2 hours 30 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 2 - Day 29						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Recap	Recap	To revise the day's learning	Reinforcement	Trainer's Guide	20 minutes
3	Building Web Application Using J2EE	Java Database Connectivity (JDBC)	To use JDBC statement in Java	Trainer-led discussion	- Trainer's Guide - Participant Handbook - Presentation	2 hours 30 minutes
4	Building Web Application Using J2EE	Java Database Connectivity (JDBC)	To use JDBC statement in Java	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide	2 hours 30 minutes
5	Building Web Application Using J2EE	Java Database Connectivity (JDBC)	To use JDBC statement in Java	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide	2 hours 30 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 1 - Day 30						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Building Web Application Using Visual Studio Dot Net	XML - Web Feeds	To use the web feeds feature in XML	Trainer-led discussion	- Trainer's Guide - Participant Handbook - Presentation	2 hours 20 minutes
3	Building Web Application Using Visual Studio Dot Net	XML – Web Feeds	To use the web feeds feature in XML	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide	3 hours
4	Building Web Application Using Visual Studio Dot Net	XML – Web Feeds	To use the web feeds feature in XML	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide	2 hours 30 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 2 - Day 30						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Building Web Application Using J2EE	JDBC Drivers and Stored Procedures	- To use JDBC drivers - To create stored procedures in Java	Trainer-led discussion	- Trainer's Guide - Participant Handbook - Presentation	2 hours 30 minutes
3	Recap	Recap	To revise the day's learning	Reinforcement	Trainer's Guide	35 minutes
4	Building Web Application Using J2EE	JDBC Drivers and Stored Procedures	- To use JDBC drivers - To create stored procedures in Java	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide	4 hours
5	Games and Activities	Games and Activities	To refresh the participants	- Group participation - Activity	Games and Activities Booklet	45 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 1 - Day 31						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Soft Skills	Way to Success	Plan your growth path	Trainer-led discussion	- Trainer's Guide - Participant Handbook	1 hour
3	Building Web Application Using Visual Studio Dot Net	.DOT Net features- Summary	To state the features of .Net and its framework	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Participant Handbook	1 hour 50 minutes
4	Building Web Application Using Visual Studio Dot Net	Practice Exercises	To practice the learnings acquired on the topic	Hands on Practice	- Trainer's Guide - Practical Guide	5 hour

Elective 2 - Day 31					
Sr. No.	Module	Session	Objectives	Methodology	Tools
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser
2	Soft Skills	Way to Success	Plan your growth path	Trainer-led discussion	- Trainer's Guide - Participant Handbook
3	Building Web Application Using J2EE	AJAX	- State the purpose of AJAX and its objects - Use AJAX technology in web applications	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Presentation - Participant Handbook
4	Recap	Recap	To revise the day's learning	Reinforcement	- Trainer's Guide - Presentation
5	Building Web Application Using J2EE	AJAX	- State the purpose of AJAX and its objects - Use AJAX technology in web applications	- Trainer-led discussion - Hands on Practice	- Trainer's Guide - Practical Guide

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 1 - Day 32						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	◆ To energise the participants to take on the day	◆ Group participation	◆ Trainer's Guide ◆ Morning Energiser	10 minutes
2	Formative Assessment - 3	Test -3	◆ To test the participants on the skills acquired in HTML	◆ Trainer-led discussion ◆ Hands on Practice	◆ Trainer's Guide ◆ Assessment Guide	2 hours
3	Soft Skill	Presentation Skills	◆ To improve your pskills	◆ Trainer-led discussion ◆ Activity	◆ Trainer's Guide ◆ Participant Handbook	1 hour
4	Soft Skill	Group Discussion	◆ To take part in a group discussion	◆ Trainer-led Discussion ◆ Activity	◆ Trainer's Guide ◆ Participant Handbook	1 hour
5	Building Web Application Using Visual Studio Dot Net	Practice Exercises	◆ To practice the learnings acquired on the topic	◆ Trainer-led Discussion ◆ Hands on Practice	◆ Trainer's Guide ◆ Practical Guide	3 hours 50 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Elective 2 - Day 32						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Formative Assessment	Test -3	To test the participants on the skills acquired in HTML	- Theory - Practical	Assessment Guide	2 hours
3	Soft Skill	Presentation Skills	To improve your pskills	- Trainer-led Discussion - Activity	- Trainer's Guide - Participant Handbook	1 hour
4	Soft Skill	Group Discussion	To take part in a group discussion	- Trainer-led Discussion - Activity	- Trainer's Guide - Participant Handbook	1 hour
5	Building Web Application Using J2EE	Practice	To practice the learnings acquired on the topic	- Trainer-led Discussion - Hands on Practice	- Trainer's Guide - Practical Guide	3 hours 50 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Day 33						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	◆ To energise the participants to take on the day	◆ Group participation	◆ Trainer's Guide ◆ Morning Energiser	10 minutes
2	Relational Database Management System (RDBMS)	Database System – Overview and Concepts	◆ To define database ◆ To describe database architecture ◆ To state the advantages of RDBMS	◆ Trainer-led Discussion	◆ Trainer's Guide ◆ Participant Handbook ◆ Presentation	3 hours 30 minutes
3	Relational Database Management System (RDBMS)	Structured Query Language (SQL) [using Oracle]	◆ To use query language for creating tables and operations on tables	◆ Trainer-led Discussion ◆ Hands on Practice	◆ Trainer's Guide ◆ Participant Handbook ◆ Practical Guide	4 hours 20 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Day 34						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Soft Skills	Motivation	To motivate yourself to reach your goal	Trainer-led Discussion	- Participant Handbook - Trainer's Guide	1 hour
3	Relational Database Management System (RDBMS)	Computations on Table Data	- To state relations represented by tables - To manipulate table data	Trainer-led Discussion	- Trainer's Guide - Participant Handbook - Practical Guide	2 hours 30 minutes
4	Relational Database Management System (RDBMS)	Procedural Language - PL/SQL	To use PL-SQL query for data manipulation	Trainer-led Discussion	- Trainer's Guide - Participant Handbook	2 hours 30 minutes
5	Relational Database Management System (RDBMS)	Procedural Language - PL/SQL	To use PL/SQL query for data manipulation	Hands on Practice	- Trainer's Guide - Practical Guide	1 hour 50 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Day 35						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Recap	Recap	To revise the day's learning	- Trainer-led Discussion - Reinforcement	Trainer's Guide	30 minutes
3	Relational Database Management System (RDBMS)	Cursors, Triggers and Functions	To use events in database with cursors and triggers	Trainer-led Discussion	- Trainer's Guide - Participant Handbook	2 hours 30 minutes
4	Relational Database Management System (RDBMS)	Cursors, Triggers and Functions	To use events in database with cursors and triggers	Hands on Practice	- Trainer's Guide - Practical Guide	2 hours 30 minutes
5	Relational Database Management System (RDBMS)	Cursors, Triggers and Functions	To use events in database with cursors and triggers	Hands on Practice	- Trainer's Guidea - Practical Guide	2 hours 20 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Day 36						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Relational Database Management System (RDBMS)	Relational Database Design	Describe organisation of data in tables	Trainer-led Discussion	- Trainer's Guide - Participant Handbook	3 hours
3	Relational Database Management System (RDBMS)	Relational Database Design	Describe organisation of data in tables	Hands on Practice	- Trainer's Guide - Practical Guide	4 hours
4	Games and Activities	Games and Activities	To refresh the participants	Group participants Activity	Trainer's Guide	50 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Day 37						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Relational Database Management System (RDBMS)	Query Processing	To process queries using SQL	Trainer-led Discussion	- Trainer's Guide - Presentation - Participant Handbook	3 hours 30 minutes
3	Relational Database Management System (RDBMS)	Query Processing	To process queries using SQL	Hands on Practice	- Trainer's Guide - Practical Guide	4 hours
4	Recap	Recap	To revise the day's learning	- Trainer-led Discussion - Reinforcement	Trainer's Guide	20 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Day 38						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Basics of Hardware and Networking	Understanding Electronic Communication	To state the functions and features of electronic communication	Trainer-led Discussion	- Trainer's Guide - Presentation - Participant Handbook	7 hours 20 minutes
3	Recap	Recap	To revise the day's learning	Reinforcement	Trainer's Guide	30 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Day 39						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Basics of Hardware and Networking	Operating System Fundamentals	To state the functions of Operating System	Trainer-led Discussion	- Trainer's Guide - Presentation - Participant Handbook	7 hours 20 minutes
3	Recap	Recap	To revise the day's learning	Reinforcement	Trainer's Guide	30 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Day 40						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Basics of Hardware and Networking	Local Area Network (LAN)	To state the features and function of LAN	Trainer-led Discussion	- Trainer's Guide - Presentation - Participant Handbook	5 hours 50 minutes
3	Recap	Recap	To revise the day's learning	Reinforcement	Trainer's Guide	30 minutes
4	Soft Skill	Critical Thinking	To stimulate critical thinking	- Trainer-led Discussion - Group Participation - Activities	- Trainer's Guide - Participant Handbook	1 hour 30 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Day 41						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Formative Assessment	Test-4	To test the participants on skills acquired in LAN and SQL	- Theory - Practical	- Trainer's Guide - Assessment Guide	2 hours
3	Project Work	Building Web application	- To evaluate a problem definition - To evaluate system requirements and design software applications - To describe software and hardware requirements - To divide the project into module and decide the mile stones of the project - To develop software application for the given problem - To present the project application developed	- Workshop - Trainer-led Discussion - Group Participation	- Trainer's Guide - Participant Handbook	5 hours 50 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Day 42						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Project Work	Web application project	Develop software application for the given problem	- Workshop - Trainer-led Discussion - Group Participation	- Trainer's Guide - Participant Handbook	7 hours 50 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Day 43						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Project Work	Web application project	Develop software application for the given problem	- Workshop - Trainer-led Discussion - Group Participation	- Trainer's Guide - Participant Handbook	7 hours 50 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Day 44						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Project Work	Web application project	Develop software application for the given problem	- Workshop - Trainer-led Discussion - Group Participation	- Trainer's Guide - Participant Handbook	7 hours 50 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Day 45						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	◆ To energise the participants to take on the day	◆ Group participation	◆ Trainer's Guide ◆ Morning Energiser	10 minutes
2	Project Work	Web application project	◆ Develop software application for the given problem	◆ Workshop ◆ Trainer-led Discussion ◆ Group Participation	◆ Trainer's Guide ◆ Participant Handbook	7 hours 50 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Day 46						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Project Work	Web application project	Develop software application for the given problem	- Workshop - Trainer-led Discussion - Group Participation	- Trainer's Guide - Participant Handbook	7 hours 50 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Day 47						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	◆ To energise the participants to take on the day	◆ Group participation	◆ Trainer's Guide ◆ Morning Energiser	10 minutes
2	Project Work	Web application project	◆ Develop software application for the given problem	◆ Workshop ◆ Trainer-led Discussion ◆ Group Participation	◆ Trainer's Guide ◆ Participant Handbook	7 hours 50 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Day 48						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Project Work	Web application project	Develop software application for the given problem	- Workshop - Trainer-led Discussion - Group Participation	- Trainer's Guide - Participant Handbook	7 hours 50 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Day 49						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Soft skill	Interview Preparation	Prepare yourself for an interview	- Multimedia based learning - Trainer-led discussion	- Multimedia (K-Yan) - Trainer's Guide - Participant Handbook	2 hours
3	Summative Assessment	Project Presentation	To test and evaluate participants on the skills acquired during the training course	- Group participation - Presentation by participants	- Trainer's Guide - Assessment Guide	5 hours 50 minutes

Sr. No.	Module	Session	Objectives	Methodology	Tools	Time
Day 50						
1	Morning Energiser	Morning Energiser/ Good Sayings/ Stretching Exercises/ Prayer/ Moral Stories	To energise the participants to take on the day	Group participation	- Trainer's Guide - Morning Energiser	10 minutes
2	Summative Assessment	Project Presentation	To test and evaluate participants on the skills acquired during the training course	- Group Participation - Presentation by participants	- Trainer's Guide - Assessment Guide	7 hours 50 minutes

Day - 1	Morning Energiser	10 minutes
1.1		

Resources to be used	Morning Energisers booklet
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Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 1	Ice breaker	50 minutes
1.2		

Do	Introduce yourself by mentioning your name and telling the class about one of your likes and dislikes.
Say	Let’s play a simple game to help us get to know each other. Each participant will get 30 seconds to introduce themselves.
Notes for Facilitation	◆ Form a circle with one half of the participants. ◆ With the other half form the outer circle. ◆ The inner circle will face the outer circle. ◆ Each participant in the inner circle will have to spend 30 seconds talking to a participant in the outer circle; try to get to know each other and move on to the next participant.

Day - 1	Introduction to Computers and Windows Operating Systems	1 hour
1.3		

Resources to be used	Multimedia (K-Yan) Participant Handbook
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Say	This session introduces you to the basics of computing and Windows Operating System.
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Do	Run the multimedia unit.
Ask	Can you name some applications and uses of the computer?
Expected Responses	Used in banks, offices, etc.
Say	There are many other applications such as train and air ticket booking, in schools and colleges.
Do	Discuss some of the applications and uses with the class. Continue with the multimedia unit.
Notes for Facilitation	◆ Explain what are input and output device. ◆ Ask them if they know about any more input devices such as joystick, lightpen, scanner, etc. ◆ Explain the features of the keyboard and the mouse. ◆ Show the various sets of keys on the keyboard. ◆ Describe the working of the mouse clearly. ◆ Demonstrate the procedure of "Turning ON" and "Shutting DOWN" the computer. ◆ Demonstrate the basics of File Management System. ◆ Demonstrate the use of short cut keys and icons.

Day - 1	Word Processor (MS Word 2007)	1 hour 30 minutes
1.4		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook
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Say	In this session we will be learning about the features and uses of a Word Processor of the MS Office.
Do	◆ Play the multimedia unit. ◆ Pause the multimedia and explain the interface of Word 2007.

Notes for Facilitation	◆ Demonstrate the use of short cut keys and icons. ◆ Demonstrate the editing and formatting features of Word 2007. ◆ Demonstrate page layout and page settings features of Word 2007. ◆ Demonstrate inserting images and Word Art in Word 2007. ◆ Demonstrate the procedure for saving and opening a document file in Word 2007.
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Day - 1	Spreadsheet (MS Excel 2007)	2 hours 30 minutes
1.5		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook
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Say	In this session we will be learning how to create and use a spreadsheet application using MS Excel 2007.
Ask	Has anyone here used any spreadsheet application before?
Expected Responses	“Yes, to create a database of my friends.” “Yes, to create and update my bank book.”
Say	That’s correct. You can use spreadsheet application for creating a database of information, do calculations, and create charts and many more things. In this session, let us take a look at some of the features of MS Excel 2007.
Do	Show the multimedia unit. Pause and explain Worksheet, Cells, Address of a cell, Columns and Rows.
Notes for Facilitation	◆ Demonstrate entering text in a cell, and editing and formatting that text. ◆ Demonstrate entering different Data Types in Excel. ◆ Demonstrate Fill Data feature in Excel. ◆ Demonstrate the use of functions. ◆ Demonstrate copying text and formula in Excel. ◆ Demonstrate the Sort and Filter feature in Excel. ◆ Demonstrate creating Charts in Excel. ◆ Demonstrate Printing in Excel. ◆ Demonstrate the procedure of Saving and Opening a worksheet in Excel 2007.

Day - 1	Presentation (MS Power Point 2007)	1 hour 30 minutes
1.6		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook
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Say	In this session we will be learning how to create, view and deliver a presentation using MS PowerPoint 2007.
Ask	Has anyone here used or created a presentation before?
Expected Responses	"Yes, for my class projects."
Say	That's great! Let us look at some of the features of MS PowerPoint 2007 in this session.
Do	Run the multimedia module. Pause and explain what are Slides, Presentation, Views and Slide Show.
Notes for Facilitation	◆ Pause the multimedia and demonstrate the features with an example. ◆ Demonstrate entering text in a slide; editing and formatting text. ◆ Demonstrate the use of various slide layouts and how to choose them to create a presentation. ◆ Demonstrate the use of pre-defined templates and slide layouts to create a quick presentation. ◆ Demonstrate inserting pictures, media elements, charts and tables from Word or Excel. ◆ Demonstrate the use of slide views and notes. ◆ Demonstrate how to deliver presentation using slide show. ◆ Demonstrate the procedure of saving, opening creating handouts in Word.

Day - 1	Internet	30 minutes
1.7		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook
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Say	In this session we will be learning about the use of networking and Internet.
Ask	You must have heard the terms “World Wide Web”, “Internet” or “e-mail”. Do you know how to use them?
Expected Responses	“Yes! I browse the net for information.” “I use Google.com to get all my information.”
Say	That is great! The Internet, indeed, is an ocean of information—you can find information on almost every topic you want.
Do	Show the multimedia module. Pause the multimedia unit and ask questions related to the topic.
Notes for Facilitation	Make the session interactive by initiating a discussion. Let the class answer and see to it that all students participate. If the Internet connection is available: ◆ demonstrate using the web browser to browse the Internet; ◆ demonstrate the use of search engines; ◆ demonstrate creating an e-mail account and using it.

Day - 2	Morning Energiser	10 minutes
2.1		

Resources to be used	Morning Energisers booklet
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Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 2	Recap	20 minutes
2.2		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook
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Notes for Facilitation	Give a quick recap of the previous day’s topics in a question-answer format. Make it interactive and make sure all the students participate.
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Day - 2	MS Office–Practice	3 hours
2.3		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook
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Say	This is a practice session for MS Office. Make the most of it and practise what you have learnt in MS Word, Excel and PowerPoint in yesterday’s session.
Do	Give the participants an assignment for each: MS Word, MS Excel and MS PowerPoint.

Notes for Facilitation	Assignments for practice: 1. Prepare your own résumé and time table for the course using MS Word. 2. Make a database of your batchmates with: telephone numbers, addresses, mark sheet, income-expense statement, cricket scoreboard using MS Excel. 3. Create a presentation of with 3 to 4 having details of their education qualifications; why have they joined this course; and what they would like to do after the course.
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Day - 2	Algorithm and Flowchart	2 hours
2.4		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook
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Say	In this session we will be learning the basics of programming. Before we start with any programming language let us understand the principles of programming.
Ask	What is programming?
Expected Responses	“Write a task to be done using programming language”
Say	That’s right. It is, in fact, a set of sequential instructions for the computer.
Do	Introduce the session briefly. Ask questions and encourage the participants to respond.
Do	Play the multimedia module. Pause at screen 1.
Say	An algorithm is a stepwise solution for any problem. It is a finite set of unambiguous instructions performed in a prescribed sequence to achieve a task. Algorithms are the basis for most computer programming.
Do	Continue to play the multimedia module. Pause at screen 2 and give them an activity for writing an algorithm.
Do	Continue to play the multimedia unit. Pause at screen 5.
Say	A flowchart is a pictorial representation of step-by-step instructions.
Do	Give out assignments for creating a flowchart.

Say	Can you now explain what a flowchart is and why it is required?
Do	Wait for the responses and explain the uses of flowchart. Explain how flowcharts and algorithms are useful while writing a program code.
Notes for Facilitation	Encourage all participants to answer in class. You may also provide clues or pose leading questions regarding the topic. Give some practical examples and mathematical tips for writing algorithm and flowcharts: 1. Write an algorithm for reaching the Training School from your home. 2. Write a flowchart for the different routes and modes of transport available to reach the Training School from your home. 3. Write an algorithm to tell your friend who has just arrived at the railway station, as to how to reach your house. 4. Create an algorithm and flowchart to check if the number entered by the user is "odd" or "even". Check the algorithm and flowchart assignments, completed by the students and give individual feedback. You can ask one of the participants to solve the assignment on the board for the class.

Day - 2	Basics of C Programming	1 hour
2.5		

Resources to be used	◆ Participant Handbook ◆ Presentation
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Say	In this session we will look at the basic features and significance of the C language.
Do	Take the class through the presentation and explain why the C language is widely used– its scope, history and significance.

Notes for Facilitation	Explain the following topics with the help of examples: ◆ Basic features and characteristics of the C language ◆ Basic structure of a C program ◆ C character set – the valid characters and the reserved keywords ◆ C identifiers and constants; its types ◆ Rules for writing C identifiers and constants while programming ◆ Variables in C ◆ Data Types in C – explain how to declare variables and initialize them ◆ Different operators available in C – assignment operators, arithmetic operators, expressions with arithmetic operators and unary operators ◆ Increment and decrement operators and their types–pre and post operators ◆ Relational operators ◆ Logical operators ◆ Precedence of operators. Summarise the following: ◆ Data Types in C ◆ How to declare variables ◆ Expressions with arithmetic and relational operators ◆ Different types of logical operators and how to use them.
Do	Give out the following assignments: ◆ Construct and declare variables with different Data Types and initialize them. ◆ Write expressions using C operators. ◆ Work on an assignment on pre and post–decrement/increment operators. ◆ Work on an assignment on relational and logical operators. Discuss the solutions with the class.

Day - 2	First C Program	1 hour 30 minutes
2.6		
Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook ◆ Practical Guide	

Say	Before we start programming in C, let us first understand the basic structure and syntax of a C program.
Do	Play the multimedia unit.
Say	In this session, we will learn how to write a simple program in C.
Do	Open the first C program given in the Practical Guide. Ask the class to do the same.
Notes for Facilitation	Explain the first C program with the help of the following: ◆ #include is a pre-processor directive that directs the C compiler to include the Header File stdio.h, etc. into the current program, ◆ main() is a function that must be called to execute every program, ◆ Printf() is a function that is used to display output on the screen ◆ \n is used so that the cursor on your screen can move to the new line.
Do	Demonstrate how to write the first C program in the Turbo C editor. Compile and run the program. Show the output window to the class.
Ask	Note down all your queries in your notebook; we shall discuss them at the end of this session.
Do	Move on to your next C program.
Say	Look at the program carefully and try to understand the syntax. Notice how all the Data Types are implemented in C, how variables are initialized and declared while programming in C.
Do	Write the second C program in the Turbo C editor. Compile and run the program. Show the output window to the class.
Notes for Facilitation	Explain the second C program in the following manner: ◆ Include header files in your program. ◆ Start the main function. ◆ Declare variables by specifying Data Types. ◆ Assign values to the variables that you declared. ◆ Display the output on screen using print function.
Ask	Did you understand the meaning of %d, %f, %c etc that are being used in the printf() function ?
Notes for Facilitation	All these are Format Specifiers . %d is used to display integer values , % f is used to display floating point values, %c is used to display character values.

Day - 3	Morning Energiser	10 minutes
3.1		

Resources to be used	Morning Energisers booklet
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Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 3	Recap	10 minutes
3.2		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook
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Notes for Facilitation	Do a quick recap of the previous day’s topics. Have an interactive session, in a question-answer format and making sure the whole class participates. Questions for recap: ◆ Explain the syntax of printf() function and scanf() function. ◆ What does the “&” symbol before the variable name in scanf() stand for? ◆ What is the format specifier used for integer data and float data? ◆ How many main() can a ‘C’ program have? ◆ How do we end a statement in a ‘C’ program? Revise the module on tips to write a ‘C’ program.
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Day - 3	Turbo C Editor	1 hour 30 minutes
3.3		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook ◆ Practical Guide
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Do	Show the multimedia unit.
Do	Demonstrate how to write a C program in Turbo C editor. Compile and run the program. Show the output window to the class.
Ask	Start writing your first C program using the Turbo C editor.
Do	Now open your Practical Guide to the corresponding session.

Day - 3	Decision Making and Branching	1 hour
3.4		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook ◆ Practical Guide
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Do	Play the multimedia unit.
Say	This session will introduce you to Decision Making and Branching using the If-Else construct.
Do	Pause the multimedia and explain the structure; encourage the participants to ask questions.
Do	Run the PowerPoint presentation corresponding to this session.
Say	Explain Decision Making and Branching along with examples.
Notes for Facilitation	Ask the students to solve the programs given in the multimedia module and the Practical Guide. Help the participants in debugging in case they encounter any bugs and errors.

Day - 3	Nested If-Else	1 hour
3.5		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook ◆ Practical Guide
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Do	Show the multimedia module. Pause and explain the Nested If-Else construct. Continue with the multimedia module.
Say	Now let us try out the programs using the Nested If-Else.
Do	Do a quick recap of the topic.
Notes for Facilitation	Ask the students to solve the programs given in the multimedia module as well as the Practical Guide. Help the participants in debugging in case they encounter any bugs and errors.

Day - 3	Logical Operator–AND	30 minutes
3.6		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook ◆ Practical Guide
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Do	Run the multimedia module. Pause at Screen 2.
Say	Logical operators are also called Boolean operators as they return Boolean values: TRUE or FALSE. There are three logical operators in C: AND – denoted by the double ampersand (&&) OR – denoted by the double vertical lines () NOT – denoted by the exclamation (!)
Do	Run the multimedia module.

Say	Please open your Practical Guides and look at the first program. This program will check equality of three numbers demonstrating the use of AND operator (&&).
Notes for Facilitation	Ask the students to solve the programs given in the multimedia module as well as the Practical Guide. Help the participants in debugging in case they encounter any bugs and errors.

Day - 3	Logical Operators–OR and NOT	30 minutes
3.7		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook ◆ Practical Guide
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Do	Run the multimedia module. Pause at Screen 2 that displays the true table. Explain the true table.
Notes for Facilitation	Ask the students to solve the programs given in the multimedia module as well as the Practical Guide. Help the participants in debugging in case they encounter any bugs and errors.

Day - 3	Switch Case	35 minutes
3.8		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Ask the participants to open their Practical Guides.
Notes for Facilitation	The students are required to solve the programs given in both the multimedia module as well as the Practical Guide. Help the participants in debugging in case they encounter any bugs and errors.

Day - 3	The While and the Do-While Loops	1 hour
3.9		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook ◆ Practical Guide
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Say	In this session we will learn about Loops. Loops allow executing a set of instructions repeatedly, until a certain condition is reached.
Ask	What is the difference between the While and Do-While loops?
Expected Responses	"In While the condition is checked at the top of the loop and in the Do-While the condition is checked at the end of the loop."
Say	That's correct! The While is a type of entry control loop, where the condition is checked at the top of the loop. The Do-While is a variation of the While, where the condition is checked at the bottom of the loop. Therefore, the Do-While always executes at least once and once inside the loop and iterates until condition becomes false.
Notes for Facilitation	The students are required to solve the programs given in both, the multimedia module as well as the Practical Guide. Help the participants in debugging in case they encounter any bugs and errors.

Day - 3	The For Loop	1 hour 35 minutes
3.10		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Say	In this session we will learn about Loops that allow executing a set of instructions repeatedly until a certain condition is reached.
Do	Play the multimedia module. Pause at screen 4 and explain the Memory Bytes occupied by each Data Type.
Notes for Facilitation	The students are required to solve the programs given in both, the multimedia module as well as the Practical Guide. Help the participants in debugging in case they encounter any bugs and errors.

Day - 4	Morning Energiser	10 minutes
4.1		

Resources to be used	Morning Energisers booklet
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Notes for Facilitation	Please refer to the "Morning Energisers" booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 4	Recap	20 minutes
4.2		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook
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Notes for Facilitation	Do a quick recap of the previous day's topics. Have an interactive session, in a question-answer format making sure the whole class participates. Answer the queries (if any) noted down by the students in the previous day's session.
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Day - 4	Introduction to Functions	30 minutes
4.3		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook
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Say	Let us see how to write our own functions in C.
Ask	What is the need for writing functions?
Expected Responses	"The code for a procedure can be written once and be <i>called</i> many times."

Say	That's right. Functions are small routines or set of instructions that can carry out specific tasks. Writing functions avoids re-writing the same block of code over and over again to carry out a task exactly in the same way at various stages of a program.
Do	Play the multimedia module. Pause at Screen 4 and explain return value of a function. Continue with the multimedia module.
Notes for Facilitation	The students are required to solve the programs given in both, the multimedia module as well as the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 4	Writing C Functions	1 hour
4.4		

Resources to be used	◆ Multimedia (K-Yan) ◆ Practical Guide ◆ Participant Handbook
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Do	Show the multimedia module. Let us see how to declare and define functions in this session. Pause at Screen 5 and ask the students to try the program. Continue with the multimedia unit and summarise the topic.
Notes for Facilitation	The students are required to solve the programs given both in the multimedia module as well as the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 4	Function Categories	1 hour
4.5		

Resources to be used	◆ Multimedia (K-Yan) ◆ Practical Guide
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Say	Based on the arguments and return values, functions can be categorised into: 1. Functions with no arguments and no return values 2. Functions with arguments and no return values 3. Functions with arguments and return values 4. Functions with no arguments and return values Let us see all the categories with examples.
Do	Show the multimedia module. Pause at Screen 2 and ask the students to try the program.
Do	Continue with the multimedia module. Explain how function with arguments and no return values work. Pause at Screen 3 and ask the students to try the program.
Say	We saw a type of function which takes arguments but does not return a value back to the caller function. Let us continue with the multimedia unit and see the next type.
Do	Pause the multimedia and ask the trainees to solve the program for this type shown in the multimedia unit. Discuss the output with the students.
Say	This is a commonly used function category since it can facilitate a two-way communication. Arguments can be passed from the "caller" function to the "called" function and the result data can be returned to the caller function. Let us now see the last type – Functions with no arguments and Return Values.
Notes for Facilitation	The students are required to solve the programs given in both, the multimedia module as well as the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 4	Function Calls	1 hour
4.6		

Resources to be used	◆ Multimedia (K-Yan) ◆ Practical Guide
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Ask	What is a Function Call?
Expected Responses	"Executing the statements defined in the function"
Say	Yes, when we <i>call</i> a function it executes the statements defined within the body of the function and returns control back to main function.
Do	Explain the two types of function calls – Call by value and Call by reference. Start the multimedia module.
Notes for Facilitation	◆ Explain the difference between call by Value and call by Reference. ◆ Explain Reference Operator and Pointer. ◆ Note down a few code snippets using Reference Operator and Pointer variables on the board. The students are required to solve the programs given in both, the multimedia module as well as the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 4	Introduction to Arrays	1 hour
4.7		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook ◆ Practical Guide
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Ask	How will you define a single variable to hold a list of similar values?
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Expected Responses	"Group variables"
Say	Yes, group them to an Array. Let us now see what an Array is and how it is different from the normal variable.
Ask	What is an Array of Pointers?
Expected Responses	"Collection of addresses"
Say	That's correct. Array of Pointers is a collection of address location. The Array elements of a Pointer Array will store addresses.
Notes for Facilitation	◆ Explain one-dimensional Array and how to declare and define Array. ◆ Explain in detail how to access Array elements. ◆ Explain Index or Array Subscript. The students are required to solve the programs given in both, the multimedia module as well as the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 4	Two Dimensional Arrays	1 hour
4.8		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook ◆ Practical Guide
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Do	Play the multimedia unit. Pause at screen 4 and explain.
Say	Two dimensional Array is a matrix of rows and columns that are stored in the contiguous memory.
Notes for Facilitation	◆ Explain multi-dimensional Array and how to declare and define them. ◆ Explain in detail how to access Array elements in a two-dimensional Array. ◆ Explain Index or Array Subscript in multi-dimensional Array. The students are required to solve the programs given in both the multimedia module as well as the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 4	More on Arrays	30 minutes
4.9		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Ask	What is an Array?
Expected Responses	"An Array is a list of data item of similar type."
Say	Indeed, an Array is a group of similar Data Type. ◆ A single dimensional Array has one dimension or size. ◆ A multi-dimensional Array has more than one dimension. ◆ A two-dimensional Array is a type of multi-dimensional Array which is a matrix of rows and columns.
Notes for Facilitation	The students are required to solve the programs given in both the multimedia module as well as the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 4	Strings	30 minutes
4.10		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook ◆ Practical Guide
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Do	Run the multimedia module. Pause at screen 3 and ask the students to try the program examples in the multimedia unit.
Ask	How are elements in a character Array or String accessed?
Expected Response	"Same way as in numeric Arrays"

Say	That's right! Elements in a character Array or a String are accessed in the same way as in numerical Arrays. Using the Format Specifier %s the entire String can be accepted or displayed.
Notes for Facilitation	◆ Show the String functions and demonstrate the use of a few of them. ◆ The students are required to solve the programs given in both, the multimedia module as well as the Practical Guide. ◆ Help the participants in debugging in case they encounter any bugs or errors.

Day - 4	Structures	1 hour
4.11		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook ◆ Practical Guide
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Do	Play the multimedia unit. Ask the students to try the programs in the multimedia unit.
Notes for Facilitation	◆ Explain the Structure-Code. ◆ Explain the difference between Arrays and Structures. ◆ Demonstrate Structure Variable as an argument in a program. ◆ Explain the use of fflush() function. The students are required to solve the programs given in both, the multimedia module as well as the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 5	Morning Energiser	10 minutes
5.1		

Resources to be used	Morning Energisers booklet
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Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 5	Recap	20 minutes
5.2		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook
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Notes for Facilitation	◆ Recapitulate the previous day’s topic in a question-answer format. ◆ Make it interactive and encourage the whole class to participate. ◆ Take questions and clarify doubts related to the previous session.
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Day - 5	Concept of OOP	30 minutes
5.3		

Resources to be used	◆ Participant Handbook ◆ Practical Guide ◆ Presentation
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Do	Run the PowerPoint presentation.
	◆ Explain Subject-Oriented Programming and its unique features. ◆ Explain the concept of Objects and Classes. Show them the diagram as it illustrates: A class with 3 objects is shown and the objects can communicate with each other through messages. Each object is unique in its own with data and functions that manipulate that data.

Notes for Facilitation	◆ Explain Inheritance; how reusability of code is done while one class acquires the properties of another class. ◆ Explain Polymorphism; how a single function is used again and again by overloading and overriding it according to the behaviour of the object. ◆ Explain the concept of Data Abstraction and Encapsulation; how data is protected from external interference or misuse. ◆ Explain the difference between Procedure-Oriented programming approach and Object-Oriented approach .
Do	Ask participants to cite some more examples based on the concept.
Notes for Facilitation	The students are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 5	Programming with C++	1 hour
5.4		

Resources to be used	◆ Participant Handbook ◆ Practical Guide ◆ Presentation
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Do	Run the PowerPoint presentation.
Notes for Facilitation	◆ Share the history of C++ programming. ◆ Illustrate the basic structure of a C++ program. There are 6 sections in the structure of a C++ program. ◆ Show the participants how to work out the program given as an example: Adding two numbers in C++. ◆ Explain the first and second section of the structure of C++ program. ◆ Explain the third, fourth and fifth section of the structure of C++ program in detail. The students are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 5	Objects and Classes	1 hour
5.5		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 5	Member Functions	30 minutes
5.6		

Resources to be used	◆ Participant Handbook ◆ Practical Guide ◆ Presentation
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Do	Run the PowerPoint presentation.
Notes for Facilitation	Introduce the Member Functions in C++. Explain how they are defined. Explain: ◆ The access specifier–Private; it describes the scope of that particular member function ◆ The access specifier–Public; it describes the scope of that particular member function ◆ The access specifier–Protected; it describes the scope of that particular member function.
Do	Explain the program corresponding to Member Functions. It calculates the area and circumference of a circle using Objects and Classes. Ask the participants to focus on the coding and declaration, definition of functions along with their scope.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 5	Constructors and Destructors	1 hour
5.7		

Resources to be used	◆ Participant Handbook ◆ Practical Guide ◆ Multimedia (K-Yan) ◆ Presentation
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Do	Run the multimedia module.
Say	We saw how we can initialize member variables using a Constructor. Remember, a Constructor is a Member Function with the same name as its Class. By default, the Compiler defines a Destructor and executes it at run time.
Ask	How do we declare a Destructor?
Expected Responses	"With the tilde symbol"
Say	That's right! A Destructor is declared with the same name of the class, preceded by a tilde symbol. Destructors are used to release the memory allocated for the object.
Notes for Facilitation	◆ The participants are required to solve the programs given in both, the multimedia as well as the Practical Guide. ◆ Help the participants in debugging in case they encounter any bugs or errors. ◆ Do a quick recap with the help of the presentation. Answer the participants' queries and clarify their doubts.

Day - 5	Operator Overloading and Function Overloading	2 hours
5.8		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook ◆ Practical Guide ◆ Presentation
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Do	Run the multimedia module.
Notes for Facilitation	◆ Discuss Function Overloading. ◆ Explain Operator Overloading. ◆ What is this overloading behaviour referred to as in OOP?
Do	The participants are required to solve the programs given in both, the multimedia as well as the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors. Do a quick recap with the help of the presentation. Answer the participants' queries and clarify their doubts.

Day - 5	Inheritance	1 hour 30 minutes
5.9		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook ◆ Practical Guide ◆ Presentation
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Do	Show the multimedia unit. Pause at screen 2.
Notes for Facilitation	◆ Explain Inheritance in OOP. ◆ Use diagrams to explain Base Class and Derived Class. ◆ Explain the program code and output of the program in detail.

	◆ Elucidate: • the syntax for base class and derived class; • the significance of the scope resolution operator:: used in the program codes; • how the functions (of base class and derived class) are accessed through object with dot operator in the main function.
Notes for Facilitation	The participants are required to solve the programs given in both, the multimedia as well as the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors. Do a quick recap with the help of the presentation. Answer the participants' queries and clarify their doubts.

Day - 6	Morning Energiser	10 minutes
6.1		

Resources to be used	Morning Energiser booklet
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Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 6	Recap	20 minutes
6.2		

Resources to be used	Presentation
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Notes for Facilitation	Do a quick recap of the previous day’s topics through a quiz.
Do	◆ Divide the team in to two teams and use the presentation for the quiz questions. ◆ Give one minute for each question. ◆ Pass the question to the second team if the first team is unable to respond within a minute. ◆ Keep scores on the board. ◆ Announce the winners. ◆ Make it interactive and encourage whole class participation.
Notes for Facilitation	Answer the participants’ queries and clarify their doubts from the previous day’s session.

Day - 6	Understanding Protected Members	1 hour
6.3		

Resources to be used	◆ Participant Handbook ◆ Practical Guide ◆ Presentation
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Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.
Notes for Facilitation	The students are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 6	Pointers in C++	1 hour
6.4		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook ◆ Practical Guide
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Do	◆ Run the multimedia module. ◆ Recall pointers done in C. Make the session interactive by putting down code snippets on the board and asking questions to the class. ◆ Ensure that the participants are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in both, the multimedia as well as the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 6	Pointers in C++ – Practice	1 hour
6.5		

Resources to be used	◆ Participant Handbook ◆ Practical Guide ◆ Presentation
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Do	Do a quick recap using the Presentation. Ask the participants to continue with the exercises from the Practical Guide.
Notes for Facilitation	The students are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors. Have an interactive question-answer session while clarifying doubts.

Day - 6	Introduction to Data Structure	30 minutes
6.6		

Resources to be used	◆ Trainer's Guide ◆ Participant Handbook ◆ Presentation
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Do	Show the PowerPoint presentation of this session to all the trainees.
Notes for Facilitation	◆ Explain Data Structures with the help of a diagram. ◆ Explain Linear Data Structures. ◆ Explain Stack. ◆ Explain Queues. ◆ Explain Enqueue and Dequeue Operations on Queues. ◆ Explain Linked List Data Structure. ◆ Explain Non-Linear Data Structures. ◆ Explain Tree Data Structure. ◆ Explain Graph Data Structure.

Day - 6	Data Types	30 minutes
6.7		

Resources to be used	◆ Trainer's Guide ◆ Participant Handbook ◆ Presentation
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Do	Run the PowerPoint presentation.
Notes for Facilitation	◆ Start with the definition of Data Types. ◆ Use a table to describe the range and size (in bytes) of all the fundamental Data Types. ◆ Explain Fundamental Data Types with examples. ◆ Explain Derived Data Types with examples.

Day - 6	Operations on Data Structures	1 hour
6.8		

Resources to be used	◆ Trainer's Guide ◆ Participant Handbook ◆ Presentation
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Do	Run the PowerPoint presentation.
Notes for Facilitation	◆ Explain the operations that can be performed on Data Structures. ◆ Explain the Traversing Operation with an example. ◆ Explain the Searching Operation along with the types of Searching. ◆ Explain Insertion operation on the Data Structures. ◆ Explain Deletion Operation on the Data Structures. ◆ Explain Sorting Operation on the Data Structures. ◆ Explain Merging Operation on the Data Structures.

Day - 6	Bubble Sort	1 hour
6.9		

Resources to be used	◆ Trainer's Guide ◆ Participant Handbook ◆ Presentation
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Say	Bubble Sort is an algorithm to arrange items in order.
Notes for Facilitation	The students are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 6	Searching	1 hour
6.10		

Resources to be used	◆ Participant Handbook ◆ Practical Guide ◆ Presentation
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Do	Run the PowerPoint presentation.
Notes for Facilitation	◆ Introduce the term Linear Search. ◆ Explain the algorithm for Linear Search. ◆ Explain Binary Search. ◆ Write down the steps for the Binary Search algorithm. The algorithm helps to implement Binary Search while programming in C++.

Day - 7	Morning Energiser	10 minutes
7.1		

Resources to be used	Morning Energiser booklet
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Notes for Facilitation	Please refer to the "Morning Energisers" booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 7	Recap	20 minutes
7.2		

Resources to be used	Presentation
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Do	Do a quick recap of the previous day's topics in a question-answer format. Refer to the following questions.
Notes for Facilitation	Recap questions and answers: Q: What is the Accessibility Rule for Protected Inherited Base Class Members? A: The Protected Inherited Base Class Members can be accessed in the base class but inaccessible from outside the base class or derived class. Q: What is a Pointer Variable? A: A Pointer Variable is a variable that contains the address of another variable in a memory location. Q: What are the two C++ Operators which work on Pointers? A: Address Operator or Reference Operator (&) and Indirection Operator (*) Q: What is Data Structure? A: The logical model of a particular organisation of data is known as Data Structure. The Data Structure stores data in such a way that data can be accessed appropriately. Q: Name some commonly used Data Structure. A: The most commonly used Data Structures are arrays, stack, queues, linked list, trees and graphs.

	Q: What are the two types of Data Structures? A: Linear and Non-Linear Q: Classify the commonly used Data Structures – Arrays, Sets, Stack, Queues, Linked lists, Trees, Graphs – into linear and non-linear Data Structure. A: Linear: Linked list, Stack, Queues, Array Non-linear: Tree, Sets, Graph Q: What are the common operations which can be performed on these Data Structure? A: Traversing, searching, insertion, deletion, sorting Q: What is Merging? A: Merging is the process of combining several sorted lists (arrays) into single sorted list (array). Q: What is Bubble Sort? A: Bubble Sort is nothing but an algorithm to arrange items in order. Q: Explain the process of Bubble Sort algorithm. A: Bubble Sort is an algorithm that repeatedly compares two adjacent elements and places the elements in position during the process, to give an ordered list of elements.
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Day - 7	Stacks	1 hour
7.3		

Resources to be used	◆ Participant Handbook ◆ Practical Guide ◆ Presentation
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Do	Run the PowerPoint presentation.
Notes for Facilitation	◆ Introduce Stacks and the principle on which it works. ◆ Discuss the two operations that can be performed on Stacks and explain to them that "push" means to insert the elements and "pop" means to delete the elements from the Stack.
Ask	Note the difference between push and pop and where these operations take place.

Notes for Facilitation	◆ Explain the terms Overflow and Underflow. ◆ Show the algorithm for the insertion of an element to the Stack; and that when the Stack memory is full it returns the Overflow Condition, otherwise it adds 1 element to the top of the Stack. ◆ Show the algorithm for the deletion of an element from the Stack. If the Stack memory is empty and still we tend to delete any element, it returns an underflow condition, otherwise it deletes an element from the top of the Stack. ◆ Discuss the need for using Stacks and that we can implement Stacks using linked lists also.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 7	Queues	1 hour
7.4		

Resources to be used	◆ Participant Handbook ◆ Presentation
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Do	Run the PowerPoint presentation.
Do	Explain the term Circular Queues with the help of a diagram.
Ask	Note the types of queues and understand the difference between them.
Notes for Facilitation	◆ Explain the Ascending Priority Queue in which the priority is given to the elements which have the least priority number. Similarly, explain Descending Priority Queue. ◆ Explain Dequeue. It is a double-ended Queue (often written as deque, pronounced "deck") is a queue where insertion and deletion can be performed at both end and it is often called Head Tail Linked List. ◆ Explain the Restricted Input and Output Dequeues with the help of diagrams.

Day - 7	Types and Applications of Queues	1 hour
7.5		

Resources to be used	◆ Participant Handbook ◆ Practical Guide ◆ Presentation
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Do	Run the PowerPoint presentation.
Notes for Facilitation	◆ Give an introduction to Circular Queues with the help of a diagram. ◆ Explain them the Ascending Priority Queue ◆ Explain Dequeue. ◆ Explain the Restricted Input and Output Dequeues with help of diagrams. ◆ Explain the applications of the Queues.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 7	Linked List	1 hour
7.6		

Resources to be used	◆ Participant Handbook ◆ Practical Guide ◆ Presentation
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Do	Run the PowerPoint presentation.
Notes for Facilitation	Give a brief introduction to Linked Lists and how it is represented.
Ask	Look at the presentation and note down the algorithm for Singly Linked List.

Notes for Facilitation	◆ Explain the algorithm to the students. ◆ Tell the students all about Singly Linked List. ◆ Explain where and how to insert nodes into Linked List. ◆ Explain the algorithm for inserting the nodes in detail. ◆ Show how one can find a node in a particular Linked List. ◆ Discuss how we can delete a particular node for the Linked List. ◆ Explain how to print all the elements of the list on the screen. ◆ Explain how the algorithm works to destroy the Linked List. ◆ Explain how we can use Linked List in our programs. ◆ Discuss the difference between arrays and Linked Lists.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 7	Types of Linked Lists	1 hour
7.7		

Resources to be used	◆ Participant Handbook ◆ Presentation
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Do	Run the PowerPoint presentation. Ask the class to take notes of the important points.
Notes for Facilitation	◆ Discuss Singly Linked List. ◆ Explain the Circular Linked List with the help of a diagram. ◆ Explain the Doubly Linked List; tell them it has two pointers, therefore it can traverse backward too. ◆ This slide helps to understand the Doubly Linked List more clearly with the help of diagrams.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 7	Trees	1 hour
7.8		

Resources to be used	◆ Participant Handbook ◆ Presentation
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Do	Run the PowerPoint presentation.
Notes for Facilitation	◆ Give a brief introduction to Trees. ◆ Show that Tree has a finite number of nodes: the first node from where all the nodes are connected is called the root node. ◆ Ask them to see the diagram carefully and understand how the degree of Tree degree of nodes and height of nodes are calculated. ◆ Show the diagram of the Tree and explain how the values are calculated. ◆ Explain what a Binary Trees is; when the Trees is called a Binary Trees; and explain the hierarchy. ◆ Ask the students to look at the diagram and try to understand the difference between this (Binary Trees) and other Trees seen before. ◆ Ask them to compare the Trees and understand their different types. ◆ Help them understand the relation between the number of leaf nodes and nodes of degree 2. ◆ Tell them the difference between complete and full binary Tree. A full binary Tree of depth k is a binary Tree of depth k having $2^{k+1} - 1$ nodes, $k \geq 0$. ◆ Show them how to represent a Binary Trees.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 7	Operations on Binary Tree	1 hour
7.9		

Resources to be used	◆ Participant Handbook ◆ Presentation
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Do	Run the PowerPoint presentation.
Notes for Facilitation	◆ Give an introduction to Binary Tree Traversal. ◆ Talk about the arithmetic expressions using Binary Tree. ◆ Show how the preorder Traversing is done. ◆ Show the post order Traversing. ◆ Explain the difference between all types of Traversing.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 7	Graphs	30 minutes
7.10		

Resources to be used	◆ Participant Handbook ◆ Presentation
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Do	Run the PowerPoint presentation.
Notes for Facilitation	Talk about the types of Graphs, viz. directed and undirected.
Ask	Look at the diagram carefully and understand a complete Graph as opposed to an incomplete graph. You may take down the diagram in your notebooks.

Notes for Facilitation	◆ Explain the diagram and tell them how to calculate the number of edges in complete directed and complete undirected graphs. ◆ Explain what a complete Graph is. ◆ Discuss adjacent and incident Graphs. ◆ Tell the students about subgraphs and paths. ◆ Help them understand subgraphs with the help of diagrams. ◆ Explain simple path and cycle. ◆ Show the different types of Graphs. ◆ Explain connected components of a Graph. ◆ Show images of connected components in a Graph. ◆ Explain the degree of a Graph. ◆ Explain adjacency matrix. ◆ Give examples of adjacency matrix. ◆ Discuss what operations can we perform on Graphs. ◆ Explain the spanning trees in detail for better understanding of the Graphs. ◆ Explain the Depth First search. ◆ Explain Breadth First search.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 8	Morning Energiser	10 minutes
8.1		

Resources to be used	Morning Energisers booklet
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Notes for Facilitation	Please refer to the "Morning Energisers" booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 8	Recap	20 minutes
8.2		

Resources to be used	◆ Participant Handbook ◆ Presentation
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Do	Do a quick recap of the previous day's topics in a question-answer format.
Notes for Facilitation	◆ Run the presentations (if required) for the recap. ◆ Answer the participants' queries and clarify their doubts from the previous day's session.

Day - 8	Introduction to HTML	2 hours 30 minutes
8.3		

Resources to be used	◆ Participant Handbook ◆ Presentation
Notes for Facilitation	◆ Introduce the class to HTML; its significance and scope. ◆ Take them through the terminologies used in Web design. ◆ Discuss some basic components of HTML. ◆ Give an in-depth explanation of some Document Structure Tags.

	◆ Show them the basic syntax of the Structure Tags. ◆ Explain Block-Level Elements in HTML. ◆ Explain Text Level Elements. ◆ Discuss some Form Elements. ◆ Look at some more Text Level Elements such as Anchor Element, Image Tag and their attributes. ◆ Tell them about Applet and Font Tags. ◆ Show them how to add Horizontal Rules, Special Characters and Comments. ◆ Give them an explanation of Meta Tag and Meta Data. ◆ Give them an in-depth explanation of URLs. ◆ Discuss all attributes of Hyperlink and Anchor Tags. ◆ Show the syntax for each and every tag element along with the given examples. Ask the participants to cite some more examples based on the given examples.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 8	Creating an HTML Page	45 minutes
8.4		

Resources to be used	◆ Presentation ◆ Participant Handbook ◆ Practical Guide
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Do	Open the Practical Guide.
Notes for Facilitation	Before the class starts coding in HTML, make sure they understand the basic HTML components. Also explain the syntax while going through the program.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 8	Creating Lists	45 minutes
8.5		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Continue with the practice session. Ensure that the participants are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 8	Images, Colors and Background	30 minutes
8.6		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Continue with the practice session. Ensure that the participants are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 8	Tables and Frames	30 minutes
8.7		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook ◆ Practical Guide
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Do	Continue with the practice session from the Practical Guide. Ensure that the participants are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 8	Forms	30 minutes
8.8		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Continue with the practice session. Ensure that the participants are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 8	Cascading Style Sheets (CSS)	1 hour 50 minutes
8.9		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Continue with the practice session. Ensure that the participants are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day 9 to Day 32

Elective 1: Visual Studio Dot NET

Day - 9	Morning Energiser	10 minutes
9.1		

Resources to be used	Morning Energisers booklet
Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.

Day - 9	Test 1	2 hours
9.2		

Resources to be used	Assessment Guide
Notes for Facilitation	Conduct a test for the topic C programming, refer to the questions in the Assessment Guide.

Day - 9	Introduction to .NET Framework	1 hour
9.3		

Resources to be used	◆ Participant Handbook ◆ Presentation
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Do	Run the PowerPoint presentation.
Say	◆ Give an introduction to .NET Framework. ◆ Talk about the main components of .NET Framework. ◆ Explain the features of CLR. ◆ Explain Intermediate Language and .NET Framework Class Library. ◆ Discuss the classification of Types. ◆ Explain .NET technologies. ◆ Detail the applications of .NET Framework.

Day - 9	VB .NET and its Functionality	1 hour
9.4		

Resources to be used	◆ Participant Handbook ◆ Presentation
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Do	Run the PowerPoint presentation.
Notes for Facilitation	◆ Give an introduction to VB .NET and its functionality. ◆ Give an overview of Visual Basic (VB) and its concepts used in .NET Framework. ◆ Discuss code writing mechanics, naming conventions in VB. ◆ Discuss Variables and Sub Procedures in VB. ◆ Explain Loop Structure, Object and Object Basics.

Day - 9	Programming Using VB .NET	2 hours
9.5		

Resources to be used	◆ Participant Handbook ◆ Practical Guide ◆ Presentation
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Do	Run the PowerPoint presentation.
Notes for Facilitation	◆ Introduce the concept of Integrated Development Environment (IDE). ◆ Explain Visual Basic Data Types, the different kinds of Data Types; its storage size and value range. ◆ Give a detailed description of Loops. ◆ Draw the class's attention to the syntax and condition of different Loops. ◆ Explain Control Structure and Selection Structure. ◆ Explain the methods of Control Structure. ◆ Explain how to use Loops and Control Structures in programs. ◆ Use a program to explain Case Selection Structure. ◆ Explain how to create Visual Basic .NET projects and show how to analyse Project Structures.

	◆ Explain Assemblies. ◆ Explain Reference; and how to add a Reference. ◆ Explain Namespaces, setting Project Properties using the Development Environment Features. ◆ Demonstrate and explain Debugging Applications and Compiling in Visual Basic .NET. ◆ Summarise IDE, Data Types, VB .NET, OOP, Control Structure and Case.
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Day - 9	Programming Using VB .NET	1 hour 40 minutes
9.6		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Open the Practical Guide.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 10	Morning Energiser	10 minutes
10.1		

Resources to be used	Morning Energisers booklet
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Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 10	Grooming	2 hours 30 minutes
10.2		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook ◆ Presentation
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Do	Play the multimedia.
Notes for Facilitation	Discuss the topic of “personal grooming”. Make it into an interactive session; invite the participants to talk about their views on the importance of personal grooming in the professional front.
Notes for Facilitation	Sensitise the participants about the following: ◆ Aspects of personal grooming that reflect on the overall personality of a person ◆ Maintaining hygiene and healthy habits ◆ Choice of clothing at workplace ◆ Hair and make-up ◆ Making an impression: effect of clothing and grooming on other people ◆ The grooming checklist ◆ Preparation before an interview, event, etc.
Do	Ask the participants to fill out the questionnaire as per their understanding.

Day - 10	Creating Forms in VB .NET	2 hours
10.3		

Resources to be used	◆ Trainer's Guide ◆ Participant Handbook ◆ Presentation
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Do	Run the PowerPoint presentation.
Notes for Facilitation	◆ Explain how to create forms in VB .NET. ◆ Explain the function of VB .NET, FunctionName (Arguments), and objective of MsgBox. ◆ Discuss the procedure and language of VB.NET. ◆ Explain Syntax and Data Types in VB.NET. Discuss the different types of Data Type, its CLR Type Structure, Size and Value Range. ◆ Explain user-defined Data Types. ◆ Explain the types of Variables in VB .NET. ◆ Explain Method, its types and the difference between Functions and Subroutines. ◆ Discuss the properties and types of properties in VB.NET. ◆ Explain Exception Handling and its types. ◆ Summarise the important points discussed in this module.

Day - 10	Creating Forms in VB .NET	1 hour 30 minutes
10.4		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 10	Recap	20 minutes
10.5		

Resources to be used	Participant Handbook Presentation
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Notes for Facilitation	◆ Recapitulate the day's topics in a question-answer format. ◆ Make it interactive and encourage the whole class to participate. ◆ Take questions and clarify doubts related to the previous session.
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Day - 10	Creating Forms in VB .NET	2 hours
10.6		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 11	Morning Energiser	10 minutes
11.1		

Resources to be used	Morning Energisers booklet
Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.

Day - 11	Discipline	2 hours 30 minutes
11.2		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook ◆ Presentation
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Do	Run the multimedia. Pause and give an explanation or allow discussion wherever required.
Notes for Facilitation	Have an interactive session on: ◆ The importance of discipline to be able to complete given tasks; ◆ How to be self-disciplined.
Notes for Facilitation	Discuss the following: ◆ Willpower is the training and control of oneself and one's conduct, usually for personal improvement. ◆ The Five Pillars of Self-Discipline: acceptance, willpower, hard work, industriousness and persistence. ◆ Being industrious doesn't necessarily mean doing the work that's challenging or difficult. It simply means putting in the time. Run the presentation and summarise the session.
Do	Ask the participants to fill out the questionnaire as per their understanding.

Day - 11	Form Controls	2 hours
11.3		

Resources to be used	◆ Participant Handbook ◆ Presentation
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Do	Run the PowerPoint presentation.
Notes for Facilitation	◆ Give an introduction to Web Form. ◆ Explain Web Controls and their categories. ◆ Explain the different types of Web controls used. ◆ Explain Label in Basic Web Controls. ◆ Explain Textbox Controls. ◆ Explain Button and its different types. ◆ Explain Picture Box, Image Button and Link Button. ◆ Explain Hyper Link. ◆ Explain Radio Button and Radio Button List. ◆ Explain Checkbox and Checkbox List. ◆ Explain DropDown List. ◆ Explain the ListBox. ◆ Explain the Panel. ◆ Explain the Placeholder. ◆ Give a detailed description of Form Controls. Highlight the different types of Web Form Controls.

Day - 11	Form Controls	2 hours
11.4		

Resources to be used	Participant Handbook Presentation
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Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.
Note for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 11	Recap	20 minutes
11.5		

Resources to be used	◆ Participant Handbook ◆ Presentation
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Do	Revise the key topics from what has been discussed so far. Refer to the questions given below.
Notes for Facilitation	1. Which is the Parent Class of the Web Server Control? 2. Can you set which type of comparison you want to perform by the CompareValidator control? 3. What is the difference between HTML and Web Server Controls? 4. What is the difference between a Label Control and a Literal Control? 5. Which Control will you use to ensure that the values in two different Controls match? 6. Which class is inherited when an ASP .NET Server Control is added to a Web Form? 7. What are the three common properties of all Validation Controls. 8. What are Web Server Controls in ASP .NET? 9. How can you implement the Postback property of an ASP .NET Control? 10. What are User Controls and Custom Controls? 11. What are the Validation Controls? 12. What is Auto Post Back? Answer the queries and clarify the doubts of the participants. Run the presentation if required.

Day - 11	Form Controls	1 hour
11.6		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 12	Morning Energiser	10 minutes
12.1		

Resources to be used	Morning Energiser booklet
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Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 12	Time Management	2 hours
12.2		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook ◆ Activities and Games booklet ◆ Presentation
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Do	Play the multimedia unit. Pause and explain wherever there is an explanation required.
Notes for Facilitation	◆ Discuss the myths about time we generally have. ◆ Discuss what activities the participants consider as “time-wasters”. ◆ Explain what “time management” means.
Do	Conduct a group activity by choosing a relevant one from the Activities and Games booklet.
Notes for Facilitation	Tell the class about how time management is important in our day-to-day life for achieving success. Lack of time management leads to non-fulfilment of tasks and duties in time which further leads to stress and mismanagement. Following are some tips on time management: ◆ Making activity logs help us to remember all the tasks to be done. ◆ Planning helps us to chalk out those activities which are necessary to achieve our goals.

	◆ Scheduling helps us to organise the activities to be done and allot appropriate time to each activity. Highlight how prioritising helps us to manage our time efficiently by concentrating more on important activities. Run the presentation for summarising the session.
Do	Ask the participants to fill out the questionnaire as per their understanding.

Day - 12	OOP Using VB .NET	1 hour 30 minutes
12.3		

Resources to be used	◆ Participant Handbook ◆ Presentation
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Do	Run the PowerPoint presentation.
Notes for Facilitation	◆ Discuss the concepts of VB .NET and OOP. ◆ Explain OOP. ◆ Discuss OOP in Visual Basic .NET. ◆ Discuss Classes. ◆ Discuss Object. ◆ Explain NameSpace. ◆ Explain Abstraction. ◆ Explain Encapsulation. ◆ Explain Overloading and Overriding. ◆ Give a description of Inheritance. ◆ Explain Polymorphism.
Notes for Facilitation	Summarise the day's learning and encourage queries from the participants.

Day - 12	Recap	20 minutes
12.4		

Resources to be used	◆ Participant Handbook ◆ Presentation
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Notes for Facilitation	Revise the key topics from what has been discussed so far. Answer their queries and clarify the doubts of the participants. *Run the presentation if required.
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Day - 12	OOP Using VB .NET	2 hours
12.5		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 12	OOP Using VB .NET	2 hours
12.6		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Continue with the practice session. Ensure that the participants are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 13	Morning Energiser	10 minutes
13.1		

Resources to be used	Morning Energisers booklet
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Notes for Facilitation	Please refer to the "Morning Energisers" booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 13	Attitude at Work	2 hours
13.2		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook ◆ Presentation
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Do	Play the multimedia module. Pause and explain the key topics.
Say	Result means the outcome or consequence of something and the right attitude will enable them to achieve their goal. Similarly, "result-oriented management style" helps an organisation to achieve success.
Notes for Facilitation	Discuss what the participants think about result-orientation. ◆ Result-orientation means knowing which results are important and striving to achieve them by channelising appropriate resources.
Do	Ask the participants to fill out the questionnaire as per their understanding.
Notes for Facilitation	Things to be kept in mind while being result-oriented: ◆ While working in a team, people with a result-oriented approach tend to be disruptive, as it involves interdependencies. ◆ With the mindset of achieving results, it doesn't mean that some steps from the procedure of achieving the result be abandoned. ◆ There are various pitfalls to be avoided while being result-oriented. *Run the presentation and summarise the session.
Do	Ask the participants to fill out the questionnaire as per their understanding.

Day - 13	Component-Based Programming	1 hour 30 minutes
13.3		

Resources to be used	◆ Participant Handbook ◆ Presentation
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Do	Run the PowerPoint presentation.
Notes for Facilitation	◆ Give an introduction to Component-based programming. ◆ Discuss what Components are used for. ◆ Discuss .NET Components. ◆ Explain Component Jargon and give an example of N-tier design. ◆ Explain how to create a Component. ◆ Explain Properties and State. ◆ Explain Data Access Components. ◆ Explain Object Data Source. ◆ Explain working with ASP .NET. ◆ Discuss Command Line Development with an example. ◆ Discuss GUI Apps; explain how to run it and provide an example.
Say	Summarise the key topics discussed in the session. Take queries and clarify the doubts of the participants.

Day - 13	Component-Based Programming	2 hours
13.4		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 13	Recap	20 minutes
13.5		

Resources to be used	Presentation
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Do	Revise the key topics from what has been discussed so far. Answer their queries and clarify the doubts of the participants.
Notes for Facilitation	Questions for recap: 1. What is meant by 3-tier Architecture? 2. What are .NET Components? 3. What is Data Access Component? 4. What is the second level of Component Jargon? 5. Why do we use Components in ASP .NET? 6. Are Classes Components or not? 7. Are DLLs Components or not? *Run the presentation if required.

Day - 13	Component-Based Programming	2 hours
13.6		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 14	Morning Energiser	10 minutes
14.1		

Resources to be used	Morning Energisers booklet
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Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 14	Programming with ASP .NET	4 hours
14.2		

Resources to be used	◆ Participant Handbook ◆ Presentation
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Do	Run the PowerPoint presentation.
Notes for Facilitation	◆ Give an introduction to ASP. ◆ Give an overview of ASP .NET with an example. ◆ Explain the architecture of ASP. NET. ◆ Explain the programming model, i.e., Controls and Events. ◆ Explain ASP .NET Object model. ◆ Explain ServerSite Control and Automatic Compilation. ◆ Explain the programming basics. ◆ Explain Server Control Syntax and State Maintaining. ◆ Explain Page Events, Page Event Life Cycle and Server Control Events. ◆ Explain Server Control events, Event Arguments and Server Controls. ◆ Explain HTML Controls. ◆ Explain Web Controls and List Controls, Checkbox List and Radio Button List. ◆ Explain Validation Controls. ◆ Summarise the key topics discussed in the session.

Day - 14	Recap	25 minutes
14.3		

Resources to be used	Presentation
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Do	Revise the key topics from what has been discussed so far. Answer their queries and clarify the doubts of the participants.
Notes for Facilitation	Questions for recap: 1. What is ASP .NET? 2. What is .NET Framework? 3. What is Response Redirect? 4. What is Data Adapter and Data Set? 5. What is Page Events? 6. What is Page Event Cycle? 7. What is the difference between ADO and ADO .NET? 8. What is the difference between Web Server Control and HTML Control? *Run the presentation, if required.

Day - 14	Programming with ASP .NET	1 hour
14.4		

Resources to be used	Practical Guide
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Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 14	Programming with ASP .NET	2 hours 25 minutes
14.5		

Resources to be used	Practical Guide
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Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 15	Morning Energiser	10 minutes
15.1		

Resources to be used	Morning Energisers booklet
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Notes for Facilitation	Please refer to the "Morning Energisers" booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 15	Recap	20 minutes
15.2		

Resources to be used	Presentation
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Do	Give a quick recap of the topics covered in the previous day, in a question- answer format.
Notes for Facilitation	Questions for recap: 1. What does HTML stand for? 2. What does URL stand for? 3. What does CGI stand for? 4. Which is not Server Side Program? 5. State whether the following statements are True or False: ◆ ASP.NET is designed as a Server-Side Technology. ◆ The Common Language Runtime (CLR) is not an Execution Environment. ◆ The .NET Framework class library (FCL) provides the core functionality of .NET Framework architecture. ◆ HTML forms do not allow Web application developers to design standard input pages. ◆ A basic HTML page is a little like a word-processing document. ◆ HTML consists of tags and data. ◆ An HTML file is a text file containing small markup tags. *Run the presentation if required.

Day - 15	ASP .NET Web Application	2 hours
15.3		

Resources to be used	◆ Participant Handbook ◆ Presentation
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Do	Run the PowerPoint presentation.
Notes for Facilitation	◆ Give an introduction to Web Applications. ◆ Discuss ASP. NET Web Applications. ◆ Discuss where Visual Studio fits in. ◆ Discuss the elements of ASP .NET Web Applications. ◆ Explain Project Management. ◆ Explain NET Web Application Projects and Deployment. ◆ Discuss the User Interface in ASP .NET and ASP .NET Page Framework. ◆ Explain Web-based Components. ◆ Explain Web Application Data Access. ◆ Explain Web Application Data Access. Explain security, performance, tracing and optimisation of ASP .NET Web Application. ◆ Explain the Windows Forms and Web Forms Recommendations. ◆ Detail the features of Windows Forms.
Do	Summarise the key topics; take questions and clarify the doubts of the participants.

Day - 15	ASP .NET Web Application	2 hours
15.4		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 15	Games and Activities	30 minutes
15.5		

Resources to be used	Activities and Games booklet
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Notes for Facilitation	Refer to the "Activities and Games" booklet and pick out a game and an activity.
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Day - 15	ASP .NET Web Application	3 hours
15.6		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 16	Morning Energiser	10 minutes
16.1		

Resources to be used	Morning Energisers booklet
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Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 16	ASP .NET IDE and Web Forms	2 hours 30 minutes
16.2		

Resources to be used	◆ Participant Handbook ◆ Presentation
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Do	Show the PowerPoint presentation of this session to the trainees.
Notes for Facilitation	◆ Give an introduction to ASP .NET IDE. ◆ Explain Visual C# Tools. ◆ Discuss how the IDE exposes the Tools. ◆ Explain Editor and Windows Form Designer Windows. ◆ Explain Solution Explorer and Project Designer of IDE Tools in ASP .NET. ◆ Explain Compiler, Debugger and Error List Windows in IDE. ◆ Discuss Customizing the IDE. ◆ Discuss Web Forms. ◆ Discuss the purpose of Web Forms. ◆ Discuss the HTML Controls of Web Forms. ◆ Explain Web Form Controls and give a detailed description of different types of Controls.

Day - 16	Recap	20 minutes
16.3		

Resources to be used	Presentation
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Notes for Facilitation	Give a quick recap of the topics covered in the previous day, in a question-answer format. Encourage the participants to get their queries answered and doubts clarified. *Run the presentation is required.
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Day - 16	ASP .NET and IDE Web Forms	3 hours
16.4		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 16	ASP .NET IDE and Web Forms	2 hours
16.5		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Continue with the practice session. Ensure that the participants are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 17	Morning Energiser	10 minutes
17.1		

Resources to be used	Morning Energisers booklet
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Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 17	Making Web Forms	2 hours
17.2		

Resources to be used	◆ Participant Handbook ◆ Presentation
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Do	Run the PowerPoint presentation.
Notes for Facilitation	◆ Give an introduction to the traditional form-based Web Apps. ◆ Explain making a WebForm. ◆ Explain Web Forms, Abstraction with an example. ◆ Discuss Project Layout and design WebForm. ◆ Discuss Web Controls versus HTML Controls, and implementing events. ◆ Explain ASP .NET programming model and Web-based dialogs. ◆ Explain Auto Postbacks and Postbacks. ◆ Explain Statelessness of Web Apps. ◆ Summarise WebForms.

Day - 17	Making Web Forms	3 hours
17.3		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 17	Recap	20 minutes
17.4		

Resources to be used	Presentation
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Notes for Facilitation	Give a quick recap in a question-answer format, of the topics covered through the day. Encourage the participants to get their queries answered and doubts cleared. *Run the presentation if required.
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Day - 17	Making Web Forms	2 hours 30 minutes
17.5		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Continue with the practice session. Ensure that the participants are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 18	Morning Energiser	10 minutes
18.1		

Resources to be used	Morning Energisers booklet
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Notes for Facilitation	Please refer to the "Morning Energisers" booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 18	Session Objects and Events	2 hours
18.2		

Resources to be used	◆ Participant Handbook ◆ Presentation
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Do	Show the PowerPoint presentation of this session to all the trainees.
Notes for Facilitation	◆ Give an introduction to Session Object in ASP .NET. ◆ Discuss when the Session starts and when it ends. ◆ Explain how to store, retrieve and remove Session Variables. ◆ Explain Loop through the Contents Collections. ◆ Explain Session State. ◆ Explain processing of Session State, State Server Session State, and SQL Server session state. ◆ Discuss the Session Events. ◆ Explain the Session Time-outs. ◆ Explain Cookies, Http Cookie properties, creating and reading a Cookie. ◆ Discuss Application Object in Session.

Day - 18	Session Objects and Events	3 hours
18.3		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 18	Session Objects and Events	2 hours
18.4		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Continue with the practice session. Ensure that the participants are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 18	Recap	20 minutes
18.5		

Resources to be used	Presentation
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Notes for Facilitation	Give a quick recap in a question- answer format of the topics covered through the day. Encourage the participants to get their queries answered and doubts cleared. *Run the presentation if required.
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Day - 18	Games and Activities	30 minutes
18.6		

Resources to be used	Activities and Games booklet
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Notes for Facilitation	Refer to the “Activities and Games” booklet to choose and conduct a game and an activity.
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Day - 19	Morning Energiser	10 minutes
19.1		

Resources to be used	Morning Energisers booklet
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Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 19	Team Work	2 hours
19.2		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook ◆ Presentation ◆ Activities and Games booklet
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Do	Show the multimedia unit.
Notes for Facilitation	Discuss what “team work” is all about.
Say	Team work is important to accomplish tasks at a faster pace. The four stages of team development are: Forming, Storming, Norming and Performing.
Notes for Facilitation	◆ Explain the advantages of working in a team. ◆ Explain how to become an effective team member. ◆ Ask the participants how they can contribute by being in a team. Take up a team activity for the class from the Activities and Games booklet.
Do	Ask the participants to fill out the questionnaire as per their understanding.

Day - 19	Session Objects and Cookies	2 hours 30 minutes
19.3		

Resources to be used	◆ Participant Handbook ◆ Presentation
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Do	Run the PowerPoint presentation.
Notes for Facilitation	◆ Give an introduction to the concept of Sessions. ◆ Discuss the use of Sessions. ◆ Explain the advantages and disadvantages of Sessions. ◆ Explain the Session ID diagram. ◆ Explain Solution Explorer and Project Designer of IDE Tools in ASP .NET. ◆ Explain Session State with an example. ◆ Discuss Session Tracking. ◆ Explain Session Identifier. ◆ Explain Session Variables. ◆ Explain Session Cookies. ◆ Discuss Http Cookie Properties. ◆ Explain the Cookie Setting and Cookie Reading examples.
Notes for Facilitation	Revise the key topics discussed in the session.

Day - 19	Recap	20 minutes
19.4		

Resources to be used	◆ Presentation
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Do	Give a quick recap of the topics in a question-answer format. Encourage the participants to get their doubts clarified.
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Notes for Facilitation	Questions for recap: 1. What is a Session? 2. What is the State of a Page? 3. What setting can you use to see information about the different stages of the Page Life Cycle? 4. In which life cycle phase does the page determine if it was called as the result of a Postback? 5. What Event is usually used to take actions during the Load phase? 6. What are the four kinds of state that ASP .NET manages? Which one can you not affect in any way? 7. What does the EnableViewState= "false" setting do? 8. Where would you store the value of a Counter that is incremented each time the page is loaded? 9. Suppose you wanted to ask the user to enter her name on a page, and you wanted to retain that value the entire time the user is at your site: what's the best mechanism to use? 10. What's the proper syntax for storing and retrieving that username? *Run the presentation if required.
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Day - 19	Session Objects and Cookies	3 hours
19.5		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 20	Morning Energiser	10 minutes
20.1		

Resources to be used	Morning Energisers booklet
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Notes for Facilitation	Please refer to the "Morning Energisers" booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 20	Test 2	2 hours
20.2		

Resources to be used	Assessment Guide
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Notes for Facilitation	Formative Assessment for the topic C++ and Data Structures. Refer to the questions in the Assessment Guide and conduct the test.
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Day - 20	Programming with C#	2 hours 20 minutes
20.3		

Resources to be used	◆ Participant Handbook ◆ Presentation
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Do	Run the PowerPoint presentation.
Notes for Facilitation	◆ Give an introduction to C# language. ◆ Explain Console Applications. ◆ Explain how to create C# program. ◆ Explain Syntax and Data Types.

	◆ Discuss Comments and Case Sensitivity. ◆ Explain Namespace, Body of a Method, main (), Program Statements, and C# elements. ◆ Explain how to create a console application, and create a new project. ◆ Explain how to write Program Statements, rename Source Code, compile and run application. ◆ Explain how to build Visual Studio Project, run an application, debug an application and list error.
Do	Revise the key topics covered in this session.

Day - 20	Programming with C#	1 hour 30 minutes
20.4		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 20	Programming with C#	2 hours
20.5		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Continue with the practice session. Ensure that the participants are able to apply their learning.
Say	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 21	Morning Energiser	10 minutes
21.1		

Resources to be used	Morning Energisers booklet
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Notes for Facilitation	Please refer to the "Morning Energisers" booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 21	Recap	20 minutes
21.2		

Resources to be used	Presentation
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Do	Give a quick recap of the topics in a question-answer session format. Encourage the participants to get their doubts clarified.
Notes for Facilitation	Questions for recap: 1. How many kinds of Data Types does Microsoft .NET support? 2. What is the difference between Value Type and Reference Type? 3. What are the benefits of Value Type? 4. What are Primitives? 5. What are Enumerators? 6. What is Array and Class in Data Types of C#? 7. What are Delegates? 8. What are Methods and Method Arguments? *Run the presentation if required.

Day - 21	Data Types and Classes in C#	2 hours 30 minutes
21.3		

Resources to be used	◆ Participant Handbook ◆ Presentation
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Do	Show the PowerPoint presentation of this session to all the trainees.
Notes for Facilitation	◆ Give an introduction to Data Types in C#. ◆ Discuss the different types of Data Types used in C#. Explain Value Types and Primitives. ◆ Explain Enumerators and Structs. ◆ Explain Reference Types Editor and Windows Form Designer Windows. ◆ Explain Reference Types that include the different Data Types. Explain the terms String, Array, Class, Interface, and Delegate. ◆ Explain Classes with an example. ◆ Discuss the common Class Members, Static Members and Fields. ◆ Explain Methods and Method Arguments.
Do	Revise the key topics discussed in the session.

Day - 21	Data Types and Classes in C#	2 hours 30 minutes
21.4		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 21	Data Types and Classes in C#	2 hours 30 minutes
21.5		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Continue with the practice session. Ensure that they are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 22	Morning Energiser	10 minutes
22.1		

Resources to be used	Morning Energisers booklet
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Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 22	Control Statements and Loops in C#	2 hours 30 minutes
22.2		

Resources to be used	◆ Participant Handbook ◆ Presentation
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Do	Run the PowerPoint presentation.
Notes for Facilitation	◆ Give an introduction to Control Statements in C# .NET. ◆ Explain Unconditional Statement; and discuss If statement with an example. ◆ Explain Switch Statement with an example. ◆ Explain Loops and highlight While loop. ◆ Explain Do loop, While loop, and the For loop with examples. ◆ Explain For Each loop and Branching Statement.
Do	Revise the key topics discussed in this session.

Day - 22	Control Statements and Loops in C#	2 hours 30 minutes
22.3		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 22	Recap	20 minutes
22.4		

Resources to be used	◆ Presentation
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Do	Give a quick recap of the topics in a question-answer format. Encourage the participants to get their doubts clarified.
Notes for Facilitation	Questions for recap: 1. What are the two types of Control Statements? 2. What is the difference between Conditional and Unconditional Statements? 3. State Switch Statement with an example. 4. What are Loops in Control Statements? 5. What is the difference between Do loop and While loop? 6. What is Branching Statement? *Run the presentation if required.

Day - 22	Control Statements and Loops in C#	2 hours 30 minutes
22.5		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 23	Morning Energiser	10 minutes
23.1		

Resources to be used	Morning Energisers booklet
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Notes for Facilitation	Please refer to the "Morning Energisers" booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 23	Using Control Statements and Loops in C#	2 hours 30 minutes
23.2		

Resources to be used	^a Participant Handbook ^a Presentation
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Do	Run the PowerPoint presentation.
Notes for Facilitation	^a Explain Statements. ^a Explain the Types of Statements in C#. ^a Explain the Statement Lists and Blocks. ^a Explain Labeled Statements and Goto statements. ^a Explain Expression and Control statements. ^a Explain If Statement. ^a Explain If-Else statements. ^a Explain If-Else If-Else statement. ^a Explain the use of Switch statement with the help of an example. ^a Use an example to highlight the use of Break Statement. ^a Explain Loop concepts using examples. ^a Explain the concept of For Each statement. ^a Explain the concept of Goto statement in detail an example. ^a Explain the use of Return statement.
Do	Revise the key topics discussed in the session.

Day - 23	Using Control Statements and Loops in C#	2 hours 30 minutes
23.3		

Resources to be used	^a Participant Handbook ^a Practical Guide
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Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 23	Recap	20 minutes
23.4		

Resources to be used	^a Presentation
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Do	Give a quick recap of the topics in a question-answer session format. Encourage the participants to get their doubts clarified.
Notes for Facilitation	Questions for recap: 1. What are Statements? 2. What are the types of Statements used in C#? 3. What are Labelled and Go-To Statements? 4. What are If and If-Else Statements? 5. What are Loops? 6. What is the difference between While Loop and Do While Loop? 7. What is Switch Statement? *Run the presentation if required.

Day - 23	Using Control Statements and Loops in C#	2 hours 30 minutes
23.5		

Resources to be used	^a Participant Handbook ^a Practical Guide
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Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 24	Morning Energiser	10 minutes
24.1		

Resources to be used	Morning Energisers booklet
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Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 24	Communication	2 hours 30 minutes
24.2		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook ◆ Presentation
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Do	Run the multimedia module.
Say	When a person receives too many messages, or the message is complex, or the presentation of information is such that it is ambiguous, or perhaps there are linguistic barriers, it leads to ineffective communication.
Do	Ask the participants how they can overcome the “barriers to effective communication”.
Expected Responses	1. By paying attention to the speaker 2. By minimising our internal and external distractions 3. By focussing on what the speaker is saying
Say	Whenever there is some interference with the decoding of messages, it is called “communication noise”.
Do	Show the presentation and conduct the Activity 1, 2 and 3.

Notes for Facilitation	◆ Communication adds meaning to human life. It helps to build relationship and foster love and understanding. ◆ Communication skills are essential in all spheres of life; be it an interview, dealing with the project leader, working out a solution with a team, or writing a report– getting across the point effectively is all that matters. ◆ Communication, in plain terms, is the basis of all relationships. ◆ When it comes to business, communication becomes even more important. This is the pillar on which professional relationships are built. *Run the presentation to summarise the session.
Do	Ask the participants to fill out the questionnaire, as per their understanding.

Day - 24	Programming with C#	2 hours
24.3		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Ask the participants to open their Practical Guides.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 24	Recap	20 minutes
24.4		

Resources to be used	◆ Presentation
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Notes for Facilitation	Give a quick recap of the topics in a question-answer session format. Encourage the participants to get their doubts clarified. *Run the presentation if required.
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Day - 24	Programming with C#	3 hours
24.5		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 25	Morning Energiser	10 minutes
25.1		

Resources to be used	Morning Energisers booklet
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Notes for Facilitation	Please refer to the "Morning Energisers" booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 25	Programming with C#	2 hours 30 minutes
25.2		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 25	Programming with C#	2 hours 30 minutes
25.3		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Continue with the practice session. Ensure that the participants are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 25	Recap	20 minutes
25.4		

Resources to be used	◆ Presentation
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Notes for Facilitation	Give a quick recap of the topics in a question-answer session format. Encourage the participants to get their doubts clarified. *Run the presentation if required.
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Day - 25	Programming with C#	2 hours 30 minutes
25.5		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Continue with the practice session. Ensure that the participants are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 26	Morning Energiser	10 minutes
26.1		

Resources to be used	Morning Energisers booklet
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Notes for Facilitation	Please refer to the "Morning Energisers" booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 26	Database Connectivity	2 hours 30 minutes
26.2		

Resources to be used	◆ Participant Handbook ◆ Presentation
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Do	Run the PowerPoint presentation. Ask the participants to note down the key concepts and their explanation.
Notes for Facilitation	◆ Explain the term ADO. ◆ Give an introduction to ADO .NET. ◆ Explain Connections and how to declare and open a Connection. ◆ Explain the Recordset Object. ◆ Explain the concept of Command Object. ◆ Explain Input Output system of a computer. ◆ Explain Connection Objects. ◆ Explain how we can connect to a Database. ◆ Explain the Insert, Update and Delete queries. ◆ Explain the Concept Select Statement. ◆ Explain how we can retrieve the Recordset. ◆ Explain how to close a Database Connection.
Do	Revise the key concepts from this session.

Day - 26	Database Connectivity	2 hours 30 minutes
26.3		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 26	Recap	20 minutes
26.4		

Resources to be used	◆ Presentation
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Notes for Facilitation	Give a quick recap of the topics in a question-answer session format. Encourage the participants to get their doubts clarified. *Run the presentation if required.
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Day - 26	Database Connectivity	2 hours 30 minutes
26.5		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Continue with the practice session. Ensure that the participants are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 27	Morning Energiser	10 minutes
27.1		

Resources to be used	Morning Energisers booklet
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Notes for Facilitation	Please refer to the "Morning Energisers" booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 27	Problem Solving	2 hours 30 minutes
27.2		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook ◆ Activities and Games
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Do	Play the multimedia unit.
Do	Ask what the participants understand by the term "problem".
Say	A problem is a situation that needs attention. A problem can be defined by asking questions like: ◆ What is happening? ◆ Where is it happening? ◆ When is it happening? ◆ To what extent is it happening?
Do	Take up activities from the Activities and Games booklet and conduct an activity or game in class.

Notes for Facilitation	Use a simple example to illustrate the definition of “problem”. Following are the steps involved in the problem-solving process: 1. Always focus on the “solution” to the problem and not the problem. 2. Have a positive and open mind to think of all possible alternatives and solutions to the problem. 3. View the problem neutrally and think laterally, changing the direction of thinking. 4. Discuss and brainstorm – helps to generate more ideas and alternative solutions. 5. Do a “What-if” analysis to generate more alternative solutions and keep your mind open for creativity. 6. Lastly, try to always simplify things. Break-up complicated problems and problem-solutions to small activities to achieve the goal.
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Day - 27	XML	2 hours
27.3		

Resources to be used	◆ Participant Handbook ◆ Presentation
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Do	Run the PowerPoint presentation.
Notes for Facilitation	◆ Give an introduction to XML. ◆ Explain how to open a project practically. ◆ Explain how to "Create an XML File with a Text Editor. ◆ Give an introduction to an XML document. ◆ Explain how to create an XML document using XML code.

Day - 27	XML	1 hour 30 minutes
27.4		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 27	Recap	20 minutes
27.5		

Resources to be used	◆ Presentation
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Do	Give a quick recap of the topics in a question-answer format. Encourage the participants to get their doubts clarified.
Notes for Facilitation	Questions for recap: 1. What is XML? 2. How to create an XML file in a text editor? 3. What is XML document? 4. How to create an XML code using Xml document? 5. How to create an XML document? *Run the presentation if required.

Day - 27	XML	1 hour 30 minutes
27.6		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 28	Morning Energiser	10 minutes
28.1		

Resources to be used	Morning Energisers booklet
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Notes for Facilitation	Please refer to the "Morning Energisers" booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 28	Stress Management	2 hours 30 minutes
28.2		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook ◆ Activities and Games booklet ◆ Presentation
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Do	Play the multimedia module.
Ask	What does the term "stress" mean to you?
Expected Responses	1. Tension 2. Workload 3. Anything which affects us to behave in an abnormal manner.
Say	That's right! There are three stages of stress: 1. Alarm stage 2. Resistance stage 3. Exhaustion stage The ABC Strategy of Stress Control: Being Aware of the causes of stress, being Balanced while taking pressure and Controlling the negative effects of stress.

Do	Run the presentation and conduct the activities mentioned. Alternatively, you could pick out a game from the Activities and Games booklet and conduct them in class.
Notes for Facilitation	◆ Discuss how relaxation techniques can bring them back into balance when stressed. ◆ Request the participants to volunteer to come up and share some funny stories, jokes or incidents to lighten up the mood.
Do	◆ Ask the participants to fill out the questionnaire as per their understanding.

Day - 28	XML–Electronic Data Exchange	2 hours
28.3		

Resources to be used	◆ Participant Handbook ◆ Practical Guide ◆ Presentation
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Do	Run the PowerPoint presentation.
Notes for Facilitation	◆ Explain the concept of Electronic Data Interchange (EDI). ◆ Explain the types of EDI. ◆ Explain the advantages and disadvantages of EDI. ◆ Explain the role of XML. ◆ Explain the various parts of EDI in detail; also explain the software used to interpret the contents of an EDI. ◆ Explain the various EDI standards. ◆ Explain how XML and its technologies can do the work of EDI, with the help of an example.

Day - 28	Recap	20 minutes
28.4		

Resources to be used	◆ Presentation
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Do	Give a quick recap of the topics in a question-answer format. Encourage the participants to get their doubts clarified.
Notes for Facilitation	Questions for recap: 1. What is EDI? 2. What are the types of EDI? 3. What are EDI standards? 4. What are EDI parts? 5. What are the advantages and disadvantages of EDI? 6. What is XML? Give an example. 7. What is the role of XML? *Run the presentation if required.

Day - 28	XML–Electronic Data Exchange	1 hour 30 minutes
28.5		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 28	XML–Electronic Data Exchange	1 hour 30 minutes
28.6		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Continue with the practice session. Ensure that the participants are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 29	Morning Energiser	10 minutes
29.1		

Resources to be used	Morning Energisers booklet
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Notes for Facilitation	Please refer to the "Morning Energisers" booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 29	Recap	20 minutes
29.2		

Resources to be used	◆ Presentation
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Notes for Facilitation	Give a quick recap of the previous day's topics in a question-answer session format. Encourage the participants to get their doubts clarified. *Run the presentation if required.
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Day - 29	XML – DTD and Schema	2 hours 30 minutes
29.3		

Resources to be used	◆ Participant Handbook ◆ Presentation
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Do	Run the PowerPoint presentation.
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Say	◆ Explain the concept of DTD. ◆ Explain the importance of DTD. ◆ Explain the relation between XML and DTD. ◆ Explain the use of DTD. ◆ Explain how to declare elements in DTD. ◆ Explain how to declare attributes in DTD. ◆ Explain Schemas. ◆ Explain the difference between the schemas of DTD and XML. ◆ Explain how DTD is associated with XML document.
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Day - 29	XML – DTD and Schema	2 hours 30 minutes
29.4		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 29	XML–DTD and Schema	2 hours 30 minutes
29.5		

Resources to be used	◆ Participant Handbook ◆ Presentation
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Do	Run the PowerPoint presentation.
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Notes for Facilitation	◆ Explain XML. ◆ Explain the syntax of an XML document. ◆ Explain Schemas. ◆ Explain the Schemas of an XML document. ◆ Explain XML Schema Version of Bank DTD.
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Day - 30	Morning Energiser	10 minutes
30.1		

Resources to be used	Morning Energisers booklet
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Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 30	XML – Web Feeds	2 hours 20 minutes
30.2		

Resources to be used	◆ Participant Handbook ◆ Presentation
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Do	Run the PowerPoint presentation.
Notes for Facilitation	◆ Explain XML. ◆ Explain the syntax of an XML document. ◆ Explain Schemas. ◆ Explain the Schemas of an XML document. ◆ Explain the XML Schema Version of Bank DTD. ◆ Explain all the features of XML Schema.

Day - 30	XML–Web Feeds	3 hours
30.3		

Resources to be used	◆ Practical Guide
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Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 30	XML – Web Feeds	2 hours 30 minutes
30.4		

Resources to be used	◆ Practical Guide
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Do	Ask the participants to open there Practical Guides. Ensure that the participants are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 31	Morning Energiser	10 minutes
31.1		

Resources to be used	Morning Energisers booklet
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Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 31	Way to Success	1 hour
31.2		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook
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Do	Run the multimedia module.
Ask:	What does “success” mean to you?
Expected Responses	“Achieving my goals ” “Making lots of money” “Making my parents happy”
Notes for Facilitation	◆ Have an interactive session around the topic of “success”. ◆ Discuss the “Keys to success” ◆ Share some real-life anecdotes, inspiring success stories about famous people. ◆ Encourage the students to share some success stories.
Do	Ask the participants to fill out the questionnaire as per their understanding.

Day - 31	DOT NET Features–Summary	1 hour 50 minutes
31.3		

Resources to be used	◆ Presentation ◆ Participant Handbook
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Do	Run the PowerPoint presentation.
Notes for Facilitation	◆ Summarise the features of .NET

Day - 31	Practice Exercises	5 hours
31.4		

Resources to be used	Practical Guide
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Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 32	Morning Energiser	10 minutes
32.1		

Resources to be used	Morning Energisers booklet
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Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 32	Test 3	2 hours
32.2		

Resources to be used	Assessment Guide
Notes for Facilitation	Conduct a test for the topic “HTML” programming, refer to the questions in the Assessment Guide.

Day - 32	Presentation Skills	1 hour
32.3		

Resources to be used	◆ Presentation ◆ Participant Handbook
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Do	Run the PowerPoint presentation.
Say	Presentation is an act of showing and explaining the contents of a topic to an audience.
Ask	Why do you think “presentation skills” are important?
Expected Responses	1. For recognition 2. To explain something

Say	Presentation skills are the techniques of showing the contents of a topic to an audience.
Notes for Facilitation	◆ The first step to making an effective presentation is "planning". ◆ Before starting a presentation, welcome the audience, introduce yourself, get their attention and explain to them the purpose of the presentation.
Ask	What are the points that one should keep in mind while delivering a presentation?
Notes for Facilitation	◆ While delivering a presentation, one should know their material; it should be rehearsed but not memorised. ◆ About taking questions, make it clear before starting the presentation when would you like to have them. ◆ Having sound knowledge about the content, practising, giving a solid introduction, smiling, making the session enjoyable, arriving early and practising deep breathing before the session help immensely in making an effective presentation.
Do	◆ Ask the participants to fill out the questionnaire as per their understanding.

Day - 32	Group Discussion	1 hour
32.4		

Resources to be used	◆ Presentation ◆ Participant Handbook
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Do	Start the PowerPoint presentation.
Say	Group Discussion is a process where ideas and opinions about a topic are debated upon.
Ask	What do you know about "Group Discussion"?
Expected Responses	1. "A group is formed and given a topic to discuss about" 2. "For selection in a job, etc"
Say	Tell them about the different types of group discussions. A group discussion can either be topic-based or case-based.

Notes for Facilitation	The skills required in any group discussion are as follows: ◆ How well do you communicate with others; ◆ How you behave and interact within a group; ◆ How open minded are you; ◆ Your listening skills; ◆ How you put forward your views; ◆ Your leadership and decision making skills; ◆ Your analytical skills and subject knowledge; ◆ Problem solving and critical thinking skills; ◆ Your attitude and confidence.
Do	Divide the class into groups of 8-10 members.
Say	A topic will be given on the spot for you to discuss in a group.
Notes for Facilitation	Topics for group discussion: ◆ Public sector being a guarantor of job security is a myth. ◆ Are Indians less quality conscious? ◆ Government pumping money into the economy is not the solution for our economic problems. ◆ Should agricultural subsidies be stopped? ◆ Are MNCs superior to Indian companies? ◆ Advertising is a waste of resources. ◆ Privatisation will lead to less corruption. ◆ Skilled manpower shortage in India ◆ Technology creates income disparities. ◆ Government needs to introspect before pointing a finger at the private sector for corruption. ◆ What shall we do about our ever-increasing population?
Do	Start calling the groups one by one and conduct the group discussion.

Day - 32	Practice Exercises	3 hours 50 minutes
32.5		

Resources to be used	◆ Presentation ◆ Participant Handbook
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Do	Run PowerPoint presentation.
Notes for Facilitation	◆ Explain the concept of .NET with the help of a diagram. ◆ List some languages used by .NET framework.
Ask	Please note down the main elements of the topic.
Notes for Facilitation	◆ Give an overview of .NET and list its three main elements. ◆ Explain the advantages of .NET. ◆ Explain the compilation stages in .NET. ◆ Explain the .NET framework. ◆ Explain the common language runtime. ◆ Explain the class library of .NET framework. ◆ Explain ASP .NET. ◆ Explain CLS. ◆ Summarise .NET.

Day 9 to Day 32

Elective 2: Java2 Enterprise Edition (J2EE)

Day - 9	Morning Energiser	10 minutes
9.1		

Resources to be used	Morning Energisers booklet
Notes for Facilitation	Please refer to the "Morning Energisers" booklet. You may pick the energiser that is recommended for the day of the week.

Day - 9	Test 1	10 minutes
9.2		

Resources to be used	Assessment Guide
Notes for Facilitation	Conduct a test for the topic "Java programming", refer to the questions in the Assessment Guide.

Day - 9	Introduction to Core Java Programming	1 hour
9.3		

Resources to be used	◆ Presentation ◆ Participant Handbook
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Do	Run the PowerPoint presentation to explain "Core Java Programming".
Say	The programs coded in Java are more secure as Java has a high security feature. Explain JVM in detail. Explain the compilation process in Java and the syntax. Explain class, object and method in OOP. Explain the features of OOP: ◆ encapsulation ◆ inheritance ◆ polymorphism ◆ abstraction.

Day - 9	Fundamentals of Java Programming	1 hour
9.4		

Resources to be used	◆ Presentation ◆ Participant Handbook
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Do	Run the PowerPoint presentation on the "Fundamentals of Java Programming".
Notes for Facilitation	◆ Explain the different Data Types in Java. ◆ Explain the syntax of declaring variables in a Java. ◆ Ask them what is Looping and explain the different types of Looping with their syntax in Java. ◆ Explain Switch and If Conditional Statements. ◆ Explain Constructors and Destructors in Java. ◆ Illustrate Exception Handling with the help of a diagram. ◆ Explain Throwable Class with the help of an example program. ◆ Mention the common default Exceptions available in Java.

Day - 9	Objects and Classes	2 hours
9.5		

Resources to be used	◆ Presentation ◆ Participant Handbook
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Do	Run the PowerPoint presentation.
Notes for Facilitation	◆ Ask the class to recall all that they have learnt about Objects and Classes. ◆ Explain the syntax to declare Classes and Objects in Java. ◆ With the help of examples explain Class Variables, and defining and declaring Methods.

Day - 9	Introduction to Core Java Programming	1 hour 50 minutes
9.6		

Resources to be used	◆ Practical Guide
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Do	Ask the participants to try out the examples given in the Practical Guide.
Notes for Facilitation	Tell the participants that Java program can be coded using any text editor, such as Notepad. Explain the program to the participants: The signature of main() method: public static void main(String args[]) The method signature for the main() method contains three modifiers: ◆ public indicates that the main() method can be called by any Object, i.e. it is accessible everywhere and by everyone. ◆ static indicates that the main() method is a Class Method. Static signifies that the Java environment should be able to call this method without creating an instance of the Class, so this method must be declared as static. ◆ void indicates that the main() method has no return value. ◆ string args[] This method must be given an array of Strings, and the array will be called 'args'. Demonstrate how to get the output using System.out.println. ◆ out is an object of PrintStream class defined in System class. out is declared as public, static and final. ◆ println() is a method of PrintStream class. ◆ The println() method is called with out object. ◆ The out object is called with System class. println() is a method of PrintStream class. out is an object of PrintStream class defined in System class. Here, we are calling println() with out object and out object with System class. Internally the out object is connected to the "standard output stream" of underlying operating system. So, any data given to out object goes to the Operating System out stream and prints at the DOS prompt.

Day - 10	Morning Energiser	10 minutes
10.1		

Resources to be used	Morning Energisers booklet
Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.

Day - 10	Grooming	2 hours 30 minutes
10.2		

Resources to be used	◆ Multimedia ◆ Participant Handbook ◆ Presentation
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Do	Play the multimedia on Grooming.
Notes for Facilitation	Discuss the topic of “personal grooming”. Make it into an interactive session; invite the participants to talk about their views on the importance of personal grooming in the professional front.
Notes for Facilitation	Sensitise the participants about the following: ◆ Aspects of personal grooming that reflect on the overall personality of a person ◆ Maintaining hygiene and healthy habits ◆ Choice of clothing at workplace ◆ Hair and make-up ◆ Making an impression: effect of clothing and grooming on other people ◆ The grooming checklist ◆ Preparation before an interview, event, etc.
Do	Ask the participants to fill out the questionnaire as per their understanding.

Day - 10	Arrays and Strings	2 hours
10.3		

Resources to be used	◆ Participant Handbook ◆ Presentation
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Do	Run the PowerPoint presentation on Arrays and Strings.
Notes for Facilitation	Ask the students to recall the definition of Arrays they learnt in previous modules. Use an example to explain: ◆ how to declare and create Arrays in Java ◆ how to create Array references ◆ how to initialise and access Arrays ◆ how to copy data from one Array to another, in the program. Introduce the concept of Strings in Java. ◆ Elucidate the two methods to create and implement Strings, with the help of an example. ◆ Teach them how to use the length() method to calculate string length. ◆ Teach them to concatenate two strings using concat() method in the same example.

Day - 10	Arrays and Strings	1 hour 30 minutes
10.4		

Resources to be used	◆ Practical Guide
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Do	Ask the participants to look at the first question.
Notes for Facilitation	◆ Demonstrate the use of various functions in the String Class using programs. ◆ Demonstrate sorting an Array of numbers in ascending order.

Day - 10	Recap	20 minutes
10.5		

Resources to be used	Participant Handbook Presentation
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Notes for Facilitation	◆ Recapitulate the day's topics in a question-answer format. ◆ Make it interactive and encourage the whole class to participate. ◆ Take questions and clarify doubts related to the previous session.
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Day - 10	Arrays and Strings	1 hour 30 minutes
10.6		

Resources to be used	◆ Practical Guide
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Do	Ask the participants to open their Practical Guides ensure that they are apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 11	Morning Starter	10 minutes
11.1		

Resources to be used	Morning Energizer Booklet
Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.

Day - 11	Discipline	2 hours 30 minutes
11.2		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook
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Do	Run the multimedia. Pause whenever there is an explanation/discussion required.
Notes for Facilitation	Have an interactive discussion on: ◆ the importance of discipline to be able to complete given tasks; ◆ how one can be self-disciplined.
Notes for Facilitation	Discuss the following: ◆ Willpower is the training and control of oneself and one's conduct, usually for personal improvement. ◆ The Five Pillars of Self-Discipline: acceptance, willpower, hardwork, industriousness and persistence. ◆ Being industrious doesn't necessarily mean doing the work that's challenging or difficult. It simply means putting in the time. Run the presentation and summarise the session.
Do	Ask the participants to fill out the questionnaire as per their understanding.

Day - 11	Inheritance in Java Programming	1 hour 30 minutes
11.3		

Resources to be used	◆ Presentation
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Do	Start the Power Point presentation.
Say	The slide is summarising the contents of this session.
Notes for Facilitation	◆ Explain Inheritance and its need in Java. ◆ Explain the use of Inheritance, i.e. what can we inherit in a Java program: Class Inheritance and Interface Inheritance. ◆ Explain in detail the features of Java, i.e. Code Reusability, Super class, Subclass. ◆ Explain how to implement Inheritance in an interface with the help of the syntax. (Ask the students to note down the syntax.) ◆ Explain Class Inheritance with its syntax and an example. ◆ Explain the various types of Inheritance in Java. ◆ Explain single-level Inheritance. ◆ Explain multilevel Inheritance and its need. ◆ Explain hierarchal Inheritance. ◆ Explain all types of Inheritance with the help of diagrams shown on the slide. Next, explain the difference between the various types of Inheritance. ◆ This slide explains the order of construction of objects in Inheritance. ◆ Explain Method Overriding with the help of an example. ◆ Explain Final Class and Method. ◆ Explain the Final Method and Final Class and also explain the finally keyword which is used to define a block that is always executed.

Day - 11	Inheritance in Java Programming	2 hours 30 minutes
11.4		

Resources to be used	◆ Practical Guide
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Do	Ask the participants to open their Practical Guides ensure that they are apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 11	Recap	20 minutes
11.5		

Resources to be used	◆ Questionnaires
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Do	Ask the participants to take out the questionnaires.
Notes for Facilitation	Instruct the participants to fill out the questionnaires as per their understanding.

Day - 11	Inheritance in Java Programming	1 hour
11.6		

Resources to be used	◆ Practical Guide
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Notes For Facilitation	◆ Use a program to illustrate the order of calling of default constructor in super and sub class. Look at the code and refer to the comments are understand the code. ◆ Take a program and show how to access the super class method, super class parameterized constructor and super class instance variable by using super keyword from sub class. ◆ While trainees compile the program code, ask if they encounter any problems/errors. Discuss the bugs, if any.
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Day - 12	Morning Energiser	10 minutes
12.1		

Resources to be used	Morning Energisers booklet
Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.

Day - 12	Time Management	2 hours
12.2		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook ◆ Activities and Games booklet ◆ Presentation
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Do	Play the multimedia unit. Pause and explain wherever there is an explanation required.
	Discuss the myths about time generally we have.
Notes for Facilitation	Discuss what activities the participants consider as “time-wasters”. Explain what “time management” essentially means.
Do	Conduct a group activity by choosing a relevant one from the Activities and Games booklet.
Notes for Facilitation	Tell the class about how time management is important in our day-to-day life for achieving success. Lack of time management leads to non-fulfilment of tasks and duties in time which further leads to stress and mismanagement. Following are some tips on time management: ◆ Making activity logs help us to remember all the tasks to be done. ◆ Planning helps us to chalk out those activities which are necessary to achieve our goals. ◆ Scheduling helps us to organise the activities to be done and allot appropriate time to each activity. Highlight how prioritising helps us to manage our time efficiently by concentrating more on important activities. Run the presentation for summarising the session.
Do	Ask the participants to fill out the questionnaire as per their understanding.

Day - 12	Files and Streams	1 hour 10 minutes
12.3		

Resources to be used	◆ Presentation ◆ Participant Handbook
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Do	Run the PowerPoint presentation.
Notes for Facilitation	◆ Explain how the files are processed in Java and also its File Class which is a part of the java.io package. ◆ Explain how to create a File in Java with its syntax. ◆ Emphasise and explain the need for checking the existence of a File. We need to check the existence of a File since sometimes the File is not created and we try to load the contents and it returns an error. So to reduce these errors we have to check that whether a File exists or not.

Day - 12	Recap	10 minutes
12.4		

Resources to be used	◆ Questionnaires
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Do	Ask the participants to fill out the questionnaires as per their understanding.
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Day - 12	Files and Streams	2 hours 30 minutes
12.5		

Resources to be used	◆ Practical Guide
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Do	Ask the participants to open their Practical Guides ensure that they are apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 12	Files and Streams	2 hours
12.6		

Resources to be used	◆ Practical Guide
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Do	Continue with the practice session. Ensure that the participants are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 13	Morning Energiser	10 minutes
13.1		

Resources to be used	Morning Energisers booklet
Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.

Day - 13	Attitude at Work	2 hours
13.2		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook ◆ Presentation
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Do	Play the multimedia module. Pause and explain the key topics.
Say	Result means the outcome or consequence of something and the right attitude will enable them to achieve their goal. Similarly, “result-oriented management style” helps an organisation to achieve success.
Notes for Facilitation	Discuss what the participants think about result-orientation. ◆ Result-orientation means knowing which results are important and striving to achieve them by channelising appropriate resources.
Notes for Facilitation	Things to be kept in mind while being result-oriented: ◆ While working in a team, people with a result-oriented approach tend to disrupt it, as it involves interdependencies. ◆ With the mindset of achieving results, it doesn’t mean that some steps from the procedure of achieving the result be abandoned. ◆ There are various pitfalls to be avoided while being result-oriented. *Run the presentation and summarise the session.
Do	Ask the participants to fill out the questionnaire as per their understanding.

Day - 13	Types of Streams	2 hours
13.3		

Resources to be used	◆ Presentation ◆ Participant Handbook
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Do	Open the PowerPoint presentation.
Notes for Facilitation	◆ Explain the term and uses of Input/Output Streams. ◆ Explain the Class <code>java.io.OutputStream</code> which is the Base Class for all the I/O Stream. ◆ Explain the types of Streams. ◆ Explain how Byte Stream is used for Binary Data and Character Stream is used for unicode characters. ◆ Talk about the Input and Output Streams. ◆ Explain the concepts of Node Stream and Filter Stream. ◆ Explain the Chaining in I/O Streams. ◆ Explain the Control Flow in Input/Output operations of Streams. ◆ Explain to the students the Control Flow in I/O operations. With the help of a diagram use arrows to explain the diagram to the students. ◆ Explain Byte Streams and Buffered Streams in detail using examples. ◆ Explain Standard Streams in detail. There are basically three types of Standard Streams, viz., <code>system.in</code>, <code>system.out</code> and <code>system.err</code>
Say	Please note down all the Streaming Operations and study them carefully.
Notes for Facilitation	Explain the Data Output Stream and Data Input Stream in detail. In data Stream the input and output of data takes place and in this Stream the I/O operations on objects are operated. Explain that closing a Stream when it's no longer needed is very important—so it is important that your program should use a Finally Block to guarantee that both Streams will be closed even if an error occurs.

Day - 13	Types of Streams	2 hours 30 minutes
13.4		

Resources to be used	◆ Practical Guide
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Do	Ask the participants to open their Practical Guides ensure that they are apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 13	Recap	20 minutes
13.5		

Resources to be used	◆ Presentation
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Notes for Facilitation	Revise the day's learning. Consider the below-mentioned questions and have an interactive session with the class. Q1. Explain Byte and Character Streams. Q2. Explain Control Flow of I/O operation using Streams. Q3. Draw the diagram to show I/O Stream chaining. Q4. Explain the I/O Stream concept. Q5. Show the class hierarchy for the Reader Class. *Run the presentation if required.
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Day - 13	Types of Streams	1 hour
13.6		

Resources to be used	◆ Practical Guide
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Do	Ask the participants to open their Practical Guides ensure that they are apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 14	Morning Energiser	10 minutes
14.1		

Resources to be used	Morning Energisers booklet
Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.

Day - 14	Abstract Windowing Toolkit (AWT)	4 hours
14.2		

Resources to be used	◆ Presentation ◆ Participant Handbook
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Do	Use the PowerPoint presentation to explain AWT.
Notes for Facilitation	◆ Introduce AWT and how we can develop Graphical User Interface in Java, to the Class. ◆ Share the history of AWT and its use in Java Development Kit. ◆ Discuss the hierarchy of AWT. ◆ Give an introduction to Containers and components. ◆ Discuss AWT components and their hierarchy with the help of a diagram. ◆ Ask the participants to note down the points. ◆ Demonstrate how to create components in Java. ◆ Explain how to create windows and frames in Java.
Ask	Note down the examples and try them in the practice sessions.

Notes for Facilitation	◆ Explain how to create frames in Java. ◆ Explain how to add Panels, Buttons, Labels, List Checkboxes, Choices and Text Fields in a Java window. ◆ Explain how to arrange the components in a Java applet. ◆ Explain the layout manager used in Java applets with its different types. ◆ Explain Flow Layout with the help of an example and show how a flow Layout looks. ◆ Pause at the diagram and explain it to the participants. ◆ Explain the Border Layout with the help of an example. Its source code is also explained; a diagram of the border layout is shown on the slide. ◆ Explain the grid layout in Java AWT along with its source code. ◆ Introduce Graphics in Java. Show how we can create different Graphics like Line in AWT.
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Day - 14	Recap	25 minutes
14.3		

Resources to be used	◆ Presentation
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Notes for Facilitation	Revise the day's learning. Consider the below-mentioned questions and have an interactive session with the class. Q1. What is AWT? Explain in brief. Q2. Explain standalone GUI application. Q3. What packages should a Java program import to deal with any GUI applications? Q4. What are the three categories AWT is divided into? Q5. How can we build a GUI? *Run the presentation if required
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Day - 14	Abstract Windowing Toolkit (AWT)	3 hours 25 minutes
14.4		

Resources to be used	◆ Practical Guide
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Do	Ask the participants to open their Practical Guides ensure that they are apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 15	Morning Energiser	10 minutes
15.1		

Resources to be used	Morning Energisers booklet
Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.

Day - 15	Recap	20 minutes
15.2		

Resources to be used	◆ Practical Guide
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Say	◆ Revise the previous day's learning. ◆ Write down the following questions on the board. Request two participants to solve the questions while the rest of the class helps them in completing the program codes. Q1. Write a program to create a Frame without Extending. Q2. Write a program to create a Frame by Extending Frame Class
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Day - 15	Threads in Java	2 hours
15.3		

Resources to be used	◆ Presentation ◆ Participant Handbook
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Do	Run the PowerPoint presentation.
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Notes for Facilitation	◆ Explain the concept of threads in Java with the help of the diagram. ◆ Explain the life of a thread, i.e. how the thread is created executed and destroyed. ◆ Explain the priorities of a thread. ◆ Explain the scheduling of threads. ◆ Explain daemon thread. ◆ Explain that Java does not run multiple interface ◆ Explain the runnable interface. ◆ Explain some thread group methods such as: activeCount, enumerate, getMaxPriority, getName, getParent. ◆ Explain to the students the group methods of thread clearly and stress on the point that constructors are used to associate threads with thread groups.
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Day - 15	Threads in Java	2 hours
15.4		

Resources to be used	◆ Practical Guide
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Do	Ask the participants to open their Practical Guides ensure that they are apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 15	Games and Activities	30 minutes
15.5		

Resources to be used	◆ Activities and Games booklet
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Notes for Facilitation	Pick up any game from the Activities and Games booklet and conduct it in class.
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Day - 15	Threads in Java	3 hours
15.6		

Resources to be used	◆ Practical Guide
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Do	Ask the participants to open their Practical Guides ensure that they are apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 16	Morning Energiser	10 minutes
16.1		

Resources to be used	Morning Energisers booklet
Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.

Day - 16	Networking	2 hours 30 minutes
16.2		

Resources to be used	^a Presentation
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Do	Open the PowerPoint presentation on the topic of Networking.
Notes for Facilitation	Explain: ^a Java networking model; ^a Socket programming; ^a IP addresses in Java; ^a How to use TCP sockets for clients and servers; ^a How to use UDP sockets for clients and servers; ^a The concept of Server Sockets in Java; ^a The concept of Server Side Socket operations; ^a The different types of Socket Operations performed on Server Side; ^a Data input and output Streams; ^a URL connections in Java programs; ^a The Get and Post Requests sent and received by the URL connection; ^a How to close Sockets, i.e. how to close the server and the client Sockets; ^a The methods called to close the sockets.

Day - 16	Recap	20 minutes
16.3		

Resources to be used	^a Presentation
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Notes for Facilitation	Revise the day's learning. Consider the below-mentioned questions and have an interactive session with the class. Q1. What is Network Programming? What are the 3 requirements to establish a Network? Q2. Explain the Java Networking model. Q3. Explain the concept of IP addresses in Java. Q4. What are Java Server Socket? Explain the Server Side Socket Operations. Q5. What is Socket Programming? Explain in brief. *Run the presentation if required.
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Day - 16	Networking	3 hours
16.4		

Resources to be used	^a Practical Guide
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Do	Ask the participants to open their Practical Guides ensure that they are apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 16	Networking	2 hours
16.5		

Resources to be used	- a Practical Guide - a Participant Handbook
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Do	Continue with the practice session. Ensure that the participants are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 17	Morning Energiser	10 minutes
17.1		

Resources to be used	Morning Energisers booklet
Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.

Day - 17	SWING	2 hours
17.2		

Resources to be used	◆ Presentation ◆ Participant Handbook
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Do	Start the PowerPoint presentation on the topic of the day.
Note for Facilitation	◆ Introduce the term Java Foundation Class (JFC). ◆ Explain the features of JFC. ◆ Explain the SWING component set. ◆ Explain the SWING architecture, i.e. Model View Controller (MVR) architecture. ◆ Explain separable Model Architecture of SWING. ◆ Explain the luggable look and feel of a SWING. ◆ Explain 2D graphics in Java and state its importance. ◆ Explain how we can drag and drop in Java. Copy and paste operations are important in almost all applications. Explain how to drag and drop SWINGs. ◆ Explain Accessibility API. ◆ Give a brief introduction to internationalization in Java. ◆ Explain the undo framework API. ◆ Explain the flexibility in deployment of Java programs. ◆ Explain to the students that if they want a program to run within a browser window, they can create it as an Applet and run it using Java plug-in, which supports a variety of browsers, such as Internet Explorer, Firefox, and Safari.

Day - 17	SWING	3 hours
17.3		

Resources to be used	◆ Practical Guide
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Do	Ask the participants to open their Practical Guides ensure that they are apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 17	Recap	30 minutes
17.4		

Resources to be used	◆ Questionnaire
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Note for Facilitation	Revise the day's learning by testing them on the key topics. Refer to the test sheet below and discuss the answers later in class.
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Test

1. Swing GUI uses how many kinds of classes?

- a. 1
- b. 2
- c. 4
- d. 6

☐
☐
☐
☐

The GUI Classes are visual descendants of JComponent and are called Jclasses.

- a. True ☐
 - b. False ☐
2. *According to the MVC architecture, a component can be modelled into how many separate parts?*
 - a. 1 ☐
 - b. 2 ☐
 - c. 4 ☐
 - d. 6 ☐
3. *Which MVC part stores the data used to define the component?*
 - a. Model ☐
 - b. View ☐
 - c. Controller ☐
4. *In MVC, the role of the controller is heavily dependent on the implementation of the view.*
 - a. Yes ☐
 - b. No ☐
5. *The Java 2D API is implemented in the same java.awt package.*
 - a. True ☐
 - b. False ☐

6. Which software mediates the user experience?
- a. Expert system ☐
 - b. Assistive technologies ☐
 - c. Passive technologies ☐
 - d. Active technologies ☐
7. If you want your program to run within a browser window, you can create it as an applet.
- a. True ☐
 - b. False ☐
8. The feature that allows developers to build applications that can interact with users worldwide in their own languages and cultural conventions is _____.
- a. Externalization ☐
 - b. Accessibility ☐
 - c. Drag and drop ☐
 - d. Internalization ☐
9. The part of MVC that deals with the behaviour of the components when a user interacts with it is called _____.
- a. Model ☐
 - b. View ☐
 - c. Controller ☐
10. Which architecture is based on MVR architecture?
- a. Applet ☐
 - b. AWT ☐
 - c. Windows ☐
 - d. Swings ☐

11. *Java 2D is built upon the AWT.*

a. True

☐

b. False

☐

12. *All graphics operations in Java are performed using java.awt.Graphics object.*

a. True

☐

b. False

☐



Answers: Recap

Answer	
1.	b
2.	a
3.	c
4.	a
5.	a
6.	a
7.	b
8.	a
9.	d
10.	c
11.	d
12.	a
13.	a

Day - 17	SWING	2 hours 20 minutes
17.5		

Resources to be used	◆ Practical Guide
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Do	Ask the participants to open their Practical Guides ensure that they are apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 18	Morning Energiser	10 minutes
18.1		

Resources to be used	Morning Energisers booklet
Notes for Facilitation	Please refer to the "Morning Energisers" booklet. You may pick the energiser that is recommended for the day of the week.

Day - 18	Handling GUI Generated Events	2 hours 30 minutes
18.2		

Resources to be used	◆ Presentation
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Do	Run the PowerPoint presentation to the corresponding topic.
Say	Explain the event handling process.
Ask	Please understand and take down the diagram.
Notes for Facilitation	◆ Explain the Event Source in detail. ◆ Explain Event Handler and Listener in detail. ◆ Explain the Mouse Motion Listener methods. ◆ Explain the Windows Listener methods. ◆ Explain the hierarchy of Event Objects. ◆ Explain the different types of Event Objects. ◆ Explain creating GUI applications with Event handling. ◆ Explain Adapter Classes for Event Handling in detail and how to use an Adapter Class. ◆ Explain how to handle the common event-handling issues in detail.

Day - 18	Handling GUI Generated Events	3 hours
18.3		

Resources to be used	◆ Practical Guide
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Do	Ask the participants to open their Practical Guides ensure that they are apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 18	Handling GUI Generated Events	2 hours 20 minutes
18.4		

Resources to be used	◆ Practical Guide
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Do	Continue with the practice session. Ensure that the participants are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 19	Morning Energiser	10 minutes
19.1		

Resources to be used	Morning Energisers booklet
Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.

Day - 19	Teamwork	2 hours
19.2		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook ◆ Presentation ◆ Activities and Games booklet
Do	Show the multimedia unit.
Notes for Facilitation	Discuss what “team work” is all about.
Say	Team work is important to accomplish tasks at a faster pace. The four stages of team development are: Forming, Storming, Norming and Performing.
Notes for Facilitation	◆ Explain the advantages of working in a team. ◆ Explain how to become an effective team member. ◆ Ask the participants how they can contribute by being in a team. Take up a team activity for the class from the Activities and Games booklet.
Do	Ask the participants to fill out the questionnaire as per their understanding.

Day - 19	AWT and SWING	2 hours
19.3		

Resources to be used	◆ Presentation ◆ Participant Handbook
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Do	Run the PowerPoint presentation.
Notes for Facilitation	◆ Explain how to select the component for GUI. ◆ Explain the features of AWT. ◆ Explain the features of SWINGS in detail. ◆ Explain the pros and cons of AWT. ◆ Explain the pros and cons of SWINGS. ◆ State the difference between the AWT and SWINGS.
Notes for Facilitation	◆ Discuss Menus and Toolbars. ◆ Give an introduction to Text Control. ◆ Give an introduction to Creating Frames in Java. ◆ Explain Frame and Dialog Boxes in Java. ◆ Explain Data Selection and Display Control. ◆ Explain the concept of Timers in Java.

Day - 19	Recap	20 minutes
19.4		

Resources to be used	◆ Presentation
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Do	Revise the day's learning by testing them on the key topics. Refer to the questions below.
Say	Q1. What are Applets? Q2. What are Swings? Q3. What are Timers? *Run the PowerPoint presentation if required.

Day - 19	SWING Components	1 hour 30 minutes
19.5		

Resources to be used	◆ Practical Guide
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Do	Ask the participants to open their Practical Guides ensure that they are apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 19	SWING Components	2 hours
19.6		

Resources to be used	◆ Practical Guide
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Do	Continue with the practice session. Ensure that the participants are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 20	Morning Energiser	10 minutes
20.1		

Resources to be used	Morning Energisers booklet
Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.

Day - 20	Test 2	2 hours
20.2		

Resources to be used	Assessment Guide
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Notes for Facilitation	Formative Assessment for the topic C++ and Data Structures. Refer to the questions in the Assessment Guide and conduct the test.
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Day - 20	Applets	2 hours 30 minutes
20.3		

Resources to be used	◆ Presentation
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Do	Run the PowerPoint presentation.
Notes for Facilitation	◆ Introduce the term applets and its advantages. ◆ Explain that Swing classes in applets can be used in the same way as AWT classes. ◆ Illustrate the applet hierarchy and what components this class inherits. ◆ Explain the syntax for declaring the applet. ◆ Explain how to create an applet. ◆ Explain Compilation and execution of applets. ◆ Explain the structure of an applet. ◆ Explain how to run an applet in a Java program. ◆ Show the steps to the two different ways to run applets. ◆ Explain the difference between applets and applications.

Day - 20	Applets	2 hours 30 minutes
20.4		

Resources to be used	◆ Practical Guide
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Do	Ask the participants to open their Practical Guides ensure that they are apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 20	Recap	20 minutes
20.5		

Resources to be used	◆ Presentation
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Do	Revise the day's learning by testing them on the key topics. Refer to the questions below.
Notes for Facilitation	Q1. What are applets? Q2. List the advantages of applets. Q3. Explain the hierarchy of applet class. Q4. Write the syntax for applet class. Q5. What is the difference between an applet and an application?

Day - 20	Games and Activities	30 minutes
20.6		

Resources to be used	Activities and Games booklet
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Notes for Facilitation	Refer to the "Activities and Games" booklet to choose and conduct a game and an activity.
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Day - 21	Morning Energiser	10 minutes
21.1		

Resources to be used	Morning Energisers booklet
Notes for Facilitation	Please refer to the "Morning Energisers" booklet. You may pick the energiser that is recommended for the day of the week.

Day - 21	Recap	20 minutes
21.2		

Resources to be used	◆ Presentation
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Do	Revise the key topics from what has been discussed so far. Refer to the questions given below.
Say	Q1. Explain Applets in detail. Q2. What is Exceptional Handling? Explain with the help of a program. Q3. Explain the use of Finally keyword with the help of a program. *Run the presentation if required.

Day - 21	JavaScript	2 hours
21.3		

Resources to be used	◆ Presentation
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Do	Run the PowerPoint presentation.
Notes for Facilitation	◆ Introduce the term JavaScript. ◆ Explain what JavaScript can do along with its application areas. ◆ Explain how JavaScript is embedded into the Webpages. ◆ Introduce the variables in JavaScript, both Local and Global. ◆ Introduce the functions in JavaScript. ◆ Explain the syntax of defining JavaScript Functions. ◆ Illustrate the procedure to call a function with Arguments. ◆ Explain JavaScript Function Literals along with the syntax. ◆ Explain the functions with a Return Value. ◆ Show an example of the “functions with a return value”. The code shown on the slide multiplies two numbers using multiply() function. ◆ Explain the examples; relate them with the syntax given on the previous slides. ◆ Explain the code to display “Hello World!” using functions. Here the display message() function is used to serve the purpose .

Day - 21	JavaScript	3 hours
21.4		

Resources to be used	◆ Practical Guide
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Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.
Notes for Facilitation	◆ Use an example to explain how to display messages on the WebPage. ◆ Help the participants to debug in case they encounter any problems or errors.

Day - 21	JavaScript	2 hours 30 minutes
21.5		

Resources to be used	◆ Practical Guide
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Do	Continue with the practice session. Ensure that the participants are able to apply their learning.
Notes for Facilitation	◆ Use an example to concatenate 2 Strings in JavaScript. ◆ Help the participants to debug in case they encounter problems or errors.

Day - 22	Morning Energiser	10 minutes
22.1		

Resources to be used	Morning Energisers booklet
Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.

Day - 22	JavaScript – Operators and Conditional Statements	3 hours
22.2		

Resources to be used	◆ Presentation ◆ Participant Handbook
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Do	Run the PowerPoint presentation.
Notes for Facilitation	◆ Introduce the Operators in JavaScript. ◆ Explain Arithmetic Operators in detail. ◆ Explain Assignment Operators in detail. ◆ Explain Comparison Operators in detail. ◆ Explain Logical Operators in detail. ◆ Explain the special characters that can be used in JavaScript. ◆ Illustrate what are JavaScript Statements with an example. ◆ Explain the JavaScript Code. ◆ Discuss comments in JavaScript. ◆ Explain JavaScript Blocks. ◆ Explain JavaScript Conditional statements. ◆ Explain the syntax of If statement along with an example. ◆ Explain the syntax of If-Else statement. Give two examples. ◆ Explain the syntax of Nested-If-Else statement along with examples. ◆ Explain Switch statement; mention and give two examples. ◆ Explain what Break statement is in JavaScript. State its syntax. ◆ Explain Continue statement. ◆ Explain For-In statement with an example.

	◆ Explain Try-Catch-Throw statement with its syntax. ◆ Explain Loops in JavaScript. ◆ Explain For-Loop in detail along with its syntax and an example. ◆ Explain While loop in detail. State its syntax and give examples. ◆ Explain Do-While loop in detail, stating syntax and examples.
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Day - 22	JavaScript – Operators and Conditional Statements	2 hours 30 minutes
22.3		

Resources to be used	◆ Practical Guide
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Do	Ask the participants to open their Practical Guides ensure that they are apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 22	JavaScript – Operators and Conditional Statements	2 hours 20 minutes
22.4		

Resources to be used	◆ Practical Guide
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Do	Continue with the practice session. Ensure that the participants are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 23	Morning Energiser	10 minutes
23.1		

Resources to be used	Morning Energisers booklet
Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.

Day - 23	JavaScript – Objects	2 hours 30 minutes
23.2		

Resources to be used	◆ Presentation
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Do	Run the PowerPoint presentation
Notes for Facilitation	◆ Introduce JavaScript Objects. ◆ Explain the naming Objects and built-in objects in detail. ◆ Explain string Objects, their syntax, properties along with examples. ◆ Introduce Methods in JavaScript. ◆ Introduce charAt() method in JavaScript along with an example. ◆ Introduce charCodeAt() method in JavaScript along with an example. ◆ Introduce concat() method in JavaScript along with an example. ◆ Introduce indexOf() method in JavaScript along with an example. ◆ Introduce lastIndexOf() method in JavaScript along with an example. ◆ Introduce Events in JavaScript and explain the various types of Events. ◆ Explain with an example Illustrating Events. ◆ Introduce Event Handlers along with their functionalities.

Day - 23	JavaScript – Objects	2 hours 30 minutes
23.3		

Resources to be used	◆ Practical Guide
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Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.
Notes for Facilitation	Take a program and illustrate a function to display event model of JavaScript; check the form validity on 'onsubmit' event. Help the participants to debug the programs in case they encounter any problems or errors.

Day - 23	Recap	15 minutes
23.4		

Resources to be used	◆ Presentation
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Do	Revise the key topics from what has been discussed so far. Refer to the questions given below.
Say	Q1. Create an object to demonstrate Objects in JavaScript. Q2. Create an object and its properties to demonstrate Objects in JavaScript. *Run the presentation if required.

Day - 23	JavaScript – Objects	2 hours 30 minutes
23.5		

Resources to be used	◆ Practical Guide
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Do	Continue with the practice from the Practical Guide. Ensure that they are able to apply their learning.
Say	◆ Use a program to illustrate a function in JavaScript to display event model of JavaScript. ◆ Help the participants to debug in case they encounter any problems or errors.

Day - 23	Morning Energiser	10 minutes
24.1		

Resources to be used	Morning Energisers booklet
Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.

Day - 23	Communication	2 hours 30 minutes
24.2		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook ◆ Presentation
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Do	Run the multimedia module.
Say	When a person receives too many messages, or the message is complex or the presentation of information is such that it is ambiguous, or there are linguistic barriers, it leads to ineffective communication.
Do	Ask the participants how they can overcome the “barriers to effective communication”.
Expected Responses	1. By paying attention to the speaker 2. By minimising our internal and external distractions 3. By focussing on what the speaker is saying
Say	Whenever there is some interference with the decoding of messages, it is called “communication noise”.
Do	Show the presentation and conduct the Activity 1, 2 and 3.
Notes for Facilitation	◆ Communication adds meaning to human life. It helps to build relationship and foster love and understanding. ◆ Communication skills are essential in all spheres of life; be it an interview, dealing with the project leader, working out a solution with a team, or writing a report– getting across the point effectively is all that matters. ◆ Communication, in plain terms, is the basis of all relationships. ◆ When it comes to business, communication becomes even more important. This is the pillar on which professional relationships are built.

Notes for Facilitation	*Run the presentation to summarise the session.
Do	Ask the participants to fill out the questionnaire, as per their understanding.

Day - 23	JavaScript – Objects	2 hours
24.3		

Resources to be used	◆ Presentation
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Do	Ask the participants to open their Practical Guides ensure that they are apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 23	JavaScript – Objects	2 hours
24.4		

Resources to be used	◆ Practical Guide
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Do	Continue with the practice from the Practical Guide. Ensure that they are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 23	JavaScript – Objects	1 hour
24.5		

Resources to be used	◆ Practical Guide
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Do	Continue with the practice from the Practical Guide. Ensure that they are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 23	Recap	20 minutes
24.6		

Resources to be used	◆ Presentation
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Notes for Facilitation	Revise the key topics from what has been discussed so far. Run the presentation if required.
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Day - 25	Morning Energiser	10 minutes
25.1		

Resources to be used	Morning Energisers booklet
Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.

Day - 25	Java Servlets	3 hours
25.2		

Resources to be used	◆ Presentation
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Do	Run the PowerPoint presentation.
Notes for Facilitation	◆ Introduce Java Servlets. ◆ Explain the Servlet Process Flow through a diagram. ◆ Describe the advantages of Servlets. ◆ Explain Java Servlet API. ◆ Explain Servlet Lifecycle through a diagram. ◆ Illustrate how to load a Servlet Class. ◆ Explain how to define the init() method of the Servlet. ◆ Explain how to define the service() method of the Servlet. ◆ Explain how to define the doGet() method of the Servlet. ◆ Explain how to define the doPost() method of the Servlet. ◆ Explain how to define the destroy() method of the Servlet. While explaining the concepts with their examples, relate them with the syntax given on the previous slides.

Day - 25	Java Servlets	2 hours 30 minutes
25.3		

Resources to be used	◆ Practical Guide
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Notes for Facilitation	Explain a program illustrating: ◆ a servlet that prints a hello message; ◆ a simple counter; ◆ how to handle form data with servlet. Help the participants to debug in case they encounter any problems or errors.
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Day - 25	Recap	20 minutes
25.4		

Resources to be used	◆ Presentation
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Do	Revise the key topics from what has been discussed so far. Refer to the questions given below.
Say	Q1. Explain the Life Cycle methods of a Servlet in detail. Q2. What are the types of Servlets? Q3. Explain ServletContext in detail. Q4. Distinguish between doGet() and doPost(). Q5. Distinguish between servlet and applet. *Run the presentation if required.

Day - 25	Java Servlets	2 hours
25.5		

Do	Ask the participants to open their Practical Guides ensure that they are apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 26	Morning Energiser	10 minutes
26.1		

Resources to be used	Morning Energiser booklet
Notes for Facilitation	Please refer to the "Morning Energisers" booklet. You may pick the energiser that is recommended for the day of the week.

Day - 26	Java Servlets	2 hours 30 minutes
26.2		

Resources to be used	◆ Practical Guide
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Do	Ask the participants to open their Practical Guides.
Notes for Facilitation	◆ Using a program illustrate a Servlet to process a Web Form submitted online. ◆ Use a program to illustrate a Servlet to generate Web Form dynamically. Help the participants in debugging in case they encounter any problems or errors.

Day - 26	Java Servlets	2 hours 30 minutes
26.3		

Resources to be used	◆ Practical Guide
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Do	Continue with the practice session.
Say	◆ Use a program to illustrate a servlet to display date. ◆ Use a program to illustrate a servlet to simulate a dice roller. Help the participants to debug in case they encounter any problems or errors.

Day - 26	Recap	20 minutes
26.4		

Resources to be used	◆ Presentation
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Do	Revise the key topics from what has been discussed so far. Refer to the questions given below.
Say	Q1. Write a code for the first servlet “Hello World” and run it on the server. Q2. Write a code for a servlet that displays the current date and time local to timezone of the server running the servlet. *Run the presentation if required.

Day - 26	Java Servlets	2 hours 30 minutes
26.5		

Resources to be used	◆ Practical Guide
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Notes for Facilitation	◆ Ask the participants to open their Practical Guides. ◆ Help the participants to debug in case they encounter any problems or errors.
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Day - 27	Morning Energiser	10 minutes
27.1		

Resources to be used	Morning Energisers booklet
Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.

Day - 27	Problem Solving	2 hours 30 minutes
27.2		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook ◆ Activities and Games
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Do	Play the multimedia unit.
Do	Ask what the participants understand by the term “problem”.
Say	A problem is a situation that needs attention. A problem can be defined by asking questions like: ◆ What is happening? ◆ Where it is happening? ◆ When it is happening? ◆ To what extent is it happening?
Do	Take up activities from the Activities and Games booklet and conduct an activity or game in class.

Notes for Facilitation	Use a simple example to illustrate the definition of “problem”. Following are the steps involved in the problem-solving process: 1. Always focus on the “solution” to the problem and not the problem. 2. Have a positive and open mind to think of all possible alternatives and solutions to the problem. 3. View the problem neutrally and think laterally – that is, changing the direction of thinking. 4. Discuss and brainstorm – helps to generate more ideas and alternative solutions. 5. Do a “What-if” analysis to generate more alternative solutions and keep your mind open for some creativity. 6. Lastly, try to always simplify things. Break-up complicated problems and problem-solutions to small activities to achieve the goal.
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Day - 27	Java Server Pages (JSP)	2 hours
27.3		

Resources to be used	◆ Presentation ◆ Participant Handbook
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Do	Run the PowerPoint presentation.
Notes for Facilitation	◆ Introduce the concept of JSP. ◆ Explain how to install JSP. ◆ Explain the Servers that support JSP. ◆ Describe how JSP is processed. ◆ Explain how Java Server Pages work. ◆ Explain JSP Scripting Elements. ◆ Explain the structure of a JSP Page and illustrate how to create the first JSP page. ◆ Explain the advantages of using JSP technology.

Day - 27	Java Server Pages (JSP)	2 hours
27.4		

Resources to be used	◆ Practical Guide ◆ Participant Handbook
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Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply the day's learning.
Notes for Facilitation	◆ Illustrate how to build a JSP Page to display date and time. ◆ Use a program to illustrate how to build a JSP page that evaluates sum of two numbers. ◆ Use a program and illustrate how to build a JSP page that evaluates table of a number. Help the participants in debugging their programs in case they encounter any problems or errors.

Day - 27	Recap	20 minutes
27.5		

Resources to be used	◆ Presentation
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Do	Revise the key topics from what has been discussed so far. Refer to the questions given below.
Notes for Facilitation	Q1. Explain Declarations in JSP through a program. Q2. Write a JSP program code that displays the date. *Run the presentation if required.

Day - 27	Java Server Pages (JSP)	1 hour
27.6		

Resources to be used	◆ Practical Guide
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Do	Continue with the practice session.
Notes for Facilitation	Use a program to illustrate how to build a JSP page that prints a calendar of the current month. Help the participants in debugging in case they encounter any problems or errors.

Day - 28	Morning Energiser	10 minutes
28.1		

Resources to be used	Morning Energisers booklet
Notes for Facilitation	Please refer to the "Morning Energisers" booklet. You may pick the energiser that is recommended for the day of the week.

Day - 28	Stress Management	2 hours 30 minutes
28.2		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook ◆ Activities and Games Booklet
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Do	Play the multimedia module.
Ask	What does the term "stress" mean to you?
Expected Responses	1. Tension 2. Workload 3. Anything which affects us to behave in an abnormal manner
Say	That's right! There are three stages of stress: 1. Alarm stage 2. Resistance stage 3. Exhaustion stage The ABC Strategy of Stress Control: Being Aware about the causes of stress; being Balanced while taking pressure finally, Controlling the negative effects of stress.
Do	Run the presentation and conduct the activities mentioned. Alternately, you could pick out a game from the Activities and Games booklet and conduct them in class.

Notes for Facilitation	Discuss how relaxation techniques can bring them back into balance when stressed. Request the participants to volunteer to come up and share some funny stories, jokes or incidents to help lighten up the mood. Ask the participants to fill out the questionnaire as per their understanding.
Do	Ask the participants to fill out the questionnaire as per their understanding.

Day - 28	JSP Objects, Actions and Directives	2 hours 30 minutes
28.3		

Resources to be used	◆ Presentation ◆ Participant Handbook
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Do	Run the PowerPoint presentation.
Notes for Facilitation	◆ Explain Page Directives in JSP. ◆ Introduce Implicit Objects in JSP. ◆ Explain JSP Escape Sequences with the help of a Table. ◆ Describe the Standard Actions in JSP through Tables. ◆ Explain JSP-Plugin Action in detail through a Table that defines its attributes. ◆ Explain JSP-Use-Bean Action in detail. ◆ Explain JSP Directives in detail. *While explaining the examples relate them to the syntax given on the previous slides.

Day - 28	Recap	20 minutes
28.4		

Resources to be used	Presentation
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Do	Revise the day's learning. Refer to the questions given below.
Say	Q1. What are the advantages of JSP over Servlet? Q2. Explain the life-cycle of JSP? Q3. What is the <code>_jspService()</code> method? Q4. What is the <code>jspDestroy()</code> method? Q5. What are the JSP Standard Actions? *Run the presentation if required.

Day - 28	JSP Objects, Actions and Directives	1 hour
28.5		

Resources to be used	Practical Guide
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Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.
Notes for Facilitation	- Use a program to illustrate how to build a JSP page that includes another page calendar. - Help the participants in debugging in case they are encounter any problems or errors.

Day - 28	JSP Objects, Actions and Directives	1 hour 30 minutes
28.6		

Resources to be used	◆ Practical Guide
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Notes for Facilitation	◆ Continue with the practice session. Ensure that they are able to apply their learning. ◆ Help the participants to debug in case they encounter any problems or errors.
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Day - 29	Morning Energiser	10 minutes
29.1		

Resources to be used	Morning Energisers booklet
Notes for Facilitation	Please refer to the "Morning Energisers" booklet. You may pick the energiser that is recommended for the day of the week.

Day - 29	Recap	20 minutes
29.2		

Resources to be used	◆ Presentation
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Do	Revise what has been taught so far. Refer to the questions given below.
Notes for Facilitation	Q1. What is a Scriptlet? Q2. What are Implicit Objects? List them. Q3. What are the different Scope Values for the <jsp:useBean>? Q4. Explain the life-cycle methods in JSP? Q5. How does JSP handle Run-time Exceptions? Q6. What is the difference between Variable declared inside a Declaration Part and Variable declared in Scriptlet Part? Q7. Is there a way to execute a JSP from the Command Line or from my own application? Q8. How can you prevent the output of my JSP or Servlet Pages from being cached by the browser? Q9. How can you use a Scriptlet to Initialize a newly Instantiated Bean? *Run the presentation, if required.

Day - 29	Java Database Connectivity (JDBC)	2 hours 30 minutes
29.3		

Resources to be used	◆ Presentation
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Do	Run the PowerPoint presentation.
Notes for Facilitation	◆ Introduce the concept of JDBC. ◆ Explain JDBC versions along with their package through a table. ◆ Explain JDBC architecture with the help of a diagram. ◆ Describe the steps for using database in Java. ◆ Explain how to load a JDBC Driver. ◆ Explain how to connect to the database. ◆ Explain how to create JDBC statements. ◆ Explain how to execute SQL statements in JDBC. ◆ Explain with an example, what is result set. Mention the syntax for creating result sets. ◆ Explain how to close the connection with the database. While explaining relate the term with the syntax given on the previous slides.

Day - 29	Java Database Connectivity (JDBC)	2 hours 30 minutes
29.4		

Resources to be used	◆ Practical Guide
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Do	Ask the students to open their Practical Guides. Ensure that they are able to apply their learning.
Notes for Facilitation	◆ Use a program to illustrate the code for creating a Database in Java. ◆ Use a program to illustrate the code for getting Column Names from a Database Table in Java. Help the participants in debugging in case they encounter any problems or errors.

Day - 29	Java Database Connectivity (JDBC)	2 hours 30 minutes
29.5		

Resources to be used	◆ Practical Guide
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Do	Continue with the practice session. Ensure that they are able to apply their learning.
Notes for Facilitation	◆ Use a program to illustrate how to create SQL Query on the Database in Java. ◆ Use a program to illustrate how to create Prepared Statement SQL Query on the Database in Java. Help the participants in debugging in case they encounter any problems or errors.

Day - 30	Morning Energiser	10 minutes
30.1		

Resources to be used	Morning Energisers booklet
Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.

Day - 30	JDBC Drivers and Stored Procedures	2 hours 30 minutes
30.2		

Resources to be used	◆ Presentation ◆ Participant Handbook
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Do	Run the PowerPoint presentation.
Notes for Facilitation	◆ Introduce the 4 types of JDBC Drivers. ◆ Describe JDBC driver types through a Table. ◆ Explain some popular JDBC Drivers. ◆ Describe some of the methods in java.sql.Connection. ◆ Explain some methods that are useful in executing SQL statements. ◆ Explain Result Set Constants through a table. ◆ Explain Prepared Statement interface. ◆ Explain JDBC-ODBC bonding through a diagram. ◆ Explain what is Metadata and how to create Metadata Database. ◆ Explain how to create Stored procedures in Java. ◆ Explain what is Transaction Control. While explaining the examples, relate them with the syntax given on the previous slides.

Day - 30	Recap	35 minutes
30.3		

Resources to be used	◆ Presentation
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Do	Revise the day's learning. Refer to the questions given below.
Notes for Facilitation	Q1. What are the steps involved in establishing a JDBC Connection? Q2. How can you load the drivers? Q3. What will Class.forName do while loading drivers? Q4. How can you create JDBC statements and what are they? Q5. How can you retrieve data from the result set? *Run the presentation, if required.

Day - 30	JDBC Drivers and Stored Procedures	4 hours
30.4		

Resources to be used	◆ Practical Guide
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Do	Ask the participants to open their Practical Guides ensure that they are apply their learning.
Notes for Facilitation	Use a program to illustrate how to code a Java program that manages student records through JDBC. Help the students to debug in case they are encounter any problems or errors.

Day - 30	Games and Activities	45 minutes
30.5		

Resources to be used	◆ Activities and Games booklet
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Notes for Facilitation	Refer to the Activities and Games booklet; pick out a game or activity and conduct it in class.
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Day - 31	Morning Energiser	10 minutes
31.1		

Resources to be used	Morning Energisers booklet
Notes for Facilitation	Please refer to the "Morning Energisers" booklet. You may pick the energiser that is recommended for the day of the week.

Day - 31	Way to Success	1 hour
31.2		

Resources to be used	◆ Multimedia (K-Yan) ◆ Participant Handbook
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Do	Run the multimedia module.
Ask:	What does "success" mean to you?
Expected Responses	"Achieving my goals " "Making lots of money" "Making my parents happy"
Notes for Facilitation	◆ Have an interactive session around the topic of "success". ◆ Discuss the "Keys to success" ◆ Share some real-life anecdotes, inspiring success stories about famous people. ◆ Encourage the students to share some success stories.
Do	Ask the participants to fill out the questionnaire as per their understanding.

Day - 31	AJAX	2 hours 30 minutes
31.3		

Resources to be used	◆ Presentation ◆ Participant Handbook
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Do	Run the PowerPoint presentation.
Notes for Facilitation	◆ Introduce the concept of Ajax. ◆ Share the history of Ajax. ◆ Explain the purpose of Ajax technology. ◆ Describe the web standards in detail. ◆ Discuss classic versus Ajax model. ◆ Explain the technologies used by Ajax. ◆ Illustrate Simple Processing with the help of a diagram. ◆ Explain data exchange in Ajax. ◆ Explain how http requests are used by Ajax. ◆ Explain XMLHttpRequest object in detail. ◆ Explain the readyStateProperty in detail with the help of a table. ◆ Explain the responseTextProperty in detail with the help of a table. ◆ Explain the Ajax framework along with the list of some popular Ajax frameworks. ◆ Explain the pros and cons of using Ajax. While explaining the examples, relate them with the syntax given on the previous slides.

Day - 31	Recap	1 hour
31.4		

Resources to be used	◆ Presentation
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Do	Revise the day's learning. Refer to the questions given below.
Notes for Facilitation	Q1. Distinguish between the classic model and Ajax model. Q2. Describe the technologies used by Ajax. Q3. Explain Data Exchange in Ajax with the help of a diagram. Q4. Explain the ReadyState Property and responseType Property. Q5. Describe Ajax Framework in detail. *Run the presentation, if required.

Day - 31	AJAX	3 hours 20 minutes
31.5		

Resources to be used	◆ Practical Guide
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Do	Ask the participants to open their Practical Guides ensure that they are apply their learning.
Notes for Facilitation	◆ Use a program to illustrate how to change text in the webpage using Ajax. ◆ Use a program to illustrate how to load an XML file using Ajax. Help the students to debug in case they encounter any problems or errors.

Day - 32	Morning Energiser	10 minutes
32.1		

Resources to be used	Morning Energisers booklet
Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.

Day - 32	Test – 3	2 hours
32.2		

Resources to be used	◆ Assessment Guide
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Notes for Facilitation	◆ Formative assessment for the topic of HTML.
	◆ Refer to the questions in the Assessment Guide and conduct the test.

Day - 32	Presentation Skills	1 hour
32.3		

Resources to be used	◆ Presentation ◆ Participant Handbook
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Do	Run the PowerPoint presentation.
Say	Presentation is an act of showing and explaining the contents of a topic to an audience.
Ask	Why do you think “presentation skills” are important?
Expected Responses	1. For recognition 2. To explain something

Say	Presentation skills are the techniques of showing the contents of a topic to an audience.
Notes for Facilitation	◆ The first step to making an effective presentation is "planning". ◆ Before starting a presentation, welcome the audience, introduce yourself, get their attention and explain to them the purpose of the presentation.
Ask	What are the points that one should keep in mind while delivering a presentation?
Notes for Facilitation	◆ While delivering a presentation, one should know their material; it should be rehearsed but not memorised. ◆ About taking questions, make it clear before starting the presentation when would you like to have them. ◆ Having sound knowledge about the content, practising, giving a solid introduction, smiling, making the session enjoyable, arriving early and practising deep breathing before the session help immensely in making an effective presentation.
Do	◆ Ask the participants to fill out the questionnaire as per their understanding.

Day - 32	Group Discussion	1 hour
32.4		

Resources to be used	◆ Presentation ◆ Participant Handbook
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Do	Start the PowerPoint presentation.
Say	Group Discussion is a process where ideas and opinions about a topic are debated upon.
Ask	What do you know about "Group Discussion"?
Expected Responses	1. "A group is formed and given a topic to discuss about" 2. "For selection in a job, etc"
Say	Tell them about the different types of group discussions. A group discussion can either be topic-based or case-based.

Notes for Facilitation	The skills required in any group discussion are as follows: ◆ How well do you communicate with others; ◆ How you behave and interact within a group; ◆ How open minded are you; ◆ Your listening skills; ◆ How you put forward your views; ◆ Your leadership and decision making skills; ◆ Your analytical skills and subject knowledge; ◆ Problem solving and critical thinking skills; ◆ Your attitude and confidence.
Do	Divide the class into groups of 8-10 members.
Say	A topic will be given on the spot for you to discuss in a group.
Notes for Facilitation	Topics for group discussion: ◆ Public sector being a guarantor of job security is a myth. ◆ Are Indians less quality conscious? ◆ Government pumping money into the economy is not the solution for our economic problems. ◆ Should agricultural subsidies be stopped? ◆ Are MNCs superior to Indian companies? ◆ Advertising is a waste of resources. ◆ Privatisation will lead to less corruption. ◆ Skilled manpower shortage in India ◆ Technology creates income disparities. ◆ Government needs to introspect before pointing a finger at the private sector for corruption. ◆ What shall we do about our ever-increasing population?
Do	Start calling the groups one by one and conduct the group discussion.

Day - 32	Practice	3 hours 50 minutes
32.5		

Resources to be used	◆ Practical Guide
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Do	Ask the participants to open their Practical Guides ensure that they are apply their learning.
Notes for Facilitation	◆ Let the students use this session to practise the programs related to the topic of "Building Web Application Using J2EE". ◆ Ensure that the participants get thorough practice; and clarify their doubts if they have any. Help the students to debug in case they are encounter any problems or errors.

Day - 33	Morning Energiser	10 minutes
33.1		

Resources to be used	Morning Energisers booklet
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Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 33	Database System – Overview and Concepts	3 hours 30 minutes
33.2		

Resources to be used	Presentation
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Do	Run the PowerPoint presentation.
Notes for Facilitation	◆ Give an introduction to databases and the importance of creating a database. ◆ Explain the need of creating a Database system. Database system was introduced to manage and update the data in the databases.
Ask	Please copy the 2-tier architecture of the database system.
Notes for Facilitation	◆ Explain the 2-tier architecture in brief. ◆ Explain the need for 3-tier architecture of a Database System. ◆ Explain the Schemas of Database, i.e. the structures used to create Databases. ◆ Explain the three views of Database, viz.; the user view, conceptual view and internal view of a Database illustrated by a diagram. ◆ Explain the concept and use of Structured Query Language. ◆ Explain the concept of Database Management System (DBMS). ◆ Explain the concept of Relational DBMS with the help of a table in which the data is stored. ◆ Explain the concept of DBMS and RDBMS in detail and explain to them that Relational DBMS is a system in which data is kept in the form of tables. ◆ Explain the difference between DBMS and RDBMS in detail.

Day - 33	Structured Query Language (SQL) [Using Oracle]	4 hours 20 minutes
33.3		

Resources to be used	Practical Guide
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Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 34	Morning Energiser	10 minutes
34.1		

Resources to be used	Morning Energisers booklet
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Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 34	Motivation	1 hour
34.2		

Resources to be used	Participant Handbook
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Notes for Facilitation	◆ Discuss what the participants understand by the term “motivation”. ◆ Discuss what factors motivate to work towards a goal. ◆ Explain why we lose motivation. ◆ Explain the “motivation cycle” with the help of a diagram. ◆ Discuss how we can motivate ourselves. Share some real-life examples of highly motivated people. Encourage the class to share some of their experiences.
Do	Ask the participants to fill out their questionnaires as per their understanding.

Day - 34	Computations on Table Data	2 hours 30 minutes
34.2		

Resources to be used	Practical Guide
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Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 34	Procedural Language – PL/SQL	2 hours 30 minutes
34.3		

Resources to be used	◆ Presentation ◆ Practical Guide
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Do	Run the PowerPoint presentation.
Notes for Facilitation	◆ Introduce and explain the term PL/SQL. ◆ Explain the difference between PL/SQL and SQL. ◆ Explain MYSQL. ◆ Explain features of Oracle. ◆ Discuss the difference between MYSQL and Oracle databases.
Say	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 34	Procedural Language – PL/SQL	1 hour 50 minutes
34.4		

Resources to be used	◆ Presentation ◆ Practical Guide
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Do	Continue with the practice session. Ensure that the participants are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 35	Morning Energiser	10 minutes
35.1		

Resources to be used	Morning Energisers booklet
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Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 35	Recap	30 minutes
35.2		

Resources to be used	◆ Presentation ◆ Participant Handbook
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Notes for Facilitation	Recap the key topics that have been discussed so far. * Refer to the presentation if required.
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Day - 35	Cursors, Triggers and Functions	2 hours 30 minutes
35.3		

Resources to be used	◆ Presentation ◆ Participant Handbook
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Do	Run the PowerPoint presentation.
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Say	◆ Give an introduction to cursors. ◆ Explain the types of cursors. ◆ Explain cursors with the help of an example. ◆ State the cursor attributes. ◆ Explain SQL cursors in detail. ◆ Explain the concept of triggers. ◆ Describe the syntax of triggers. ◆ Explain the types of triggers. ◆ Explain functions in PL/SQL. ◆ Explain the syntax of a function. ◆ Describe how to execute a PL/SQL Function.
Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 35	Cursors, Triggers and Functions	2 hours 30 minutes
35.4		

Resources to be used	◆ Presentation ◆ Practical Guide
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Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 35	Cursors, Triggers and Functions	2 hours 20 minutes
35.5		

Resources to be used	◆ Presentation ◆ Practical Guide
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Do	Continue with the practice session. Ensure that the participants are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 36	Morning Energiser	10 minutes
36.1		

Resources to be used	Morning Energisers booklet
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Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 36	Relational Database Design	3 hours
36.2		

Resources to be used	◆ Presentation ◆ Practical Guide
Do	Run the PowerPoint presentation.
Say	◆ Explain how data is organised in tables. ◆ Explain the ACID properties of DBMS. ◆ Explain Atomicity Failure. ◆ Explain Consistency Failure. ◆ Explain Isolation Failure. ◆ Explain Durability Failure. ◆ Explain the concept of Entities. ◆ Explain what Attributes are. ◆ Explain what Database Constraints are; and list some of the constraints of database. ◆ Explain Database Anomalies. ◆ Explain how to remove Anomalies. ◆ Explain Normalization. ◆ Explain some of the normalization forms in detail, with examples.
Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 36	Relational Database Design	4 hours
36.3		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 36	Games and Activities	50 minutes
36.4		

Resources to be used	◆ Activities and Games booklet
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Notes for Facilitation	Refer to the Activities and Games booklet; pick out a game or activity and conduct it in class.
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Day - 37	Morning Energiser	10 minutes
37.1		

Resources to be used	Morning Energisers booklet
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Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 37	Query Processing	3 hours 30 minutes
37.2		

Resources to be used	◆ Presentation ◆ Practical Guide
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Do	Run the PowerPoint presentation.
Say	This slide explains what functions in Database are and lists some of the functions of Database.
Notes for Facilitation	◆ Explain the sum() function. ◆ Explain the avg() function. ◆ Explain the count() function. ◆ Explain the max() function. ◆ Explain the min() function. ◆ Explain the lcase() function. ◆ Explain the ucase() function. ◆ Explain the mid() function. ◆ Explain the first() and last() function. ◆ Explain the orderby() function. ◆ Explain the sum() function. ◆ Explain the now()and format() function.
Do	Ask the participants to open their Practical Guides. Ensure that they are able to apply their learning.

Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.
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Day - 37	Query Processing	4 hours
37.3		

Resources to be used	◆ Participant Handbook ◆ Practical Guide
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Do	Continue with the practice session. Ensure that the participants are able to apply their learning.
Notes for Facilitation	The participants are required to solve the programs given in the Practical Guide. Help the participants in debugging in case they encounter any bugs or errors.

Day - 37	Recap	20 minutes
37.4		

Resources to be used	◆ Presentation
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Notes for Facilitation	Recap the day's learning. * Refer to the presentation if required.
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Day - 38	Morning Energiser	10 minutes
38.1		

Resources to be used	Morning Energisers booklet
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Notes for Facilitation	Please refer to the "Morning Energisers" booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 38	Understanding Electronic Communication	7 hours 20 minutes
38.2		

Resources to be used	◆ Presentation ◆ Participant Handbook
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Do	Run the PowerPoint presentation.
Notes for Facilitation	Give an introduction to electronic communication. Explain how electronic communication can be classified.
Ask	Please note down the important points.
Notes for Facilitation	◆ Explain in detail the types of Electronic Communication and Signals. ◆ List some of the Common Electronic devices used in a computer system. ◆ Describe a Motherboard and its basic work. ◆ Explains Random Access Memory. ◆ Discuss the types of RAM, viz, DRAM and SRAM. ◆ Describe how BIOS and a BIOS chip look. ◆ State what is ROM; what are its uses. ◆ Explain why we use ROM. ◆ Explain what Expansion Buses are. ◆ Explain what Cables are and its different types used in Networking. ◆ Give an introduction to Connectors and the types of Connectors used in computers.

	◆ Explain Printers and their different types. ◆ Explain Networking and the types of Network. ◆ Explain Networking Devices. ◆ Explain the concept of Hubs and switches and how they are different. ◆ Explain the Network Interface Card.
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Day - 38	Recap	30 minutes
38.3		

Resources to be used	◆ Presentation
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Notes for Facilitation	Recap the day's learning. * Refer to the presentation if required.
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Day - 39	Morning Energiser	10 minutes
39.1		

Resources to be used	Morning Energisers booklet
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Notes for Facilitation	Please refer to the "Morning Energisers" booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 39	Operating System Fundamentals	7 hours 20 minutes
39.2		

Resources to be used	◆ Presentation ◆ Participant Handbook
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Do	Run the PowerPoint presentation.
Notes for Facilitation	◆ Explain the concept of Operating System with the help of a diagram. ◆ Explain the functions of an Operating System. ◆ Highlight the Structuring Methods of an Operating System. ◆ List the elements of an Operating System. ◆ Explain the concept of Kernel of an OS. ◆ Explain Input Output system of a computer. ◆ Explain the File System of an Operating System. ◆ Explain the concept of Scheduling of CPU. Also explain the algorithms for CPU Scheduling. ◆ Explain the concept of Synchronization. ◆ Explain to the students why we need Synchronization. ◆ Tell them about the concept of Interrupts. ◆ Talk about the concept of Deadlock in detail, the necessary conditions for a Deadlock and the methods to prevent.

	◆ Explain how Memory Management is done in an Operating System and the concept of Virtual Memory. ◆ Explain the concept of Segmentation. ◆ Explain Fragmentation and Defragmentation. ◆ Explain the concept of Demand Paging using a diagram. ◆ Explain the concept of I/O system. ◆ Explain What Worms and Viruses; and state the difference between the two. ◆ State the different types of Systems. ◆ Explain the features of MS DOS. ◆ Discuss some common operating systems like Windows XP and Unix and Linux.
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Day - 39	Recap	30 minutes
39.3		

Resources to be used	◆ Presentation
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Notes for Facilitation	Recap the day's learning. * Refer to the presentation if required.
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Day - 40	Morning Energiser	10 minutes
40.1		

Resources to be used	Morning Energisers booklet
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Notes for Facilitation	Please refer to the "Morning Energisers" booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 40	Local Area Network (LAN)	5 hours 50 minutes
40.2		

Resources to be used	◆ Presentation ◆ Participant Handbook
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Do	Run PowerPoint presentation.
Ask	Please note down the important points for a better understanding and retention.
Notes for Facilitation	◆ Give an introduction to Networking and its classification. ◆ Explain the concepts of LAN, WAN and MAN in brief.
Ask	Please take note of the types of LAN topologies and other important points we are about to discuss.
Notes for Facilitation	◆ List the various LAN Topologies. ◆ Explain the Bus Topology in detail, with the help of a diagram. ◆ Explain how Bus Topology works, with the help of a diagram. ◆ Explain the Star Topology with the help of a diagram. (In this the main computer acts as a server and all the other computers acts as Clients.) ◆ Explain the Ring Topology in detail. ◆ Explain the Tree Topology and with the help of a diagram. ◆ Explain why IEEE standards for networking were introduced while listing the IEEE standards for networking. ◆ Explain the concept of Network Protocols in detail; and explain all the Network Protocols with their uses.

	◆ Explain the concept of Routing Protocols and its types. ◆ Explain the concept of Routed Protocols. ◆ Enlist some common Protocols with their Port Numbers. ◆ Explain the TCP model for networking with the help of a diagram. ◆ Explain, with the help of a diagram, how the four layers of TCP model were distributed in the OSI model. ◆ Explain the OSI model for networking in detail, with the help of a diagram. Explain the 7 layers of the OSI model. ◆ Explain the Client Server architecture for networking with the help of diagram. ◆ Explain to the students that all the computers (users) act as the clients and the server acts as an interface.
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Day - 40	Recap	30 minutes
40.3		

Resources to be used	◆ Presentation
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Notes for Facilitation	Recap the day's learning. * Refer to the presentation if required.
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Day - 40	Critical Thinking	1 hour 30 minutes
40.4		

Resources to be used	◆ Multimedia (K-Yan) ◆ Presentation
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Do	Start the PowerPoint presentation.
Notes for facilitation	Tell the participants about critical thinking.
Ask	What, according to you, is the purpose of critical thinking?

Expected Responses	◆ “For creative solutions” ◆ “To solve the problems effectively”
Say	Tell them about the skills required for critical thinking.
Notes for Facilitation	◆ Discuss the importance of critical thinking. ◆ Tell them how they need to think logically. ◆ Ask the participants to fill out questionnaire.

Day - 41	Morning Energiser	10 minutes
41.1		

Resources to be used	Morning Energisers booklet
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Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 41	Test – 4	2 hours
41.2		

Resources to be used	Assessment Guide
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Notes for Facilitation	◆ Carry out a formative assessment for the topic LAN and SQL.
	◆ Refer to the questions in the Assessment Guide and conduct the test.

Day - 41	Project – Building Web Application	5 hours 50 minutes
41.3		

Resources to be used	◆ Participant Handbook ◆ Assessment Guide
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Do	◆ Divide the class into groups of 4 to 5 participants. ◆ Ask each group to name its team and choose a team leader for the project.
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Notes for Facilitation	◆ Participants are to select a project topic. No two groups can have the same topic. ◆ Let each team discuss and divide the project into smaller modules. ◆ The team leaders have to plan the project and assign modules to each team member. ◆ Help the participants with the hardware and software requirements for the project work. ◆ Explain the evaluation methodology for the project. ◆ Ensure that all the team members participate in the project work. ◆ Collect the “Project Proposal” form filled by each participant in the team. Suggested project topics: 1. Online Shopping 2. Online Billing System 3. E-commerce Payment System 4. Inventory System 5. E-examination System 6. Crime Management System 7. Students Attendance Analysis System for a teaching institution
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Day - 42	Morning Energiser	10 minutes
42.1		

Resources to be used	Morning Energiser booklet
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Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 42	Project	7 hours 50 minutes
42.2		

Resources to be used	Participant Handbook
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Notes for Facilitation	Help the students in implementing their project work successfully. Ensure that all the team members participate in the project work.
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Day - 43	Morning Energiser	10 minutes
43.1		

Resources to be used	Morning Energiser booklet
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Notes for Facilitation	Please refer to the "Morning Energisers" booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 43	Project	7 hours 50 minutes
43.2		

Resources to be used	Participant Handbook
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Notes for Facilitation	Help the students in implementing their project work successfully. Ensure that all the team members participate in the project work.
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Day - 44	Morning Energiser	10 minutes
44.1		

Resources to be used	Morning Energiser booklet
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Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 44	Project	7 hours 50 minutes
44.2		

Resources to be used	Participant Handbook
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Notes for Facilitation	Help the students in implementing their project work successfully. Ensure that all the team members participate in the project work.
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Day - 45	Morning Energiser	10 minutes
45.1		

Resources to be used	Morning Energiser booklet
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Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 45	Project	7 hours 50 minutes
45.2		

Resources to be used	Participant Handbook
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Notes for Facilitation	Help the students in implementing their project work successfully. Ensure that all the team members participate in the project work.
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Day - 46	Morning Energiser	10 minutes
46.1		

Resources to be used	Morning Energiser booklet
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Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 46	Project	7 hours 50 minutes
46.2		

Resources to be used	Participant Handbook
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Notes for Facilitation	Help the students in implementing their project work successfully. Ensure that all the team members participate in the project work.
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Day - 47	Morning Energiser	10 minutes
47.1		

Resources to be used	Morning Energiser booklet
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Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 47	Project	7 hours 50 minutes
47.2		

Resources to be used	Participant Handbook
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Notes for Facilitation	Help the students in implementing their project work successfully. Ensure that all the team members participate in the project work.
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Day - 48	Morning Energiser	10 minutes
48.1		

Resources to be used	Morning Energiser booklet
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Notes for Facilitation	Please refer to the “Morning Energisers” booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 48	Project	7 hours 50 minutes
48.2		

Resources to be used	Participant Handbook
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Notes for Facilitation	Help the students in implementing their project work successfully. Ensure that all the team members participate in the project work.
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Day - 49	Morning Energiser	10 minutes
49.1		

Resources to be used	Morning Energisers booklet
Notes for Facilitation	Please refer to the "Morning Energisers" booklet. You may pick the energiser that is recommended for the day of the week.

Day - 49	Interview Preparation	2 hours
49.2		

Resources to be used	◆ Multimedia (K-Yan)
Do	Play the multimedia unit.
Say	An interview is a guided dialogue to gain some information about a person.
Ask	State some important points to be remembered while preparing for an interview.
Expected Responses	"Dressing" "Knowledge about organisation" "Knowledge about the job profile"
Say	Discuss the points that should be kept in mind to prepare for an interview.
Notes for Facilitation	◆ Stress on the importance of developing skills and qualities and strong verbal communication skills. ◆ Discuss some sample questions that are asked in an interview. ◆ Discuss telephonic and other types of interviews. ◆ Summarise the session. ◆ Assign an activity for preparing their own professional résumé.

Day - 49	Project Presentation	5 hours 50 minutes
49.3		

Resources to be used	Assessment Guide
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Do	◆ Spread the presentations over two days and give the schedule to the students. ◆ Provide some tips on presentation. ◆ Ask the team members to present their project as a team.
Notes for Facilitation	◆ Give a detailed feedback on the each project work. ◆ Evaluate the project work as per the guidelines for Summative Assessment.

Day - 50	Morning Energiser	10 minutes
50.1		

Resources to be used	Morning Energisers booklet
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Notes for Facilitation	Please refer to the "Morning Energisers" booklet. You may pick the energiser that is recommended for the day of the week.
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Day - 50	Project Presentation	7 hours 50 minutes
50.2		

Do	◆ Continue with the project presentations. ◆ Ask the team members to present their project as a team.
Notes for Facilitation	◆ Give a feedback on the project work by the students. ◆ Evaluate the project work as per the guidelines for Summative Assessment.



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(A Joint Initiative with National Skill Development Corporation)

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